



AN EXPERIENTIAL APPROACH TO TEST DESIGN AND VALIDATION

Item Type	Dissertation (Open Access)
Authors	Araneda Galarce, Sergio Andres
DOI	10.7275/33072110
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Download date	2026-01-04 01:31:45
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Link to Item	https://hdl.handle.net/20.500.14394/19086

AN EXPERIENTIAL APPROACH TO TEST DESIGN AND VALIDATION

A Dissertation Presented

by

SERGIO ANDRES ARANEDA GALARCE

Submitted to the Graduate School of the
University of Massachusetts Amherst in partial fulfillment
of the requirements for the degree of
DOCTOR OF PHILOSOPHY

February 2023

College of Education

Research, Educational Measurement and Psychometrics

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ACKNOWLEDGEMENTS

Many people were part of my life in the Research, Educational Measurement, and Psychometrics (REMP) program at UMass Amherst. First, I would like to thank my advisor Stephen Sireci for his incredible, constant support and mentoring during this whole time. His support was absolute in so many ways. I appreciate this so much, considering my origins in Chile and the needs I faced as an International Student in a foreign country during the COVID-19 pandemic. Thanks a lot, Steve. You have changed my life; I will be forever grateful for the amazing life I will be able to have now thanks to the opportunities you have opened for me.

I also want to thank my boyfriend, Robert Gooden, who has been my support and heart all this time. Without him, I would not have had the love, peace of mind, and conditions to write this Dissertation. We faced hard times together in each of our lives when I wrote this Dissertation, and I will never forget your open arms and warm heart each day we have been together. I love you.

I also want to thank all my Chilean friends, that thanks to the technology available in our times, they could be present. I want to thank Ignacio Arenas for your friendship and everlasting support. You are like a brother to me. I want to thank my friends Rodrigo Pozo, Cristobal Puig, and Andres Navarro for their support and friendship all this time. I love you guys so much! I want to thank my friend Maritza Casanova, who has been there for a good conversation and advice all this time, my dear sister of life. I also want to thank Nicolas Messina, my dear confident friend of years, mi Vida y mi Cielo. I also

want to thank Fernando Gonzalez, my dear friend, always full of good advice, open doors, and good wine. Thanks, Fernando!

I also want to thank all the good professional friends I have made during my Ph.D. studies. I must start with FEVED (Forum of Chilean Professionals in Educational Assessment). Elisa, Fernanda, Ale, Vale, Sandra, Salome, Pedro, Francisco, Claudia, María Paz, Xaviera, Loretta, and Ricardo. What a group of excellent people and Professionals!! I am so happy we could build that group together. I also want to thank all the good friends I made during the Standards study group at NCME: Maggie Beiting-Parrish, Merve Sarac, Montserrat Valdivia, Debbie de la Torre, and so many others. I enjoyed all our discussions and work together (past and future!). I want to thank the people I could meet as co-chair and chair of the Graduate Student Issues Committee (GSIC) at NCME: Scott Holcomb, Janine Jackson, Professor Kyndra Middleton, and all other members of the GSIC. You have been part of my journey, and I am happy I could meet each of you.

I also want to thank my cohort members, Ted Dashier, Dongwei Wang, and Kevin O'Rourke, who have been amazing friends during my time in the program. Best cohort ever!! I also want to thank many professors who have been part of my journey. Starting in Chile, Leonor Varas and Monica Silva, your support was essential in my career, and I will never forget that. Special mention to Rebecca Zwick, who was always ready to give her support in my applications. Thanks so much, Rebecca. Also, one special mention to Daniel McCaffrey. He was assigned to me as a mentor at one NCME conference, and we could start a very nice academic friendship that have continued after. Thanks to much for your mentoring, Daniel. I also want to thank all the faculty at UMass Amherst that are not

part of my Dissertation committee: Lisa Keller, Scott Monroe, and Craig Wells. Your classes have been so great; I appreciate your commitment and talent to teaching us. One last thanks to David Foster, who was there for help when I needed it the most, and was such a pleasure to work with. Thanks so much, Dave!

I would like to thank the College of Education at UMass Amherst. One special mention to the program that allowed Graduate students to propose First Year Seminars, the class I could offer on Game Design was one of the most memorable experiences during my time as a student. Thanks so much for that opportunity. Also, I want to thank Jennie Southgate, who was always ready to help and encourage us as students. Thanks so much for your good energy and work.

Finally, I would like to thank all the members of my Dissertation committee, Jennifer Randall, April Zenisky, and Michael Walker. I am so honored to have your attention and feedback on this Dissertation's ideas. You are a World Class group of Professionals. You have provided incredible insights and encouragement, which I highly appreciate. Thank you so much for accepting to participate in this Dissertation committee.

ABSTRACT

AN EXPERIENTIAL APPROACH TO TEST DESIGN AND VALIDATION

FEBRUARY 2023

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INGENIERO CIVIL MATEMATICO, UNIVERSIDAD DE CHILE

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In this Dissertation, I will present the theoretical basis and two proof-of-concept studies of something I will call an “Experiential” approach to Test Design and Validation. This approach finds its origins in Game design, where the effort goes around the experience elicited by a device (a game). The idea of the experiential approach is to apply the mindset of game design into test design.

First, I present a literature review of conceptual frameworks referring to “Experience” and a novel conceptual framework for the context of Educational Testing. Then, I do a literature review of scientific epistemologies and feminist epistemologies in particular, that will set the stage for an Experiential approach to Validation.

After this, I present the general idea of an Experiential approach to Test Validation. This approach includes four questions: in what ways is this validity argument partial and situated? (Partiality question), is it ethical to use this test? (Appropriateness question), is it the test adequate for its purposes? (Adequacy question), why do I prefer this test versus other ethical and adequate tests? (Preferability question). I refer to this as the P.A.A.P. minimum in Test Validation.

I continue the Dissertation with two empirical studies that will serve as proofs-of-concept for each, the design and validation part of the experiential approach. Across both studies, I will respond to two conceptual hypotheses about the Experiential approach: what is the value above and beyond other approaches? And, is it practical/feasible to implement the Experiential approach?

First, I present the work done with Professor Lisa Keller about a diagnostic test we built together for her regression class at UMass Amherst. Second, I present an analysis done on information coming from Twitter (around 1,200 tweets) about the experiences of test users and other stakeholders in Chile relative to a test used there for college admission during the COVID-19 pandemic.

I finish this Dissertation with a Discussion about the Experiential approach. Overall, the experiential approach proposes something valuable above and beyond existing methodologies and is feasible to implement.

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CHAPTER 1: INTRODUCTION

Introduction

I would like to start this dissertation by doing a narrative about how the idea of an experiential approach was born, and why I think the field of Educational Testing should consider the ideas presented in this dissertation.

I have had a personal interest and investment in Game design in my life for a long time. Since I was a kid, I was fascinated by video games, board games, and games of all kinds that we played in Chile during my infancy in the nineties. My interest in Game Design motivated me to read more about their design practices. I read Schell's book "The Art of Game Design: A book of lenses" (Schell, 2008), the work of Csikszentmihalyi about Flow and Positive Psychology (Csikszentmihalyi & Larson, 2014a), and the founding paper of the Theory of Self-Determination (Deci & Ryan, 2000); which, together with my personal experiences designing board games, allowed me to grasp the ways Game design work.

Games need to be fun. There is no point in designing a game that will not be played. Csikszentmihalyi uses the term *Autotelic* to refer to activities with no purpose but just the interest in being experienced on its own (Csikszentmihalyi & Larson, 2014a). Games need to be autotelic, and tests in education are not autotelic. However, they can be.

As I entered the program at UMass Amherst on Educational Measurement and Psychometrics, it was inevitable to compare the practices in test design with the practices in game design and see their differences. One key difference is in that game design is based on the experience elicited by a game in the players, when test design focuses on the validity of inferences. In philosophical terms, Game design needs to be

phenomenological (i.e., centered around how players live the game within, beyond what the game designer observes). Phenomenology is an approach that concentrates on the study of consciousness and the objects of direct experience. So, if we can design games with this experiential/phenomenological way, can we do the same with tests?

The idea of centering test design around the experiences that are elicited is attractive. There is certain common understanding that being tested is an unpleasant experience. A method that could, at the same time, produce good quality tests from a technical perspective and also good user experiences would have a tremendous impact on the lives of many users, and provide a valuable advantage for test developers in the competitive market of tests.

I mentioned this difference between game and test design paradigms to my advisor Steve Sireci in one of our advisor/advisee meetings. He saw its potential far beyond what I could understand at that moment. He encouraged me to write and think more on the idea, and I tried to link it also with the topics we were studying at that moment in the Validity class that Dr. Sireci teaches at UMass.

This encouragement led to an essay about the idea of *an experiential approach to test design and validation*. In this essay, I explained the possibility of centering test design around the experiences elicited by a test, copying the practice in Game design; and doing test validation around user experience research, so both test design and validation could align in their focus.

One criticism to some validation approaches is that those are not well suited for the development stages of a test. There is value then to immediately produce both a design and a validation approach for practitioners, so both are coherent and synergic.

What is experience?

In studying this experiential approach, I felt the need to dive first into what do we mean by the word *experience*. To my surprise, there was a vast literature on Human-Computer Interface (HCI) and User Experience Research about the topic. Frameworks explaining what experience is, were usually meant to clarify the challenges in technology design, but not on the need for the scientific quality of claims.

That made me realize we needed one conceptual framework to speak about user experiences in testing that was comprehensive enough to be used, not just for design but for validation. By having such a framework, we could put the ground for an experiential approach under a specific clarification for what do we mean by the word “experience” in the context of testing test. I used the term *Assessment Experience* to refer to the experience associated with being assessed with a test.

I based this framework on Dewey's ideas about experience (Dewey, 1938), including three principles relative to the experiences: we live experiences as a continuum between past experiences and expectations of future experiences, we live experiences through interactions with the environment, and we make sense of experiences. The first chapter of this dissertation correspond to the explanation of this framework.

The concept of Assessment Experience allowed me to define what a test is from this phenomenological perspective. A test is a device with a purpose, helpful to elicit experiences that can allow an observer to make inferences about individuals.

Experience and Validation

As I moved into the program, I started to write together with my advisor a paper titled “An experiential approach to test design and validation.” In this paper, we introduced a conceptual framework to define test-taker experience and a general idea for how to validate a test around user experiences. User experience is not evidence by itself necessary, however, the information from this type of studies has an attractive characteristic: those studies are very good at finding the problems with a test, or sources of invalidity. This characteristic led to a more philosophical question about experience and validation: **is the evidence coming from the study of user experience self-sufficient to find all sources of invalidity in a test?** This is an assumption throughout this Dissertation that will be later discussed and problematized.

Given this assumption about the capacity of user experience research to find all invalidities in a test, an approach to validation can be defined. If we take a mode of argumentation like what mathematicians call as “a proof by contradiction,” we can leverage on the study of user experience to sketch a validity argument. In a proof by contradiction, we argue in favor of a claim by assuming a position against it, and showing that this assumption leads to a contradiction. In an informal logic way, this mode of argumentation is known as *reductio ad absurdum*.

To take this mode of argumentation has many advantages. For example, the validity argument emerges from the users rather than being imposed by the research that was already done or imagined relevant. Another advantage is that the validity argument can use research done about user experience like focus groups or direct recordings of interactions of users, sing the lenses of Validity to analyze information that may be

already available. Finally, another advantage is that the validity argument can help to disconfirm the concerns from the users about the uses and interpretations of the test, closing the gap between Validity and Legitimacy. I referred then to this idea as the *experiential approach to validation*.

Cronbach understood a construct as “a means of organizing experience into categories” (Cronbach, 1971; p. 464), influenced by the work of Singer (1959). If a source of evidence gives an alert about a potential source of invalidity, but we cannot triangulate the problem in user experiences, is it worth studying the problem? There is a tendency to privilege statistical analyses over qualitative research, most of the time obtaining results that cannot be interpreted and need extra work to make them useful. The experiential approach in this regard has an advantage compared to other approaches, in that it is “lazy:” it only studies what is relevant from the perspective of the experiences.

After sending a first version of the paper and having it rejected, the reviews made me realize the paper “An experiential approach to test design and validation” included too many topics within. I decided to separate the paper into two: one for the conceptual framework about Assessment experience, and one about the experiential approach to validation.

As I read more about validity and community-based research, I felt the need to have authorship in the paper about validation without my advisor’s guidance. I needed a space to propose something new that could take distance and criticize the literature on test validation from my beginner’s mind perspective¹, including criticisms of my advisor’s ideas about the topic (Sireci 2013, 2020a).

¹ A beginner’s mind is also a reference to a concept in Buddhism, known as Shoshin. It refers to having an attitude of openness, eagerness, and lack of preconceptions when studying a subject, even when studying at an advanced level, just as a beginner would.

In all fairness, we should clarify that Dr. Sireci disagreed with this decision. I took that decision anyways, making sure all the ideas in this paper were original from myself. Even though I think that this was a mistake at some point, I believe there were good reasons as well to write the validation part on my own. I think it is important to recognize Dr. Sireci's crucial role in encouraging the idea of the experiential approach and guiding me in its initial stages, even if he does not appear as a co-author of some of the papers that will come from this dissertation.

I should also recognize the influence of the concept of UNDERSTANDarization that my advisor published during the time I did my studies at UMass (Sireci, 2020b). The idea of a need for a profound re-thinking of educational testing, worrying about understanding each examinee and taking ownership of each instantiation of a test, found its inspiration in that paper.

In the papers of this dissertation, the connection with the concept of UNDERSTANDarization (Sireci, 2020b) is also expressed in a general criticism of what I paraphrase from Messick as the "conundrum of individual score interpretation." In his own words:

[...] Thus, as a consequence of strategic and stylistic consistencies, test scores may mean different things for different people – not just for different population groups [...], but for different individuals as a function of personal styles and intentions. This being the case, the notion that a test score reflects a single uniform construct interpretation or that validation should seek to defend (or challenge) a single test-construct match becomes illusory. Indeed, that a test's

construct interpretation might need to vary from one type of person to another (or from one setting or occasion to another) is a major current conundrum in educational and psychological assessment. It suggests that a purview of construct validation should include delineation of the alternative construct meanings of the test scores as a function of stylistic and strategic differences and, if possible, the development of diagnostic heuristics for deciding when a particular interpretation applies. (Messick, 1989, p. 55)

We read Messick's famous chapter in Educational Measurement 3rd edition (1989) in Dr. Sireci's validity class. I always found it so contradictory that we allow a test to be flexible for users in extreme cases like people with disabilities, but not in the mild cases of "phenomenological difference" like race or ethnicity. This links as well with the ideas in the Socio-cognitive view on assessment presented by Mislevy (2018). The concept of experience, in fact, appears in Mislevy's framework as a layer between what is cognitive and what is social.

In this regard, the experiential approach continues the concept of UNDERSTANDARDIZATION (Sireci, 2020b), fleshing out why and how we can better understand our examinees. This is an advantage of the experiential approach, as it provides a way to make more visible how a test is unequal across examinees, and how the delusion of assuming that all examinees experience in the same way the same stimuli end to create invalid interpretation and uses in a test.

The experiential approach also considers the idea of a preference for tests that can elicit good or "positive" experiences. Sireci coined the term "positive assessment" for this, presenting something at NCME 2021 on the idea (Sireci & Araneda, 2021). Positive

psychology brings an exciting theory to think about test design as a task meant to create valid interpretations and uses, and positive experiences. This aspect of positive experiences entails ideas from Duolingo about making tests delightful (Wodzak, 2021) and their concept of test-taker experience (Burststein et al., 2021). The experiential approach has an advantage relative to other approaches, as is capable to integrate this type of needs (positive experiences) into test design and validation.

The role of values in test validation

As part of the value of my “beginner’s mind” perspective when reading about test validation, I felt concerned to see in Messick that there is no mention of values in his definition for validity. Messick (1989) defined validity as “an integrated, evaluative judgment of the degree to which empirical evidence and theoretical rationales support the adequacy and appropriateness of inferences and actions based on test scores or other modes of assessment” (1989, p.13). Even if Messick is one of the few defenders of the Appropriateness question in test validation (is it ethical to use a test?), Messick definition leads to an amoral evaluative judgment, the word ‘values’ is not there in the definition. This surprise was reinforced by Shepard’s comment in her famous work titled “Evaluating Test Validity” (Shepard, 1993). She mentioned that “validity investigations cannot resolve questions that are purely value choices” (p. 428). I had a strong reaction to that phrase the first time I read it, because any validation researcher can then forget about values and justify that stance by assuming that it is “out of scope” of his/her profession. That way of thinking about value choices leads to an amoral practice in my opinion. Are we, the professionals of Educational Testing, available for any value choice?

Later, I found the proper term in Lang's work to refer to Shepard's stance: Moral quietism (Lang, 2010). Moral quietism refers to "the assumption that in the absence of an absolute determination of truth—who is right and who is wrong—relativists would be unable to justify a position and thus make moral or political decisions based on it." (p. 320) Test validation is not meant to resolve questions of value choice, but is not meant either to make them invisible. Test validation, exercised in a socially responsible way, requires showing and documenting the values and political discussions arising from the uses and interpretations of a test, so a better evaluative judgement can be performed.

The experiential approach has an advantage in this regard, as is capable to visualize the values of the users of the test through UX Research. This led to the concept of *value rejection*, defined as the moment when a user feels uncomfortable on an ethical level about any aspect of an assessment experience.

The literature on test validity speaks a lot about the values of researchers, but not necessarily about the values of the communities. The experiential approach takes a different stance regarding the Appropriateness question (is it ethical to use a test?), considering inappropriate to administer a test when the communities reject the values embedded in an assessment experience. Value rejection is considered then a source of invalidity. This view about validity is partial, in that a democratic meta-value is present in the definition itself of when a test is appropriate to be interpreted or used.

In my opinion, Messick fell short when including the topic of values in his definition for test validity, in that the values of the communities should also be part of that “support” mentioned in his definition, in the same level to theoretical rationales and

empirical evidence. And not any values, but the values of the affected communities by the uses of the test.

Situated Knowledges

The problem about values in test validation led me then in a quest for methods that can understand the values of the communities affected by the uses of a test. I found a class about community-based research methodologies in the Anthropology department with Professor Julie Hemment, that fulfilled my interest. We studied many interesting topics like indigenous research methodologies, feminist empiricism, and Participatory Action Research (PAR). Along those lines, I found in one of the papers of the class the concept of Situated Knowledges (Haraway, 1988). Donna Haraway, the author of the concept, is a feminist scholar who wrote in the eighties an amazing paper about the concept that just blew my mind, so full of color and deep insights. I found something truly remarkable: an epistemological engine that could be used to support an experiential approach to validation.

Situated Knowledges is the construction of knowledge from the assumption that a researcher is corporal and situated somewhere in the world. Haraway (1988) denounces the practice of scientific research hiding in an invisible position as the "god-trick," assuming that their research can be done from a god's point of view. In acknowledging the embodiment of all scientific work, a phenomenological take becomes a priority when we do research about other human fellows.

So, this acknowledgement about the embodied nature of any scientific discovery connects well with the idea of Experience. Particularly in a feminist project of Social Sciences, the recognition of that embodiment will see its more practical side in the study

of experiences, either of people being studied or of the researcher that is constructing knowledge.

With all of this, I was capable to bring not only an approach to test validation, but a definition for test validity from a critical perspective. I initially defined Test validity as the set of Situated Knowledges constructed between a researcher and the communities affected by the uses of a test, about the Appropriateness, Adequacy and Preferability of future instantiations of a that test, the interpretations to be done with its results, and the procedures in which those instantiations will be used for. This definition reuses vocabulary from Messick's definition, but including a critical/feminist perspective into it.

This definition of test validity has the advantage of seeing test validity as a fragile property of a device that depends on context, ultimately defined by each of its instantiations, and co-constructed with the communities. This contrasts with other approaches that make general claims, speaking about "a degree" like Messick (1989) or "plausibility" like Kane (1992, 2013a), hiding the ugly realities behind those concepts: not all examinees have valid instantiations. Who is not correctly interpreted? Who suffers due to invalid uses of tests?

Test validation as the creation of experiences that well-explain.

One new concept appeared on the map after studying further the concept of Situated Knowledges: Rhetorical Argumentation (Tindale, 2004). Lang (2010) explained that Situated Knowledges matches well with the idea of an argument built around the elicited experience, so the argument can be confirmed that well-explain something, that is possible for an audience to understand it. When I read in Tindale's book that an argument can be designed around the experience the argument elicits, then something important

clicked. A genuine experiential approach to validation cannot be just about studying user experiences. It should be about creating an argument as an experience, as part of an overall Assessment Experience. Tindale provided background and basis in the literature of New Rhetoric (Perelman & Olbrechts-Tyteca, 1969) to move beyond the Toulmin Model (1958), the base of Kane's argument-based approach (Kane 1992, 2013a), capable to embrace a phenomenological take and Situated Knowledges as an epistemic engine.

This new concept of rhetorical argumentation closed the circle. The literature mentions the dialogical aspect of testing, and testing-as-a-conversation metaphor (Hubley, 2017; Markus & Borsboom, 2013). A very curious phenomenon not discussed much in the literature relates to the implicit dialogues in testing between the test developer and examinees. Can we elicit construct-irrelevant variance by the way we explain a validity argument? If an examinee does not understand why a test is valid, or if they believe a test is not valid, can this have an impact in their assessment experience? If so, this would mean that test validation can generate invalidity by how it is communicated. To own this weird phenomenon, we must include the experience that not only a test elicits, but the validity argument elicits, as part of a broader understanding of the Assessment Experience.

The concept of Rhetorical Argumentation then allowed me to define test validation from a phenomenological perspective. Test validation is the process by which a researcher builds an experience that well-explain the validity of interpretation and uses to the communities affected by the uses of a test.

The experiential approach in this regard supports the idea that the needs and values of the audience to which the test is directed should ultimately define test

validation. I added my partial consideration that the communities affected by the uses of the test should be part of that audience. This is an advantage of the experiential approach, as it can tackle the problem of test legitimacy in the eyes of society, and test validity as part of it.

Also, one advantage of the inclusion of Rhetorical argumentation is that it makes more transparent the use of rhetoric to convince the communities about the validity of interpretation and uses. With a bad intention, the knowledge in rhetoric can be used to persuade someone against their best judgement. Kane speaks about persuasive arguments in fact (Kane, 2013a), using the word persuasive rather than well-explained arguments. This opens the door for *dark rhetoric* and the creation of arguments that can be incorrect, but persuasive.

The rhetorical aspects of an explanation have a limit, in that they should promote the best judgment of the audiences about the validity of a test and not persuade at any cost. The tendency to focus on evidence that supports statements versus the lack of attention on the rebuttals of statements can be considered a dark rhetoric move, inducing the audiences to have an impression about a test against their best judgement. Hiding the positionality of the validation researcher and what it makes them partial can also be considered dark rhetoric, by omitting the historical, cultural, social, political contingency of a validation effort. The experiential approach makes visible those potential abuses of the rhetorical aspects of argumentation (sometimes disguised as “objectivity”).

One common criticism to the argument-based approach is that it is too complicated as an explanation, making it impractical. The argument-based approach proposes a way to present a validity argument that is directed to the community of

experts, and not other communities (and even for the community of experts it can be too complicated to communicate). The experiential approach has a political advantage in this regard, as it is more conscious for the need to direct an argument to many audiences; and articulate a research agenda to satisfy the needs of explanation of all the communities affected by the uses of a test, not just the community of experts. This diversity of audiences also leads to a scientific advantage compared to other approaches to test validation, leading to a “strong objectivity” (Harding, 1992). The Experiential approach is an attempt to explore a practical way to achieve this goal for arguments that well-explain the validity of a test embedded in an ecosystem of devices that conform an assessment technology. This approach is practical enough to become an improvement on the argument-based approach.

About this dissertation

I decided then to write more about the experiential approach to test design and validation, making this dissertation a series of papers with all the work I did during my Ph.D. time at UMass Amherst. I wrote three papers before starting the official period of dissertation work. One paper speaks about a conceptual framework for Assessment experiences, one about feminist epistemologies, and one speaks about the experiential approach to validation.

I wrote two extra pieces of research specific for dissertation work. I put efforts into “proof-of-concepts” to finish this dissertation defense with empirical studies. One paper is a proof-of-concept of the design part of the experiential approach. Here, the concept of *Experience specification* is the cornerstone of the design part of the

experiential approach. I made a proof of concept with Professor Lisa Keller, designing one test used for diagnostic purposes.

A second piece of research was related to the validation part. I did a research of the experiences shared by users on Twitter about a test used for college admission in Chile, analyzing the experiences, studying invalidity hypotheses and sketching an argument that can well-explain the validity of interpretation and uses for that Test in the specific year 2021 amid the COVID-19 pandemic.

In terms of research questions, I will define two conceptual hypotheses for the experiential approach:

- a) The experiential approach will provide valuable information to improve test design and validation practices above and beyond information provided by traditional methods.
- b) By implementing aspects of the experiential approach, we can evaluate its feasibility and practicality.

The final outline of the dissertation is the following. I will introduce a conceptual framework to define 'user experience.' Next, I will introduce the literature of feminist epistemologies, making one special review of the concept of Situated Knowledges (Haraway, 1988), and providing a literature review about epistemologies in Test Validation.

After this, I will introduce the concept of an experiential approach to validation. I will propose a way to validate tests around the intensive study of user experiences. This approach will have a post-structuralist definition of validity based on the concept of

situated knowledges and a more specific procedure on how to develop and validate a test around the experience elicited.

After those sections, I present two empirical studies associated to the Experiential Approach. One will correspond to a proof-of-concept for the design part of the approach, and one for the validation part. I will present both studies in one single general chapter for Empirical Studies. I will end this Dissertation with a discussion section, coming back to the conceptual hypotheses of this work.

**CHAPTER 2: LITERATURE REVIEW ABOUT USER EXPERIENCES, AND A
PROPOSAL FOR A CONCEPTUAL FRAMEWORK FOR ASSESSMENT
EXPERIENCE**

Introduction

Researchers and practitioners in educational testing are starting to consider the phenomenological aspects of testing in more intentional ways. Recent examples include research on test-taker engagement (e.g., Wise & Kong, 2005; Wise, 2017; Wise & Kuhfeld, 2021), Mislevy's (2018) socio-cognitive foundations of educational measurement, or the concept of test-taking experience proposed by researchers at Duolingo (Burstein et al., 2021). A focus on phenomenological aspects of testing, or test takers' experiences, is new to psychometrics. However, focusing on end-users' experiences is quite typical in product design, which is known as User eXperience (UX) research.

UX research is standard when developing apps and other types of technologies. Examples of methods used in UX research that have crossed over into the testing industry include empathy maps, customer journey mapping, co-design sessions, etc. Recent efforts in educational assessment have imported many of those methods, usually in the construction of learning experiences in Education. This new ethos is part of the identity of new EdTech companies and start-ups, bringing the promise of providing a valid assessment and the added value of a good user experience. Computer-based testing (CBT) provides one good example. Every time we build a CBT, we design a test and an application that involves all the same challenges people in human-computer interface study for other types of applications.

In educational test development, cognitive response processes intended to be elicited by a test represent the phenomenological aspects of a test. A full consideration of validity evidence based on response processes focuses on properly representing the

cognitive constructs measured while minimizing construct-irrelevant variance (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 2014). Testing agencies often disregard other aspects of users' experiences because the test takers are typically not the client or user of the information that supports a testing program. Thus, some speak about test-takers instead of test users. This movement towards the users who interact with testing products is in sharp contrast to how EdTech companies work, in that they are independent of any institutional user, motivated by bringing solutions to their examinees' needs and experiences.

However, with the increasing use of experience as a conceptual category, there has been little discussion on what do we mean by the word *experience*. To define experience is a challenge on its own, in which philosophers have failed to produce a strong consensus. Despite those failures, there is value in taking some time to understand what we mean with the word experience, what it encompasses, and — if possible — define a conceptual framework to refer to it and have a common vocabulary. Other fields have tried this path, and now is the turn of educational testing to build a conceptual framework to include the phenomenology of being assessed into test development and validation.

In this chapter, we propose a conceptual framework for user experiences in testing. We will refer to a person's test experience as the *assessment experience*. The proposed conceptual framework is based on other frameworks found in the literature and adapted to the needs of educational testing. In addition to methods found in UX research, we also borrow from Dewey's three main principles of experience (Dewey, 1938), to be

explained later in this paper. We proceed by first reviewing the literature on conceptual frameworks for UX. We then propose a model for using UX methods in test development. Next, we explain how the methods we use in educational assessment can be mapped into our framework and provide a short example of how to use the framework. We end with a discussion on how validation can take advantage of the rich evidence coming from UX research.

Theoretical Basis for the Concept of Experience

The study of examinee experiences requires familiarity with some philosophical concepts. The definition of “experience” can relate to two different concepts: (a) the more immediate and instantaneous stream of information a person can perceive; or (b) a narration about something that happened, which represents an integration of the raw stream of information into a more cohesive and curated narrative of what is happening or just happened. Jay (2005) noted this difference in the etymology of the word experience in German. *Erlebnis* and *Erfahrung* both translate into English as experience but correspond to quite different notions of experience. *Erlebnis* is more related to “lived experience,” with a sense of unity and lack of differentiation or objectification. It connotes a more immediate, pre-reflexive, and a personal variant of *Erfahrung*. *Erfahrung* corresponds to some cognitive judgment about the “lived experience.” It relates to some learning process, integrating discrete moments of experience into a narrative whole or an adventure. Here, there is some role of memory in the perception of the experience. *Erfahrung* is related to a “dialectical notion” of experience (Jay, 2005. p. 11).

As we describe later, both concepts of experience have relevance to test development and validation. Several authors who proposed putting users' experience at the center of a design process borrow the concepts of experience from Dewey. In his book *Experience and Education*, Dewey (1938) proposed a new type of education that acknowledged the connections between education and personal experience. In his philosophy, experience has two main principles: a principle of continuity and a principle of interaction. The principle of continuity refers to experience lived as a continuum with past experiences and expectations of future experiences. It assumes all our instantaneous experiences, both past and future, are part of the conscious process by which we process the raw stream of information we experience.

Second, the principle of interaction says the experience happens as we interact with our environment, and it assumes there is no experience without interaction. Interaction refers to the cerebral or corporal activity that happens in the specific encounter of a person and a situation. Both concepts are part of Dewey's ideas for education in that every educational experience needs to be done in a way that makes sense with past experiences, allows for growth, and does not block future educational experiences. Dewey considered the difference between a generic experience and "an" experience as a moment with a beginning and end.

Forlizzi and Battarbee (2004) proposed a theoretical framework for designing experience in interactive systems, inspired by Dewey's work. They defined an interaction-centered model in contrast to user-centered approaches found in the user experience literature. The difference is that, to characterize and cast an experience, a designer considers users as isolated versus interacting with products. They defined three

types of interactions between users and products: fluent, cognitive, and expressive. *Fluent* interactions refer to a user's well-learned interaction, unconsciously done when facing a product (e.g., riding a bicycle). A *cognitive* interaction is when the user's cognitive resources are needed when interacting with a product (e.g., when trying to understand the functioning of a new device). *Expressive* interactions help the user form a relationship with a product, like personalization in the presentation of a device or avatars. As described later, considering a testing situation as an interaction between a user and a test helps to study examinees' experiences.

Forlizzi and Battarbee (2004) described three types of experience: experience, “an experience,” and co-experience. Experience and “an experience” are the two definitions of experience mentioned earlier. Co-experience is a new idea that considers how the experience is processed and evaluated with others. As they described,

Social situations greatly influence co-experience. For example, whether running out of gas when driving to the countryside with friends is viewed as an adventure or a disaster depends on how the friends decide to interpret the situation. One person might be upset, another might point out the humorous potential, and a third might agree. (p. 263)

In the case of educational assessment, this co-construction of the experience can happen with different actors in an assessment process (e.g., peers, teachers, parents, school administrators.)

Wright and McCarthy (2010) also focused on co-experience in product design. They proposed a design paradigm called "experience-centered design," in which the

designer and user enter in a dialogical process to create an artifact. They also discussed how the user's experience is affected by the dialogue (explicit or implicit) between designer and user. Since any person can be a co-constructor of an experience, there is a possibility of creating a co-experience with the designer of a product. We refer to this as the 'dialogical experience.'

A third principle about the experience, not directly stated by Dewey but understood as a principle by subsequent authors as part of Dewey's concept of experience, refers to the idea that we always make sense of the experiences. While we experience something, there is an underlying process by which we process the information we are receiving, the interactions we are having, the connections we make with past experiences, and expectations of future experiences. The concept of "an" experience and co-experience in Forlizzi and Battarbee's (2004) framework or the dialogical experience in Wright and McCarthy (2010) both consider the idea of this third principle that humans make sense of their experiences.

A Conceptual Framework for Assessment Experience

This section draws from previous conceptions about experience to propose a conceptual framework for assessment experience. The proposed conceptual framework is a starting point for understanding examinee's experiences in educational testing, that will need further refinements.

In our proposed framework, we draw from the previous works cited to define different levels in the sense-making process. We specify five levels of sense-making:

- *Free-stream* refers to the rawest and instantaneous experience, a stream of self-talk. Think-aloud protocols are an example of a method that taps in this level, barely made sense of.
- *Immediate* refers to the experience that just happened, on which there will be some level of evaluation, of integration of the raw experience, but is still incomplete. Examinees' report right after they respond to an item corresponds to the Immediate level of sense-making.
- The *integrative* level refers to the dialectic experience without co-construction of the experience. It refers to a more constructed narrative of an experience with a beginning and end, done in isolation from other stakeholders.
- *Co-integrative* refers to a dialectic experience enriched with the co-construction with others. This level appears in focus groups or some modes in ethnographic research.
- Finally, the *dialogical* level of experience corresponds to the most structured and long-term evaluation of experience that includes the experience affected by implicit and explicit dialogues with the test designer.

Table 1 summarizes these levels of experience, ordering them in terms of the increasing complexity of construction. Table 1 also illustrates that the first two levels can be studied at the item or activity level, whereas the three higher levels refer to the overall assessment experience.

Table 1

Levels of sense-making in the Assessment Experience

Level of Sense-Making	Description	Investigative Focus
Free Stream	Self-talk, the continuous stream of information and responses to the stimuli.	Item or Activity
Immediate	An initial and incomplete integration of a short experience that just happened.	Item or Activity
Integrative	A more complete and integrative evaluation of an experience that has a clear beginning and end, expressed as a story.	Overall Testing Experience
Co-integrative	Similar to integrative, but now the story is built including the voices of other persons, integrating them in the process of making sense of the free-stream, unitary experience.	Overall Testing Experience
Dialogical	Similar to co-integrative, but now the other is the test designer in particular, both making sense of the experience in an empathetic dialogue.	Overall Testing Experience

In addition, we define five types of interaction between an examinee and a test:

- *Fluent* interactions happen in an automatic or pre-conscious way (e.g., riding a bicycle is an interaction between user and “product” that is automatic after one learns how to ride a bike).
- A *cognitive* interaction involves a cognitive process triggered by the encounter with the object. This is the primary type of interaction expected with the test in educational testing.
- An *emotional* interaction refers to emotions in the specific encounter between an individual and a test. Anxiety, disengagement, boredom, and fun correspond to an emotional interaction.

- *Expressive* interactions are manifestations of self-expression in reaction to the test. An opinion appearing in the mind of an examinee in response to a test item is an example of expressive interaction.
- Last, a *physical* interaction corresponds to interactions that happen within the body. In the context of a test, having the need to go to the bathroom, feeling hungry, cold, hot, and so forth; are examples of physical interactions. In some cases, a test that demands a physical action by the examinee as its primary expected interaction (e.g., physical abilities testing for firefighters, police, the military).

These five types of interactions borrow the fluent, cognitive, and expressive interactions proposed by Forlizzi and Battarbee (2004) and the emotional interaction from what Holbrook and Hirshman (1982) name ‘affect’ in their experiential approach to marketing. In our framework, we add the physical interaction to account for physical situations occurring during an exam, such as gut pain due to anxiety or exams involving physical responses.

The continuum of past and future experiences involves characterizing and studying the relationship between past experiences, expectations of future experiences, and the instantaneous experiences that occur while taking a test in its many “stages.”

There are six stages relevant to the assessment experience:

- *Ante-experience* refers to experiences related to a test well before an examinee starts to prepare for it.

- *Pre-experience* refers to experiences preparing for the test; that is, experiences when an examinee is sure he or she will take the test.
- *Zero-experience* refers to the experience while taking the test.
- The *inter-post experience* will refer to the experience between taking the test and receiving the results.
- *Post experience* refers to the experience right after receiving the test results.
- Finally, *long-term experience* refers to the experience after a reasonable amount of time since receiving the results of a test.

The levels of sense-making, types of interactions, and temporal stages of an assessment experience provide a vocabulary to describe examinee's experiences.

An example can help illustrate the different concepts and stages of the framework. Imagine you are going to take a test for college admission. You will prepare for this test (pre-experience), and in your preparation you will be affected by the expectations you have relative to the test (ante-experience). You will interact with materials preparing you for the test and have different reactions to them during the preparation. The messages coming from the testing agency can affect your expectations about the test itself (dialogical level of the pre-experience) and what your friends and family think about the test (co-integrative level of the pre-experience).

When you take the test, you will experience many things in response to the test. You can correctly respond to an item and react as expected (cognitive interaction in the zero-experience on the free-stream level), or you can feel confused about the stem of an item (again, a cognitive interaction). You can respond well to an item but react to the stem (e.g., if you think it was racist in some way, which would be an expressive

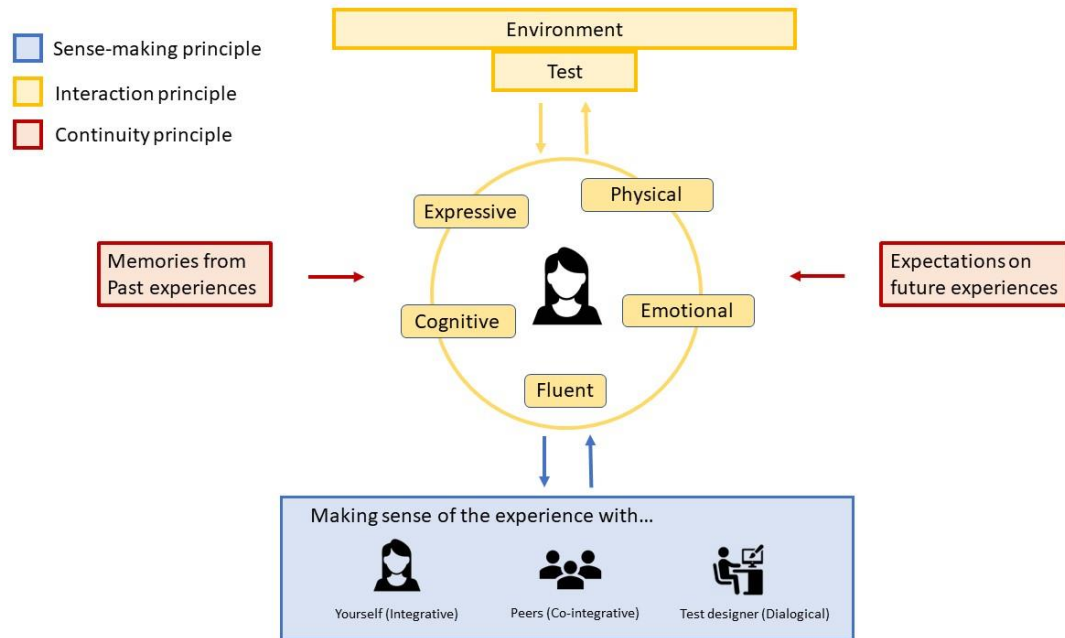
interaction), or you can be affected emotionally by the stem of an item if it speaks about something sensitive to you (emotional interaction). When you interact with a test, you may tend to first see a figure before reading the stem of a question (fluent interaction on the free-stream level), or you can have a strategy to respond to items in a certain way or a specific order (expressive interaction in the immediate level of the experience).

After you finish the test, you may have an opinion regarding the overall experience (expressive interaction on the integrative level) that you might share with your friends, which can change your perception of the assessment experience (expressive interaction on the co-integrative level). That experience can lead you to participate in a rally against standardized tests because you feel that the test is racist and unfair (physical and expressive interaction on the dialogical level).

Figure 1 presents a summary of the three principles in the conceptual framework presented here about user experience.

Figure 1

Diagram summarizing the Conceptual Framework for Assessment Experience



Methods for UX Research in Testing

The study of experience uses many varied methods. Qualitative methods such as think-aloud protocols are well-suited for the non-integrative levels of the experience. However, it is likely to be more difficult to measure other levels of experience. The GAMEX scale is a survey that tries to identify how gameful an experience was (Eppmann, Bekk & Klein, 2018). Surveys can be an effective method for gathering information about examinees' experiences.

Csikszentmihalyi and Larson (2014b) use an "experience-sampling" survey methodology to capture information about a person's immediate and integrative experience during their daily routines. This method involves using a survey to gather experience data at random moments during the day. The survey includes open-ended

questions and specific Likert-type questions about mood and relevance of the activity a person was doing. Photovoice (Wang & Burris, 1997) is another method used in ethnography by which subjects take pictures and write a text about the picture to illustrate something about their experience and the topic the ethnographer wants to investigate.

It is essential to note that experience is not a single construct in the classical sense in which the word is typically used. A theoretical reduction of all experience to a set of measurable constructs will inevitably simplify and constrain the analysis of the experience. However, when we detect some problem related to the quality of the experience elicited by a test, we will want to create instruments that are particularly suited, like in the GAMEX scale. Thus, we believe that the study of user experiences in testing should be open-ended, to find and measure later specific constructs of relevance.

To study some aspects of experience, we may need to tell examinees in advance to be particularly aware of certain occurrences such as their emotions or their cognitive difficulty when responding. This strategy requires persons aware of their experiences to help with experiential studies, like how sensitivity reviewers evaluate items on a test (Sireci & Mullane, 1994).

All analysis of direct observations of emotional states and reactions are sources of information about experience, by-passing the use of oral language. Similarly, brain scan imagery, or other devices that register information directly from the body, can be used. For example, Dew, Galassi, and Galassi (1984) used physiological measures and questionnaires to study math test anxiety. Osborne (2007) used several physiological measures, like skin conductance, heart rate, surface skin temperature, and blood pressure to study gender stereotype threat on math proficiency. Maddox (2017) analyzed talk and

gesture in the context of PIACC, tapping many types of interactions in the free stream, immediate and dialogical levels of sense-making in a problem-solving task, done in a computer and in the presence of an interviewer.

Another potentially valuable data source on examinee experience is log data from computer-based tests. These log data are often called “response process data” and can be part of the experience analysis if specific aspects of experience correspond to patterns in the log-data files (Maddox, 2017; Goldhammer & Zehner, 2017; Araneda et al., 2022). This tracing will need a triangulation, linking some direct behavior registered in log-data files to an examinee’s experience. Questions asking about the immediate experience of examinees in between items can be one way to triangulate log-data files and examinee experience.

All research related to response process can be considered as well as part of UX research (Padilla & Benítez, 2014; Ercikan & Pellegrino, 2017; Zumbo & Hubley, 2017). Think-aloud protocols, cognitive labs, ethnographic studies, focus groups and many others explored are a natural place of UX research methods.

Table 2 summarizes the different methods used to study examinee experience. Specific data collection methods apply for each combination of the level of sense-making and interaction type.

Other research methods relate to the temporal phases of an assessment experience. Funds of knowledge (Moll et al., 1992) and socio-cognitive studies (Hubley et al., 2017) are theories and methods that help to understand the zero-experience in a contextualized way. Research on score reporting (e.g., Kannan et al., 2018; Zenisky & Hambleton, 2012)

also includes a phenomenological dimension in the specific stage of receiving the results in an assessment experience. Some researchers criticizing standardized tests and their consequences show aspects of the long-term experiences of examinees (Walker-Johnson, 2009).

The Experiences of Examinees shared in Twitter: an Example from Chile

In the previous sections, we described how UX research methods could be applied to the study of examinees' experiences and presented a conceptual framework for doing so. In this section, we present a short example of how to use the conceptual framework for UX in the case of a test used for college admission selection in Chile.

We gathered information about tweets shared by examinees and others about a test used in Chile for university admission decisions, known as *Prueba de Transición* (PDT). This test happens during summer in the southern hemisphere (January), and it is the test battery used for university admission.

Table 2**Methods for Studying UX by Levels of Sense-making and Interaction type**

Level of sense-making	Type of interaction				
	Fluent	Cognitive	Expressive	Emotional	Physical
Free Stream	Direct observation, Log-data files	Think-aloud, Log-data files	Think-aloud	Direct observation of the body, face, Think-aloud	Direct observation
Immediate	Direct observation, Log-data files	Cognitive interviews	Survey-like questions after responding to an item	Survey-like questions after responding to an item	Direct observation
Integrative	Log-data files, Interviews	Specific surveys	Surveys	Surveys	Surveys
Co-Integrative	Direct observation, Ethnographic studies.	Focus groups, Surveys.	Focus groups, Surveys	Focus groups, Surveys	Focus groups, Surveys
Dialogical	Direct observation, Ethnographic studies.	Focus groups specially tailored, Surveys.	Focus groups, Social media communications with testing agencies	Focus groups, Social media communications with testing agencies	Focus groups, Ethnographic studies

In early 2021, the COVID-19 pandemic was spreading across the country as in other parts of the world, and there was a big question mark around whether doing any large-scale standardized test with the stakes and extension that the PDT test has. The Chilean government and DEMRE – Departamento de Evaluación, Medición y Registro Educacional, the testing agency in charge of designing and administering the test – decided to take the test in person. They decided on the agreement of having strict health precautions; including the use of masks, proper ventilation of classrooms, and others; however, students had to face a difficult decision of take the test in person and risk

exposure to the virus, or be safe at home and postpone their admission to universities one year.

The experiences of the examinees emerge in the content shared in social media like Twitter or Tik Tok. This situation provided a perfect opportunity to illustrate how to use the conceptual framework proposed in this paper to categorize the experiences of the examinees during their assessment process. Let us review this first tweet (translated to English):

@demre_uchile made the announcement that the test for college admission will be done in two instances, with less questions and masks. And the big question in the middle of the pandemic, with new cases in the rise and without vaccines is ... why the test is not delivered online?

We interpret this as an expressive interaction between the examinee and the test. This examinee is wondering why the test is not available in an online format. The user is directing his message to DEMRE, elucidating a sense-making process on the examinee's integrative level during his/her/their pre experience. If DEMRE responded to the tweet and a new interaction arises from that dialogue, we would also identify some interaction now on the Dialogical level.

Let us review a second tweet (translated to English):

@demre_uchile I would like to know what was the criteria to assign testing locations (Antofagasta), we live close to three testing places in the area and my daughter needs to go to another zone, needing to take public transportation and increasing her risk for COVID infection.

This tweet comes from a parent complaining about the logic in which the students were assigned to locations. The context of the COVID pandemic increased the concerns on moving far away from home and the infection risk due to the need to use public transportation. This tweet happened after the student did the test. We will locate it in the inter-post experience. We will interpret this tweet as an emotional interaction, frustration, of the student's parents.

It is unclear if this tweet is either in the integrative level of making sense or in the co-integrative, as the parent knows somehow that other available locations were closer to the one assigned to her daughter, and that might be something told by other parents as part of their co-construction of the experience. We will assume again that is at the integrative level as there is no mention of others affecting the making sense process and interaction in the user experience.

The previous two tweets can serve as examples of how to understand in a better way the experiences of the examinees and other relevant actors like parents by using the proposed conceptual framework proposed for UX in testing.

The two tweets point to some potential sources of invalidity in the test. Did students' frustration of taking the test in an in-person setting – considering the risks of infection – affect students' anxiety and general mood when doing the tests? Did that affect all the population homogeneously, or were some subgroups affected more than others?

The composition of the students who took the test that year showed an unusual change, with a jump from 11% to 17% of participation in private schools (DEMRE, 2021), usually associated with wealthy students. This jump is not

explained by a sudden change in the composition of schools but by students' decision to opt out. That result and the tweets suggest that the impact on the experiences led students not to lower scores in the test because of anxiety but to postpone their plans for higher education, having an inhomogeneous nature on its effect against populations of lower SES. The second tweet also suggests that parents' views about the risks of taking the test moderated the opting out.

This case from Chile is a perfect example of how UX research in testing informs how to enhance the examinee's experiences and generate hypotheses for potential sources of invalidity in the interpretations and uses of this test. Further inquiry is required to generate evidence about the uses and interpretations. For example, it would be interesting to see how the opt-out of students in their last year of high school was moderated by SES variables and by the distance to the testing place beyond the influence of SES variables.

The previous example illustrates how to use the conceptual framework proposed here. First, the model can map in a three-dimensional space the accountings about the experiences based on its temporal stage in an assessment experience, the level by which the experience relates with the making-sense process of the subject, and the type of interaction that is predominant in an experience. This mapping can serve as a strategy for codifying narratives and other sources of data generated by users about their experiences.

In addition to this, the conceptual framework offers a notion of what corresponds to *all* user experience. Many efforts related to user experience focus on certain stages, types of interactions, and levels of sense-making, but leaving behind many aspects

unchecked. The conceptual framework presented in this paper defines criteria to identify when we have done an exhaustive exploration of all user experience in the three-dimensional space generated by Dewey's principles for experience (Dewey, 1938).

Discussion

The needs of educational testing require a more specific conceptualization of user experience, helpful in enhancing the experience of a test and validating a test. The proposed conceptual framework here can help us understand the evidence traditionally defined as examinees' response-processes under a unifying concept that extends response-process research and includes UX research, which is very similar to what Zumbo and Hubley (2017) foresaw as the future for response-process research. Dewey's ideas about experience define the conceptual framework around three principles:

- A principle of **interaction**, assuming that experience happens in the interaction of a person with its environment and cannot happen without this interaction.
- A principle of **continuity**, assuming all experience is affected by a continuum with past experiences and expectations of future experiences.
- A principle of **sense-making**, in which we assume that we make sense of the experiences, and in this sense-making process, we can discover, re-discover, and interact in new ways.

The conceptual framework defined in this paper starts from these three principles, defining categories for each. The conceptual category of user experiences is a more comprehensive way to define the source of validity evidence, usually known as response process, and this extension can be an update to be included in future versions of the AERA, APA and NCME (2014) *Standards*.

The inclusion of a phenomenological take in testing allows us to redefine what we understand as a test. A test is a device created with purposes that elicit an experience to make inferences about an examinee based on the interactions between the examinee and a situation. Experience is what is lived in within, not automatically equivalent to a different user's experience in front of identical stimuli. The concept of UNDERSTANDarization from Sireci (2020) can be understood as making elicited experiences comparable rather than assuming experiences are comparable because stimuli are comparable.

There is a risk in the analysis of user experience data in that we, as researchers, may interpret the experiences of the examinees in a way that is not consistent with what the users would say about their own experiences. This inconsistency points to the need to double-check the interpretations and hypotheses generated from UX research and include a participatory dimension in the study of Assessment Experiences. There is a risk in that UX research in testing can find a way to study the experiences and interpret those in ways that are not correct but convenient for the narratives that the test developer needs for the subsistence of a company or the needs of an institutional user.

We actively avoided the term test-taking experience proposed by Burstein et al. (2021) because of this reason. Calling examinees test-takers and not users separates the yolk from the egg, considering their experiences, but not giving agency in the process of developing and validating a test. That means the test persists into being not beneficial for the examinees (beyond the accomplishment of bureaucratic requirements). Also, "test-taking experience" can be misleading as a name and exclude all the previous and posterior stages in the assessment experience beyond the moment itself of doing a test.

We believe examinees should be users of a test, including all the prerogatives that any user deserves when a service is provided.

There are important insights about the Dialogical level of sense-making in an assessment experience. Authors like Markus and Borsboom (2013) or Hubley (2017) have highlighted the idea of testing-as-a-conversation metaphor, understanding the inherently relational nature of any assessment experience. This aspect of an assessment experience can have surprising implications. If a validity argument is explained to a test-taker in a way that triggers him/her (considering that a lack of explanation can be a precursor of poor assessment experiences), that will create construct-irrelevant variance, and thus will create invalidity. A rationale for the way we explain the validity of interpretations and uses should also be part of the argument. A validity argument must be rhetorical and aware of the experience that is elicited by the argument itself (Tindale, 2004). Can a lack of dialogical experience be a factor underlying the appearance of disengagement?

Examinees usually are not aware of the intricacies of a validity argument; however, the messages sent in a validity argument can be passed to the stakeholders and actors in a system to create narratives with implicit messages from the test designer. The dialogical experience makes the difference between inferring about an unobservable machine and inferring about a human. A human will always be reactive to contextual elements in the task that is presented. Initiatives like Randall's (2021) anti-racist items may have a future not just for resolving potential unfairness related to race, but also for the implicit message that is sent. The message sent is "I care about you." Zumbo's abductive reasoning view on validity (Zumbo 2009, 2017) also goes in the same line,

because by explaining why an examinee received a score, there is not only a reflection about the validity of an interpretation, but also a caring intention about the experience of the examinee and his/her/their sense-making process.

We believe test developers should value the overall experiences of their examinees and consider the types of evidence that can come from UX research. We believe it is essential to include each experience so that tests will not be overly ethnocentric or represent a construct in an narrow, dominant-culture fashion. The study of user experience is an effective way to detect those issues.

In theory, it makes sense to gather data directly from examinees during all test development and operational testing stages. We have evidence from some development and validation efforts focused on validity evidence based on response processes. Our extension of this work to the totality of user experience in testing deserves further study. We hope this chapter encourages that study, and we look forward to the improvements to educational and psychological testing that will follow.

**CHAPTER 3: LITERATURE REVIEW IN TEST VALIDATION
EPISTEMOLOGIES, AND THE CONCEPT OF SITUATED KNOWLEDGES**

Introduction

Most of the discussion around test validity and validation in recent decades have been consumed by the matters of truth versus utility, the inclusion of uses and consequences as part of the definition of validity, and if what we understand as measurement in the context of psychological and educational constructs means the same to what measurement means in physical sciences (e.g., Borsboom et al. 2004; Borsboom & Wijsen, 2016; Kane, 2013b; Sireci, 2016; Wijsen et al. 2021). The legitimacy of tests comes from a structuralist assumption that there is a truth to be known that explains a person's behaviors, and thus it can be known and inferred for each individual.

However vivid this debate, the criticisms from outside the educational testing field are not necessarily related to those discussions. For example, some authors have questioned the capacity of standardized tests to correctly make inferences for each individual in populations with diverse compositions (Armour-Thomas, 1992; Bhattacharyya, Junot & Clark, 2013). Another criticism is related to the explicit rejection of the existence and uses of a test by some of the communities affected by those uses, like teachers and accountability tests (Bhattacharyya, Junot & Clark, 2013).

We can consider some of those concerns as part of the scope of unintended consequences of test use; however, the people raising those concerns do not see it as a problem of just unintended consequences. Some of those concerns correspond to the intended consequences of valid tests. In fact, some of those concerns do not come from sources of invalidity in the test. A perfectly correct test under our consensus definition still can correspond to the criticisms exposed before. In general, validation debates avoid the problematization coming from the communities about the intended uses of a test,

considering the decision to use a test in a certain way as a given in a validation effort and thus excluding any other critical judgment about the uses themselves.

Another criticism usually seen against standardized tests is their complicit role in neoliberal projects (De Lissovoy, 2013; De Lissovoy & Cedillo, 2016). This complicity is not necessarily a scientific problem but a political one. Moreover, if it is a problem, it is a problem for some and not for all. Validation usually avoids entering political debates, even when tests have an obvious political dimension, as they provide a technology to make inferences about individuals and communities, thus creating discourses about them.

All the previous relate with the interactions of testing and structures of power. Test validity can be considered value-laden (Markus, 1998; Messick, 1989; Messick, 1998), but there is no further debate on how to precisely navigate the interactions between scientific research and political implications in the case of psychological and educational tests. Is it okay to build a test to help a political project? How should validation take this into account? Does the political stance of the validity researcher affect the objectivity of a validation effort? And the legitimacy of a test? What are the relationships between validity and legitimacy? Does a neutral political stance exist in scientific research? Does a neutral political stance exist when designing or validating a test?

Another criticism brought up by authors in the United States is the passiveness – if not active omission – of the racism embedded in the constructs that a test measures (Randall, 2021). Constructs can be biased in their origins, and in that case, the state-of-the-art validation effort will not recognize it as a problem. Problems related to cultural validity (Kūkea Shultz & Englert, 2021; Solano-Flores & Nelson-Barber, 2001) also

suggest identifying construct-underrepresentation and the overall definition of a construct can be relative to a community. Should validation consider the communities' points of view and challenges in defining a construct? Is validation an effort exclusive to experts, or is there room for participatory validation? Is it still objective a test validated in a participatory way?

The previous criticisms are not related to truth vs. utility or use and consequences. Except for Dixon-Roman and Gergen (2013), the discussion has focused on other topics: the need to write an argument or not, if validity is about interpretations, interpretations related to uses, interpretation and uses, or just causal relationships; or if we should speak about educational measurement or educational assessment. However, very little has been said about the actual criticisms that standardized tests are receiving from society and other scholars in the field of Education, on the blatant interactions of tests and structures of power and oppression.

Philosophically speaking, the debate between realists, constructionists, and critical realist epistemologies has consumed the attention of our scholarly community, leaving behind all the considerations usually raised by post-structuralist / post-modern authors that better represent criticisms from the public on matters of discourse construction, structures of power, oppression, and ignorance about the phenomenology of the tests we use.

The purpose of this chapter is to extend the discussion and include the perspectives of feminist epistemologies, that have brought interesting insights that can help our field of Educational Testing. Feminist scholars have discussed for many years the interactions between knowledge generation and structures of power, and I believe

their debates can help our field of Educational Assessment to own the relationships of tests with politics and power.

Before entering feminist epistemologies, I will summarize the epistemologies present in the writings about validity and validation in our field. I will continue later with a review of Standpoint theory; then, I will move to a detailed review of the concept of Situated Knowledges (Haraway, 1988), to finally end with concrete recommendations and actions on how to include the ideas of feminist epistemologies into our practices.

Realism, Constructionism, Critical Realism and Social Constructionism.

Some practitioners consider that the philosophical debates about epistemology and theories of knowledge are not genuinely relevant for Test development and validation. I am afraid I must disagree with that view. To the very least, topics related to Epistemology, Ontology, and Moral philosophy have become a reason (or excuse) to diverge from the “Consensus” views about test validity and validation (if such a consensus really exists). I will consider the three most famous and easy to identify theories of knowledge present in the authors who have proposed ideas about test validation: Realism, Constructionism, and Critical Realism. I will add a fourth that relates with feminist epistemologies, named Social Constructionism.

It is important to remember that all the following categorizations and taxonomical work are contestable, like the definition of Impressionism: an artist usually ignores those categories. Later commenters are the ones that create those labels.

I will understand realism as “a philosophical position, a subcategory of objectivism, holding that what can be known about an object exists independently of

one's mind. It is opposed to epistemological idealism.” (“Epistemological realism,” 2021.). Jupp (2006) defines realism in the following way:

Realism takes a number of forms depending on how ‘the real’ is understood. All approaches to knowledge that endorse realism accept that a world exists that is in some respect independent of the knowing subject. This real external world may be a purely empirical reality so that what has not yet been observed will at some point appear to us. Alternatively, it may be a reality that is concealed from us in some way beyond appearances and independent of our knowledge of the world. These different ontological assumptions about what really exists have produced corresponding epistemologies – the first associated with empiricism and idealism and the second with critical realism – that are central to social research.

I will understand realism as the first option mentioned by Jupp, and critical realism as a separate epistemology to be explained later. Maul, Torres, and Wilson (2016) explain realism in the following way:

On the realist’s view, the (perhaps approximate) truth of scientific theories is generally taken to be the best explanation for their success. Different varieties of realism may or may not endorse the commitments that (a) there is a (single) natural world, which exists regardless of what any conscious being thinks or perceives (the ‘metaphysical commitment’); (b) claims about the world are to be taken at face value, as possessing truth-

values (the ‘semantic commitment’), and (c) so interpreted, true claims constitute knowledge of the world (the ‘epistemological commitment’).

Authors in testing that be recognized as realist are Borsboom et al. (2004) or Markus (Markus & Borsboom, 2013). Borsboom’s definition for validity (2004) is representative of this epistemology. A test is valid if (a) the attribute exists and (b) variations in the attribute causally produce variation in the measurement outcomes.

I will understand constructionism as the following:

[...] a perspective that considers facts, descriptions, and other features of ‘objective reality’ to be inescapably contingent and rhetorical. This is a more recent formulation of constructionism (without the ‘social’) which follows the traditional view of social constructionism as a perspective wherein people are seen as produced (constructed) through social interaction rather than through genetic programming and biological maturation. (Jupp, 2006; p. 38)

One representant of this approach is Kane’s argument-based approach (Kane, 1992, 2013). The argument-based approach to validation consists of using informal logic to sketch an argument supporting statements related to the use of a test. Usually, the argument includes two components, the Interpretation and Use argument (IUA), which is a network of propositions related to the interpretation and uses of the scores of a test; and a validity argument that presents the actual evidence in support of each of the propositions in the chains of the IUA. In the argument-based approach, there is a very explicit consideration of the rhetorical aspect validity by creating a network of

propositions that organize the validation effort around the form in which we want to present the evidence (as an argument). Kane's definition for validity is a good representative of this stance: "The proposed interpretation and use can be considered valid if the IUA is clear, coherent, and complete, its inferences are reasonable, and its assumptions are plausible." (Kane 2013, p. 451)

I will understand critical realism as the epistemology that [...] accepts epistemological relativism (knowledge is always context-bound), but argues for recognition of a reality which exists independently of human activity and which should properly be the object of inquiry. Such inquiry is not limited to what is observable (as it is for empiricists), although this is important, but should be directed towards an understanding of the deeper causal structures which shape not only reality itself but also the context and meaning of human perceptions. Critical realism thus re-asserts the importance of ontology and criteria of truth and suggests (in contrast to postmodernism) that some forms of knowledge may have more validity than others. (Jupp, 2006; p. 93)

Mislevy (2009) is one representative of this epistemology. However, his self-identification is constructivist-realist, that label corresponds better to a theoretical perspective rather than an epistemology. I will prefer the term critical realism because fields beyond educational measurement prefer the term. Mislevy (2018) recognizes his roots in authors that define themselves as critical realists, so I will also consider him one. Mislevy proposed that the socio-cognitive stance is also connected with this critical realist view, in which models approach a truth that is to be known in context.

Markus considers Messick's validity theory a dialectic of the realist and the constructionist (Markus, 1998). However, I prefer Mislavy's observation about Messick's definition for validity, defined as an "evaluative judgment," and this decision is a constructivist take on validity (Mislavy, 2009). Some authors use the terms constructivism and constructionism as equivalents, but I will prefer the term constructionist because it is more explicit in accounting for the social nature of knowledge construction.

A more recent new incursion on test validation has been suggested by Gergen and Dixon-Roman (2014), in what they call a Social Constructionist take on test validation. Social constructionism differs from constructionism in that it is more explicit on the role of discourse in the construction of the self and the world; and it is more prescriptive suggestions include in a clearer way the participation of different stakeholders in the co-construction of scientific knowledge.

Table 3 provides a summary of the previous epistemologies explained, and authors that can be considered representatives of each.

Table 3
Summary of Epistemologies in Educational and Psychological Testing,
and Authors Representatives of Each.

Epistemology	Definition (Summary)	Authors in Educational and Psychological Testing	Authors beyond testing
Realism	There is a real world that is in some respect independent of the knowing subject. Claims about the world are to be as possessing truth-values. True claims constitute knowledge of the world. Construct must exist to be measured.	Borsboom et al. (2004)	Devitt (1996)
Constructionism	Facts, descriptions, and other features of 'objective reality' to be inescapably contingent and rhetorical.	Kane (1992, 2013)	Kuhn (1970), Toulmin (1958)
Critical realism	Accepts that knowledge is always context-bound, but argues for recognition of a reality which exists independently of human activity and which should properly be the object of inquiry.	Mislevy (2009, 2018)	Bhaskar (1975)
Social Constructionism	Similar to constructionism, but highlighting the central role of discourse in the construction of self and the world.	Dixon-Roman and Gergen (2013)	Gergen (1985)

Standpoint Theory, and the Epistemic Privilege of the Oppressed

In the 1980s, second-wave feminists began to discuss the relationship between knowledge generation and how science has a significant role in constructing systems of oppression and narratives that sustain those. On one side, science itself became another field of political contestation, and on the other, a claim made by Feminist movements about their oppression may require a scientific and “objective” examination in some instances.

In the quest for better epistemologies, feminist scholars explored different ideas for knowing in a way that problematizes the relationships between knowledge generation and power. Many feminist scholars based their work on Hegel’s epistemological privilege of the oppressed (Cameron, 2005), considering that the oppressed have an “epistemic privilege,” an advantage when raising questions and investigating matters related to oppression due to their specific life experiences. Thus, women have an epistemic privilege to speak about their oppression.

Feminists scholars were able to create a political and intellectual identity, articulating a criticism against our societies that have oppressed women for centuries. Feminist scholars started to generalize what happened in feminism as a social and political movement and speak about what they call a “standpoint.” A standpoint is a shared construction of knowledge that emerges from marginalized people when they become conscious of their situation concerning socio-political structures and begin to find a voice (Bowell, 2021). Feminisms are standpoints. The three main tenants of feminist standpoint theorists are: (1) Knowledge is socially situated, (2) Marginalized groups are socially situated in ways that make it more possible for them to be aware of

things and ask questions than for the non-marginalized, (3) The research focused on power relations should begin with the lives of the marginalized. Given the previous three, the point of view of women have (or should have) a privileged position when making claims about the oppression of women.

Now, to justify the epistemic privilege of women, we need to clarify first what we mean by "women." Such a simple and easy to accomplish pre-requisite is not simple to achieve. Are trans-women included in the definition? Are Black women experiencing oppression in the same way White women experience it? One criticism of early notions of the concept of Standpoint is that it needs to be capable of navigating difference and partial affinities.

Black feminist Patricia Hill Collins (1986) brought another profound concept into the discussion: the Outsider-within. The idea came from the experiences of Black women that have served as housemaids in white families. On one side, sometimes those families create a very intimate relationship with their "Black mothers" and make them an insider of the family. Those women can demystify white families by discovering how their privilege better explains their success. Nevertheless, they are still not part of those families and are still an "outsider." Then, the status of an outsider-within. The standpoints in the borders, sometimes even in contradiction, appear many times when we think about the Standpoints we inhabit.

There can be many standpoints: the Standpoint of the proletariat, of women, of third-world women, of black women, of trans women. Furthermore, each interlock with their peer categories: class, race, gender, sexual orientation. In each case, a political process to construct a common understanding of the oppression is a prerequisite to define

a Standpoint. It is not just a non-empty intersection of shared past experiences; it is a co-construction making sense of those together in a communal way.

Hekman (1997) criticized Standpoint Theory in that an epistemic privilege of any Standpoint is liquified by the inevitable multi-dimensional (infinite-dimensional perhaps) differences within a Standpoint. She also touches ground in a more evident and general consideration of Standpoint theories, in that accepting the epistemic privileges of Standpoints can imply, in the end, an epistemological relativism.

Defenders of Standpoint theory would respond that a Standpoint can lead to a better truth, and part of the "game" of a researcher is to explore which Standpoints are better suited for knowledge production and scrutiny. However, this still recognizes a truth to be discovered. The modernist ideas from Marxist analysis appear in the concept of epistemic privilege, in contrast with the social constructionist vocations in feminist scholars.

Harding considered two interesting insights relative to those problems. First, she defines the concepts of Strong and Weak objectivity (Harding, 1992). Strong objectivity arises from the examination of the method used to back a claim (or find the truth), plus the examination and reflection of the subject that is knowing. In her own words:

[...] Thus, strong objectivity requires what we can think of as "strong reflexivity." This is because culturewide (or nearly culturewide) beliefs function as evidence at every stage in scientific inquiry: in the selection of problems, the formation of hypotheses, the design of research (including the organization of research communities), the collection of data, the

interpretation and sorting of data, decisions about when to stop research, the ways results of research are reported, and so on. [...] But a maximally critical study of the study of scientist and their communities can be done only from the perspective of those whose lives have been marginalized by such communities. (p 136).

So, to Hekman's point, even if a Standpoint has so much diversity within that may make us re-think its epistemic privilege, the fact that it is different from the knowing subject can help to do this "strong reflexivity" requirement for a Strong(er) objectivity in scientific research. In that regard, any epistemic advantage happens in the relationship to an other and not in absolute terms.

Also, in response to Hekman (Harding, 1997), she clarifies that feminist epistemologies are not focused on truth but power and its relationship with knowledge generation. The concept of Standpoint is meant to elucidate, to make transparent, how systems of oppression find their way to influence scientific research. If we are worried about truth and method, then Harding asks: whose standpoints require this? She uses the term "Administrator Standpoint" to refer to the ones that, in the process of knowledge generation, do worry about truth, method, or what the "court of reason" would say (Toulmin, 1958). In many ways, the researchers validating a test or developing a test, or the institutional test users, come from that Standpoint.

A proper understanding of this Administrative Standpoint is essential in Test development. Most of the time, practitioners are in this Administrative Standpoint when designing and validating a test. How is this Standpoint? Feminist scholars would suggest that we need to co-construct a better Standpoint and understand our relationships with

structures of power (as part of them or against them). In the encounter of other Standpoints is when we should discover more about our own Administrative Standpoint.

Situated Knowledges, and Validation as a Multi-perspective Process with the Communities

Donna Haraway is a feminist, socialist scholar who participated in the philosophical debates about epistemology and the relationships between structures of power and knowledge generation. She wrote a response to Harding's book "The science question in Feminism" (Harding, 1986), entitled "Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective" (Haraway, 1988) as her take in the debates about standpoint theories, epistemic privileges and the never-ending question: what do we mean by the word "objective."

I will cite some excerpts from her very rich paper, and sketch a list of core ideas in her work: the idea of the god-trick, the problem of vision devices, situated knowledges as a multi-vision account, and participatory research as a pre-requisite of any situated knowledge.

The god-trick, situatedness, and objectivity as candid partiality

Haraway (1988) starts her paper criticizing classical epistemologies like realism and constructionism by pointing out the disembodied nature of those philosophical perspectives. She denounces that researchers do science assuming that they are invisible, imagining themselves with the potentials of a god that can "see everywhere from nowhere." Haraway refers to this attitude as the *God-trick*.

The God-trick is present in two ways. It is present when we omit relevant contextual aspects affecting the researcher's body and the ones researched. There, the god-trick is in action. Also, Haraway considered that to accept radical relativism is a god-trick as well, in the sense that seeing from all points of view is another form of denial of the embodiment of knowledge generation. Experience is partially non-fungible.

So, a better accounting for what is "real" must include all contextual elements that can help future rational beings to understand the historical contingency of research; and in general, any other information that relates to the fact that the research is happening somewhere with persons, within bodies, in places, at some moment in time.

This last observation about the inherent situatedness of any knowledge relates to the question of objectivity, as she rejects relativism as the ultimate response to the question (i.e., it is impossible to decide what is objective) nor to define what is objective from nowhere, disembodied from the body of the knower and the known. She posed the question of objectivity in the following way:

So, I think my problem, and "our" problem, is how to have simultaneously an account of radical historical contingency for all knowledge claims and knowing subjects, a critical practice for recognizing our own "semiotic technologies" for making meanings, and a no-nonsense commitment to faithful accounts of a "real" world, one that can be partially shared and that is friendly to earthwide projects of finite freedom, adequate material abundance, modest meaning in suffering, and limited happiness. (p. 579)

In this paragraph, there are a lot of layers and ideas to unpack. In the quest for a better definition of objectivity – one that does not collude with systems of oppression – there is on one side a need to consider the historical contingency of some research. So, for example, in the case of a validity argument, the historical contingency of a validity argument can relate to the political context of using a test. Historical events that may be related to the need for a test or limitations in the methods, historical context, and the point of view of the communities affected by the uses of the test and their political processes. The Covid-19 pandemic has revealed the relevance of including this sort of meta-data in any research because the generation of questions, methods to respond to those questions, and the conclusions finally posed can be highly affected by contextual elements.

In addition, Haraway (1988) mentioned the need to examine our semiotic technologies. Semiotic is "the study of sign processes (semiosis), which are any activity, conduct, or process that involves signs, where a sign is anything that communicates a meaning that is not the sign itself to the sign's interpreter" ("Semiotics," 2021). This need to examine our semiotic technologies follows the post-structuralist criticism that science relies on language. A semiotic technology is then a way to use language, so language is coherent with the knowledge we want to generate.

Argumentation as in Kane's argument-based approach (1992, 2013) is a semiotic technology in the sense that it proposed a recipe to use symbols and language to speak about what we know. The Toulmin model (1958) is, in general, an example of a semiotic technology. This examination of the semiotic technologies in test validation should make us wonder: Is the way I am communicating the validity of a test (relative to its uses and interpretations) limiting the capacity to understand that validity?

The word “no-nonsense” is fascinating. The nonsense is a criticism of realist epistemologies, in that Haraway (1988) considered those as non-sensical. She explains this criticism in more detail, describing how male built modernism under the concomitant Cartesian gaze of the scientific as “*maître et possesseur de la nature*.” (Master and owner of nature). This gaze is the nonsense to which she is referring. The use of the word “nonsense” is a direct criticism also to capitalism. Haraway does not offer herself a replacement for this nonsense. In the context of situated knowledges, any “sense” that we could have in our scientific practices should be situated. That is why – I believe – Haraway used the word “no-nonsense,” just referring to this sense as the opposite of the nonsense of modernism but not prescribing any “sense.”

This nonsense becomes more evident in the description of the accounts of the “real” that Haraway (1988) foresees. She lists them as partially shared, friendly with the earth, accepting limited freedom and limited happiness, pointing out that the nonsense Haraway criticized is certain omnipotence in the metaphysics of scientific research of western cultures. We can do anything we want with nature (humans included as part of this nature), and there are no limits to the power we humans can yield from the knowledge we can discover—a project of unlimited happiness. Science done from narcissism, disconnected from the body. In opposition to this omnipotence, Haraway proposed an epistemology that grounds itself in connection with the body. An epistemology that puts the gaze in the situatedness and embodiment of the generated knowledge. We are just humans, after all.

This omnipotence is present in our validation practices in testing, for example, when we assume that one single validity argument is enough to convince not only a

scientific community but all other communities that may wonder if a test is valid or not. It is the same with causal relationships – we assume – can be discovered to apply to all humans. This omnipotence is also seen in the idea of UNDERSTANDARDization (Sireci, 2020b), in that an under-standardized test would include all specificities for each examinee using a universal design perspective. This is the type of omnipotence Haraway (1988) rejected in relativism, assuming we can see from any standpoint.

Haraway (1988) proposed a solution. Her solution is the concept of situated knowledges. In her own words:

The moral is simple: only partial perspective promises objective vision. All Western cultural narratives about objectivity are allegories of the ideologies governing the relations of what we call mind and body, distance, and responsibility. Feminist objectivity is about limited location and situated knowledge, not about transcendence and splitting of subject and object. It allows us to become answerable for what we learn how to see. (p. 583)

The idea is that the best way to handle the constructed nature of knowledge and avoid being part of oppressive systems in our ways to know; is to acknowledge anything related to the fact that research is situated and disclose the partiality of our gaze. Not only accept it but honor it and share it, including the limitation of that sharing process. The shift proposed by Haraway (1988) implies then that true objectivity does not come from emptying us from values when doing research but to honoring and disclosing those values. And not only values, but any other relevant aspects related to the fact that all research is situated.

One perspective from situated knowledges is that the discussions about truth versus utility are a situated matters and thus should be ultimately decided “on the ground” with the communities. At least if we worry about matters of oppression. Solano-Flores and Nelson-Barber (2001), for example, discussed the concept of “cultural validity” in that the responses of students in science questions depended on the preferred epistemologies of the culture they belonged, with one example of a community that had an epistemology closer to what we know as realism and the problems of an item written in a language proper of a constructionist perspective. So, if that epistemological style in the one writing of an item impacts the assessment and ultimately the interpretation of a score, it would make sense that any acceptability criteria for a validity argument will also depend on the epistemology of that community.

In general, criticisms to standardized tests are rooted in the mismatch between values implicit in a test and the values of the communities affected by the uses of the test. Is it appropriate to apply a test based on values different from the values of a tested person? Or with the communities affected by the decisions of test use? It is not appropriate. Moreover, this view about the appropriateness of a test, as a match between values, is a partial position.

The problem of vision devices and the blood needed to craft the eyes

Haraway (1988) framed her view about feminist objectivity in the concept of situated knowledges, as a criticism of vision as the preferred sensory system used for scientific research. She stressed the idea that any visual device (a satellite, a microscope, the human eye itself) will have entailed a “power move,” something seen and something

unseen, with the potential for a structure of oppression to hide in the actions on the unseen. In her own words:

Vision is always a question of the power to see, and perhaps of the violence implicit in our visualizing practices. With whose blood were my eyes crafted? (p. 585)

A microscope is a vision device. We know an object under this tool by examining how the electromagnetic waves reflect on the object. Tests are different technology in that we do not “see” directly, but we make inferences about observables. Nevertheless, even for a Microscope, we do not “see” all; there is a layer of inference and sense-making. In any case, for both a visual device and an inferential device, we will need “blood,” meaning that we will sacrifice some in the decisions about how to build the device.

Even if we make a test centered around equity, around the best of our values, even if we do it with the oppressed, we will create blind spots that will be the perfect room for oppression to happen. This is, in fact, a fundamental difference in Haraway from the concept of epistemic privilege. An epistemic privilege is still a privilege and can then create new marginalized.

That final question, “With whose blood were my eyes crafted?” is a fundamental question in testing. The attempts to study the consequences of uses and interpretations of a test go toward accounting for the blood, but still speaking about consequences in a disembodied way. We shift the attention from the persons affected by the consequences

to an abstract idea of consequence. How can we genuinely account for “the blood needed to craft the eyes” of a test?

No testing agency would disclose such specific information about a test because it would entail legal consequences and even the potential to close a testing program. In that regard, there is no incentive for a testing agency to disclose too much detail about who is being affected by the invalidities of a test because that immediately will become bad press and will have legal implications. The best way to avoid the problem is not to know, ignore the problem and not to research. However, the only way to truly distance from structures of oppression is not to collude with them by hiding relevant information, but to have a parsimonious accounting of who is oppressed or potentially oppressed by the devices we create, as an act of scientific integrity.

If not for an ethical preference, validation made in a testing agency making profits should consider the risks that the business is having by crafting the eyes with the blood of someone. The risk of spending one dollar in a testing program and losing it later. This links to the concepts of social responsibility (Koljatic et al., 2021) as a liberal strategy for global corporations to mitigate social risk (Kytte & Ruggie, 2005). This way of thinking about the problem would be a capitalistic way to understand the need for situated knowledges.

However, situated knowledges is explicit as a tactic against capitalism, patriarchy, and other structures of oppression. It is not politically innocent or neutral. By acknowledging the embodiment of research, by giving agency to the subjects/objects of study, by making us aware of the blood used to craft the eyes; a researcher is pursuing knowledge, undressing, and dismantling structures of oppression at the same time. This

characteristic of dismantling structures of oppression is a quintessential element of any feminist epistemology.

Situated knowledges as a multi-perspective account for what is “real.”

The concept of situated knowledges points then to knowledge fully aware of the embodied nature of the researcher and the researched, considering not only a target truth but all the contingent aspects of "the real", including the fact that researchers and subjects are all human corporal beings. Haraway (1988) explained that a situated knowledge is an open conversation that would have at least double-vision, open for the problematization of the communities on the assumptions in any argument or other mode of communication about the validity of a test. Table 4 is a reproduction of a table Haraway used to capture situated knowledge essentially.

Table 4

Comparison between some of the Classical Paradigms for Scientific Research and Situated Knowledges

De-located Knowledges	Situated Knowledges
Universal rationality	Ethnophilosophies
Common Language	Heteroglossia
New organon	Deconstruction
Unified field theory	Oppositional positioning
World system	Local knowledges
Master theory	Webbed accounts

From Haraway (1988, p. 588), header modified.

I interpret from this table that the concept of situated knowledges applied in educational testing will imply that test validity itself should be considered a situated knowledge. If an argument makes sense for a community as an acceptable way to discuss the validity of a test, then it will be appropriate to use this semiotic technology and structure an IUA for that community. However, we cannot prescribe this a priori as the validity of the uses and interpretations of a test will highly depend on the contextual elements present for each instantiation of the test.

The idea of webbed accounts posed by Haraway (1988) suggests that the validity argument should be an open conversation with communities that dialogue between them and with the test developer. The traditional validity argument approach as a series of studies or a technical manual is too static to be considered a webbed account or a situated knowledge. I interpret the term “webbed accounts” as a network of participants that collaboratively create knowledge, allowing contradictions and different points of view. However, they are together looking for a truthful account of the “real,” including the possibility of accounts relative to different life experiences.

The argument-based approach (Kane, 2013) mentions an appraisal stage, in which there is a review of the IUA by some external person. There is plenty of room to enhance this appraisal stage and give agency to the communities affected by the uses of a test. A situated knowledge approach would take a proactive action towards the generation of that dialogue and allow the validity argument to evolve with the concerns raised by the communities in an appraisal phase.

Participatory research as a core element in situated knowledges.

Haraway (1988) clarified what it means to do research in an embodied way in social sciences, coherent with the idea of situated knowledges. She mentioned the need for knowing subjects to consider their known subjects as agents in the research process. She posed this requirement for situated knowledges in the following way:

Situated knowledges require that the object of knowledge be pictured as an actor and agent, not as a screen or a ground or a resource, never finally as slave to the master that closes off the dialectic in his unique agency and his authorship of "objective" knowledge. The point is paradigmatically clear in critical approaches to the social and human sciences, where the agency of people studied itself transforms the entire project of producing social theory. (p. 592)

This echoes with the concepts of epistemic privilege and how topics related to oppression require the participation of examinees and communities affected by the uses of the test in the validation process. Not as data, but as agents.

Messick (1989) included the idea of listening to the voices of different stakeholders; however, he never mentioned the possibility of examinees being agents in the validity/evaluative judgment being constructed. On the other hand, Dixon-Roman and Gergen (2013) proposed this participatory need in their postmodern, pragmatic, social constructionist take about testing.

How Can We Improve our Validation Practices from the Perspectives of Feminist Epistemologies?

The previous considerations and philosophical debates about validating the uses and interpretations of a test can be very interesting. However, if we cannot connect those with practice, all the effort in clarifying our ideas by knowing more about the philosophical debates is somehow irrelevant. Validation practice is often very limited and affected by the long-standing traditions of validity as correlation with a criterion, trade-offs between validity and reliability, a fragmentation of validity into different types, and a shameful lack of actualization in many practitioners validating tests (Newton & Shaw, 2014). It is crucial then to explain in a more didactic way concrete recommendations for practitioners.

I will give my recommendations from the insights mentioned in the previous sections, and when possible, provide examples of other practitioners who have done work in the suggested direction.

Including the positionality of the validity researcher: Positionality refers to researchers' detailed explanations of their identity (gender, race, ethnicity, sexual orientation, epistemological choices, ontologies) and how this identity interacts with the research (Holmes, 2020). Positionality is a typical step in qualitative research in social sciences, particularly in anthropology, where the relationship between researcher and researched can highly influence the research results. In the case of test validation, the tradition has been for the validity researcher to be hidden and avoid direct communication with the subjects (except for response process research). However, the case made by Haraway (1988) and other feminist scholars about the situatedness nature

of research brings the question of how the identity of the validity researcher can be part of the partiality in the validation effort.

A Co-construction of an Administrative Standpoint: As Harding (1997) pointed out, every time we care about truth and method, it makes sense to clarify why we are taking this stance. As with any other Standpoint, a co-construction within the team making the validation effort, for example, can help to gain awareness of our role and relationship with structures of power and oppression. The construction of an Administrative Standpoint can be considered as a collective positionality statement.

Context and Discourses as an object of inquiry: There are elements beyond the researcher's identity that also impact the research and are part of acknowledging the embodied nature of research. One example of this was provided by Flórez-Petour (2015), in which she situated the public policies in Chile related to K-12 school accountability and a test taken for that purpose. She used discourse analysis to understand how the use of this test was situated historically, ideologically, and within a system.

Inquiry on structures of power: Part of the message in situated knowledges is to become aware of the structures of power that can surround some knowledge. In our case, structures of power will have many relations with a test and the validity researcher validating it. Randall's (2021) set of questions for anti-racist testing and validation is one example of this proactive questioning of the structures of power from the lenses of critical race theory. There is also a movement named "quantCrit" that includes the ideas of critical race theory, particularly in quantitative methods (Garcia et al., 2018).

User Experience (UX) and the phenomenological dimension of testing: Some researchers who have tried to integrate the ideas of situated knowledges into their research have taken a phenomenological approach, and have considered the necessity of studying the experiences of the subjects in the research. Their point of view “from the inside.” User eXperience research (UX) in testing has been done by EdTech companies (Wodzak, 2021), bringing the design practices from human-computer interface into educational assessment. Araneda and Sireci’s (2021) conceptual framework for UX in testing can serve as a guide to include more sources of positive and negative evidence into a validity argument.

Participatory research methodologies: Haraway (1988) clearly stated how situated knowledges require that subjects be considered agents in the research process. Many participatory research methods have been developed, particularly in anthropology (Bennet, 2004). Many of those methodologies have Freire’s pedagogy of the oppressed work in mind (Freire, 1968). Some examples of participatory methods in validation can be found in Flórez-Petour and Olave-Astorga (2020) or Kūkea-Shultz and Englert (2021).

Assembled arguments and Actor-Network Analysis: Part of the idea of webbed accounts in opposition to master theories in situated knowledges will require to account for the standpoints contingent on a validation effort. Addey et al. (2020) presented a more concrete example for the validity argument in PISA, showing how an argument may require room for problematization and eventually the generation of a validity argument for each country. Addey et al. brought ideas from Actor Network Theory (ANT) as a framework for understanding the “webbed accounts” that Haraway mentioned in her piece.

Addey took a sociological perspective to study how OECD implemented the PISA for development (PISA-D). This initiative consisted of the inclusion of emerging economies in countries doing PISA. In that process, some countries abandoned the process because of the invalidities of the test; in which, for example, an item asked a question that mentioned ATMs, but was presented to children in India who have never seen one (p. 9). Addey's case is not prescriptive on how the validation effort should have been done, except for considering the opinions of the dissident voices in the process.

Rhetorical Argumentation: Lang (2010) makes the case in favor of connecting Situated Knowledges with the ideas developed by Tindale (2004) about what he calls "Rhetorical argumentation." In his book, he explains that any argument should be able to be correct in its explanation and consider an effort to be well understood. Rhetorical argumentation implies participating and tuning the argument to the experience elicited by the argument in a targeted audience. Haraway herself considered that part of the problem was understanding this rhetorical part of knowledge generation. We can apply this idea in our validation practices by explaining the validity argument to the communities affected by the uses of the test and structuring the validity research agenda around how to well explain the argument for them.

Conclusions

There is an enormous need for continuing the Social Constructionist project proposed by Dixon-Roman and Gergen (2013, 2014) on test validation. The works done by feminist epistemologists are a great starting point for a validity theory that can own its relationship with structures of power and politics.

Situated knowledges and Standpoint Theory offer a new perspective to enrich the Social Constructionist take. Rouse (1996) argued that Situated Knowledges could not be considered even an epistemology per se but more of a manifesto of the politics of knowledge generation. He uses many words to describe Situated Knowledges: post-epistemology, moral-epistemology, epistemic-politics. This break with epistemology itself may be a pre-requisite to tackle the core modernist legacy in our scientific practices (or our intentions to be scientific) that can percolate even in the idea of epistemology. The concept of objectivity as candid partiality offers new ideas to the problem of epistemic relativism.

In summary, situated knowledges bring two main topics into test validation. First, there is a need to recognize the situatedness of the work done by a validity researcher in more explicit ways. I propose then that every validity argument should start by responding to the following question: in what ways is this validity argument partial and situated? I will call this the *partiality question* in test validation.

Randall's recommendation for an anti-racist stance (Randall, 2021) in the definition of a construct can be considered, in fact, an early response to the partiality question in validation. The categories she used to group the questions that a practitioner can ask while developing a test and defining a construct are indeed very similar to the points mentioned in this paper. In the partiality question I propose in this paper, an author interested in taking a more specific Standpoint can have the chance to explain what this Standpoint is and how it will appear throughout the validation effort.

Second, situated knowledges point to conceive validation as an argument-process instead of an argument-product, and make the validation a dialogical and participatory

process with the communities affected by the uses of a test. In the argument-based approach (Kane, 2013), there is a consideration of an appraisal phase in which an external actor reviews the validity argument, but nothing else is said about how this can be done. If we want to consider validity as a situated knowledge, the validation of the test must be participatory, in which examinees and other groups are considered co-constructors.

The concept of objectivity as an embodied, candid partiality is an interesting path for the future in educational assessment. Instead of justifying a split between politics and scientific work under a false concern for objectivity – which is suspiciously convenient from a political perspective to hide the values behind that split – we should try a full disclosure of the partialities in validation, that will allow future researchers to have better accounts of the validity of the interpretation and uses of a test. This candid, embodied partiality also helps to include any validity researcher with a political standpoint in his/her/their research.

The relationship between scientific work and politics is terribly evident in Educational Testing. We must incorporate a proper account of the political dimension of our work and craft as an act of scientific integrity, and as a cynical strategy to keep testing legitimated in the eyes of society. We are doing science, “science,” about other humans just like us. This only fact should make us think and suspect about any direct incorporation of the methodologies of natural sciences in social sciences. As said before, we are just humans. Or, in the words of Haraway, we are all cyborgs (Haraway, 2006).

CHAPTER 4: TOWARDS A VALIDATION APPROACH CENTERED ON USER EXPERIENCES

Introduction

Before starting this chapter, I would like you to re-experience one instance in which you took a test and felt that the experience was positive for you. Bring into your memory with as much detail as possible everything related to that experience: the way you felt, contextual elements, and thoughts you had when responding to the test. Please take a moment to bring those memories into your mind. I will come back to this exercise through the reading of this chapter.

Becoming aware of the turbulent flow of information that constitutes experience can be so illuminating. Phenomenology refers to an approach and philosophy that concentrates on studying consciousness and the objects of direct experience. Critical concepts of phenomenology are imagination, the situatedness of the human subject in society and history, and structures of experience.

What if we see Educational Testing through the lenses of phenomenology? What if we study the phenomenology of being assessed? Testing is one of the more ubiquitous technologies in western culture, happening in almost all educational systems worldwide. Anxiety, exhilaration in success, frustration when we are denied, joy when we are understood; we can share and empathize with those experiences. What can we discover about a test if we see them through those lenses?

There is a long tradition in the literature of test validation about the relevance of studying the inner processes of examinees when taking a test. The considerations about response process evidence in Cronbach and Meehl's (1955) famous article about construct validity, the socio-cognitive foundations of Mislevy (2018), or the many studies

using think-aloud protocols to discover the nuances of culture and language in reaction to a stem (Ercikan, 2002; Solano-Flores & Nelson-Barber, 2001); all have indirectly acknowledged the relevance of understanding the assessment experience from within, the *experiential* aspect of testing. I will use the term experiential as a synonym for ‘applied phenomenological’ (van Manen, 2011).

However, this tradition limited the experiential aspects of testing to cognitive interactions between the examinee and a test, in what we know as response process evidence in test validation (Ercikan & Pellegrino, 2017; Padilla & Benitez, 2014; Zumbo & Hubley, 2017). The interest in the experiential aspects of testing has expanded to UX (User eXperience) research and approaches to test development that considers in more empathetic and holistic ways the experiences of the examinees (Wodzak, 2021). Methods coming from UX research like A/B tests, co-design sessions, journey maps, have permeated the practices of some test developers exploring innovative methods for research and development.

Part of the motivation in Game-based assessment (GBA) relates to the experiential. The inspiration of practitioners in Game-based assessments (e.g., DiCerbo 2014, 2017; Kim & Shute 2015; Mislevy et al., 2016; Ventura & Shute 2013) started from an understanding that classical modes of assessment elicit unpleasurable experiences for their examinees. Games bring the opportunity to change this and transfer the fantastic experiences that a ludic attitude can bring into the development of any device.

Research related to engagement can also be considered part of this experiential aspect of testing (Wise and DeMars, 2005; Wise & Kuhfeld, 2020, 2021). The work done

by Wise and his colleagues have shown that the concerns about Engagement are more serious than just a bad experience: validity is in the line. The interpretation of test scores can be affected by the misinterpretation of a quick response. Engagement is just the tip of the iceberg of the relationship between the experiential and validity. In what ways experience and validation relate?

Experience and validation

There is an intuition that the information about the experiences can be very rich and informative. Cronbach and Meehl (1955) mention in a very informal way that “One of the best ways of determining informally what accounts for variability on a test is the observation of the person’s process of performance.” (p. 10). Cronbach also understood a construct itself as “a means of organizing experience into categories” (Cronbach, 1971; p. 464). So, the direct study of the experiences is a natural place to check if a test is tapping a construct.

One insight made by Human-Computer Interface (HCI) researchers is that we can make sense of our experiences with others. Forlizzi and Battarbee (2004) included the concept of co-construction, creating a model for user experience in interactive products based on Dewey’s concept about experience (Dewey, 1938). See Araneda (2022a) [Chapter 2] for a proposal of a conceptual framework of user experience in testing based on the same concepts. Co-construction with peers and the test designer is critical for ‘user experience.’

If we make other stakeholders like teachers, parents, and community members take a test and see their reactions to it, we can see beyond construct-irrelevant variance

and discover a misfit in the content domain detected by teachers, discover the funds of knowledge students bring through what the parents report, or discover potential consequences associated with the uses and interpretations of a test. User experience is rich and complex, with many facets across different stages of an assessment experience, different types of interactions between an examinee and a test, and different levels of sense-making of those experiences (Araneda, 2022a) [Chapter 2].

There is an interesting question in this perspective on user experience: **Is it possible to detect all sources of invalidity in a test in the extensive study of user experiences?** If the answer is correct, all invalidity would be traceable in UX research. Given that the construct itself is defined from experience and that user experience is so complex and extensive, it may be possible that UX research is self-sufficient to find all invalidities and thus capable of being the cornerstone of an approach to validation. This will be an hypothesis to explore in this dissertation.

Under the assumption that all invalidity is traceable in UX research, we can study the plausibility of statements by doing a parsimonious revision of the experiences of the users, hypothesize on how and for whom those statements can be false, checking with further statistical analysis and evidence if those statements can resist serious falsification attempts. I will call this approach an *experiential approach* to validation.

In this approach, the validation effort concentrates on the extensive study of user experiences as a pivotal step to finding invalidity hypotheses that are either disproven or solved by doing changes in the test. Cronbach (1980) mentioned, “The job of validation is not to support an interpretation, but to find out what might be wrong with it. A proposition deserves some degree of trust only when it has survived serious attempts to

falsify it” (p. 103). The experiential approach is a straightforward implementation of this view.

This procedure is a falsificationist way to validate a test (Popper, 1959). Popper’s proposition about scientific inquiry refused to fall into the traps of inductive reasoning by limiting ourselves to deductive reasoning and the study of propositions that can be realistically disproven with empirical evidence or other sources of evidence.

In informal logic, a double negation (this interpretation/use is valid if not invalid) may be insufficient to back a claim. We can surpass this limitation if we gather all the evidence and discoveries of the first stage of investigation and present what was discovered later in the more classical “p implies q,” à-la-Toulmin way like the one proposed by Kane in the argument-based approach (Kane 1992, 2013). By making the validation effort a two-step procedure, any Interpretation and Use argument (IUA) is built in an emergent way from users’ experiences.

I will dive into the experiential approach’s details and make a case for its value as an alternative to test validation. I will explain first the concept of validity in which I will base the approach, explaining the epistemological core that justifies such a procedure. I will explain later the procedure I envision for this approach, based on the stages sketched in Kane’s argument approach, but including the participation of the communities in its process. I will discuss at the end the differences of this approach with Messick’s ideas about consequences and the appropriateness of a test.

The Concept of Validity in the Experiential Approach

An experiential turn in test development and validation goes far deeper than creating good user experiences. The spirit behind phenomenology enters in the realms of epistemology and ethics, defining new ways to see test validity. What is the concept of validity in the experiential approach to validation?

For starters, I will understand a test as a device with a purpose that elicits experiences helpful in making inferences about an individual. This definition implies that I will understand a test as a technology, and thus its validity will be a concern not only of philosophy of science but of philosophy of technology. This definition also implies that uses and interpretations of a test are inevitably part of the validity of a test as a device made to help a scientific community's research and beyond.

I will understand **test validity** as a the set of Situated Knowledges constructed by a researcher and a community about the appropriateness, adequacy and preferability of future instantiations of a test, the interpretations to be done with those instantiations; and the procedures in which those instantiations will be used for.

I will say that an *interpretation or use is valid for a community* if that community accepts an argument about the validity of a test. This definition presupposes an acceptability criterion defined by the community. One instantiation of a test is valid for a community if the logic sketched in the validity argument is likely correct for that instantiation.

I will say that an *interpretation or use is valid* (without specifying a community) if it is valid for the community of all rational entities in the future. I will refer to this

group as the universal audience. I will say that one instantiation of a test is valid if the argument accepted by the universal audience applies in specific for this instantiation.

Let us unpack the definitions made here. Validity here is defined as a *Situated Knowledge*. Situated Knowledges is a concept coming from feminist epistemologies and developed by Donna Haraway (1988), in which knowledge is constructed in an embodied way in opposition to what Haraway denounces as the “God-trick.” The God-trick refers to scientific classical “objective” work written as if the researcher can disappear from the writing, detaching himself/herself/themselves from the subjects and places, denying all the aspects of the scientific research that derive from being an embodied researcher. The God-trick is the delusion of seeing “everywhere from nowhere.”

Haraway (1988) foresaw a phenomenological way to approach knowledge generation. She made the case by criticizing vision as a sensory system in scientific research and proposing the olfactory system as a metaphor of a phenomenological take on knowledge generation. Vision implies distance from what is seen, and olfactory requires experiencing the phenomenon being studied. The nuances of Situated Knowledges are complex, but in concrete, a Situated Knowledge is an open conversation with the subjects of research considered as agents and participants in the process of knowledge generation; allowing at least “doble-vision” in the accountings of their experiences on what can be considered as “real.”

Under the mindset of Situated Knowledges, objectivity is understood as **embodied, candid partiality**. This definition for objectivity means that, instead of taking an impossible value-free position to become rational, rationality starts from a candid disclosure of what can make us partial and situated. This definition poses a new question

in test validation: in what ways is the validity argument partial and situated? I will refer to this as the partiality question in test validation. See Araneda (2022b) [Chapter 3] for an explanation about the connections of Situated Knowledges and test validation.

Inheriting from Messick in the Experiential Approach

In the definition of validity of the experiential approach, the words ‘appropriateness’ and ‘adequacy’ follow the definitions that Messick (1980, 1989) gave to them in the context of educational testing. *Appropriateness* corresponds to the ethical question of whether we should use a test or not because of ethical considerations around the test. *Adequacy* corresponds to the scientific part of test validation, the scientific correctness of the definition of the test relative to the construct that it is supposed to measure. The word ‘appropriate’ will be used exclusively for ethical considerations.

Another novelty in test validity presented here is the idea of preferability among tests. The preferability of a test refers to why we prefer one type of assessment versus other types of assessment that are also valid (adequate and appropriate). Here, any societal goal associated with a test can create a preferability criterion, usually defended in a theory of action (Bennet, 2010). Any specification on the test to create experiences of a certain quality, like an engaging experience, a culturally sustainable experience, an experience coherent with a learning progression, can create preferability criteria on a test as well.

Situated Knowledges implies a Rhetorical Turn in Test Validation

Lang (2010) makes the case that Situated Knowledges fit well with the concept of rhetorical argumentation developed by Tindale (2004). Tindale proposes that, in informal logic, an argument can be better constructed from its rhetorical aspects as an ethical need

to check that arguments are adequately understood and not just correct. An argument constructed around the experience it generates in the audience, so the argument is well understood. Rhetorical argumentation fits well with the experiential approach in that the test and its validity argument can be constructed around the experience they generate.

This rhetorical turn in informal argumentation makes the process of argument construction bi-directional. A researcher explains the argument to an audience so the argument can be better understood, and the audience is allowed to participate in the process of argument generation, including their concerns and problematizations. Both Haraway (1988) and Tindale (2004) agree that scientific research in social sciences should be participatory because of the epistemological take of doing research in an embodied and situated way.

This rhetorical turn corresponds with the ideas from Dixon-Roman and Gergen (2014) on their social constructionist proposal for test validation and the ideas from Addey et al. (2021) about assembled arguments. In the concept of validity proposed here, when we assume that validity is a Situated Knowledge, we implicitly assume that validation is a process rather than a product; that must be done with the communities in a participatory way.

One observation made by Haraway (1988) is related to what she called "semiotic technologies" (p. 579). A semiotic technology is a procedure for communicating knowledge among ourselves. An argument like in the Toulmin model (Toulmin, 1958) is a semiotic technology. In the tradition of educational measurement, argumentation is a golden standard to communicate the case about claims related to a test and its interpretations and uses. This standard may be the consensus in the United States;

however, the experiential approach is prepared to change the semiotic technology used in the validation effort to other forms of knowledge construction. Some authors have criticized the adversarial nature of arguments in scientific research (Tindale, 2004). I will continue to use the term "Validity argument" with the caveat that we may require to take a different semiotic technology, including the possibility of leaving an adversarial tone entirely.

One interesting idea from Tindale (2004) is the definition of a *universal audience*. I will understand the universal audience as the group of all rational beings in the future. The concept of a universal audience is equivalent to the concept of a "Court of Reason" mentioned by Toulmin (1958, p.8). The vision now explained about test validity has a strong flavor of relativism. It is important to clarify then that a Situated Knowledge rejects relativism and assumes itself as limited and partial, basing its legitimacy as objective work by acknowledging its partiality and limits candidly.

The validity of the test, its interpretations, and its uses are defined by the universal audience's judgment, which is not relative. What can become "relative" is our vision about how the universal audience thinks about test validity. This inevitable partiality is candidly shared by explaining what we imagine the universal audience will require to properly understand the test's validity and its interpretations and procedures. In the experiential approach, an argument will be the default semiotic technology to explain the interpretation's validity and procedures to a universal audience. I will assume that the Universal Audience will require to include in the argument all the evidence generated with the communities affected by the uses of a test to satisfactorily explain to them the validity of the interpretation and uses.

The P.A.A.P. Minimum in the Experiential Approach

Given the previous definition about validity, a validity explanation in the experiential approach will be sketched around the response to four questions:

Partiality: in what ways is this validity explanation partial and situated?

Appropriateness: are the future instantiations of a test appropriate (ethical) in the view of the community to which I am directing this explanation? When used to make interpretations or in the procedures in which the test will be used.

Adequacy: are the future instantiations of a test adequate (scientific correctness) in the view of the community to which I am directing this explanation? When used to make interpretations or in the procedures in which the test will be used.

Preferability: why do we prefer this test versus other tests that are also appropriate and adequate for the community to which I am directing this explanation?

I will call this set of questions the P.A.A.P. Minimum in test validity. Part of the proposal in the experiential approach is to include certain minimums in test validation. The experiential approach uses the P.A.A.P. minimum to guide its validation effort.

For the experiential approach in specific, I will respond to the P.A.A.P. minimum in the following way:

Partiality: a selection of studies like researchers' positionality, discourse analysis, problematizations on structures of power; proposed initially by the validity researcher and opened for the communities to include their questions about the partiality of the validity explanation.

Appropriateness: an instantiation of a test will be appropriate if it is consistent with the values of the examinee taking the test and the community in which the examinee belongs. An instantiation will be appropriate for the universal audience if the communities affected by the uses of a test agree through a democratic deliberation about the appropriateness of that instantiation.

Adequacy: an instantiation of a test will be adequate if the community accepts the levels of construct under-representation and construct irrelevant interactions between the examinees and the test. Also, in the case of procedures, the test will be adequate if the variability of repeated instantiations of the test matches the acceptability criteria of the community for how reliable the instantiations of the test should be.

Preferability: an instantiation of a test will be preferable if the expectations negotiated with a community about any externalities of a test (societal goals, experience requirements, others) are satisfied for that instantiation.

All the previous are just a starting point, that can and should be problematized with the communities to which the validity explanations are directed. Validity is ultimately conceived as a Situated Knowledge, thus a conversation with alternative views is encouraged and expected.

Defining types of invalidities

The way in which we will respond to the questions in the P.A.A.P. minimum implies then five types of invalidities:

Value rejection: This will happen when there is a rejection to the test on an ethical level.

Construct under-representation: This will happen when the expectation about a test or part of it regarding the construct to be measured/assessed is not met.

Construct-irrelevant interaction: This will happen when an interaction outside the universe of acceptable interactions happens in an individual encounter with a test.

Purpose misfit: This will happen when a test fails on its purpose as a device, considering, in particular, the reliability of a score and how any conception of error of measurement does not fit with the needs in the context of use of the test.

Failed test expectation: This will happen when the expectation in specific instantiations or many instantiations of a test are not fulfilled, and a community would have preferred not to do a test if that expectation is not met. This situation can happen, for example, when a societal goal is not achieved; or when an experiential requirement is not met.

Table 5 summarizes the definition of those types of invalidities on different levels: a test relative to the universal audience, relative to a community, and one single instantiation.

The types of invalidity correspond to different levels of agency in the examinees: individual, community, and all communities affected by the uses of the test. There is a clarification on how exactly we know that an agent raised a concern, if as an individual decision or a communal one. For the universal audience, I will assume that it will expect a democratic deliberation among the communities affected by the uses of the test.

All the previous sources of invalidity are expected to be found in the extensive study of user experiences; that goes beyond the classical response process research to

more “thicker” modes of research, across all the stages in the experiences of the examinees and including the experiences of many groups. As validity is conceived a Situated Knowledge; the definition of types of invalidities must be available for change and problematizations generated by the communities.

Test validity as a Situated Knowledge is then a sort of explanation generator, that is prepared to write many explanations for many audiences; and those explanations may have different levels of quality and acceptability on each case. The work of test validation then is not to produce one single outcome (“the argument” in the argument-based approach) but to create experiences well explaining the validity of a test as part of the assessment experience. Those explanations are based in a compendium of studies that explore the questionings about the validity of a test from many communities; shared with those communities using appropriate semiotic technologies.

Table 5

**Summary of Definitions of Invalidity in the Experiential Approach, relative to
Different Audiences.**

Type of invalidity	Relative to the universal audience	Relative to one community	Relative to one single instantiation
<i>Appropriateness</i>			
Value rejection	There is an ethical concern among the communities affected by the uses of the test, deciding democratically.	There is an ethical concern for a community, decided through their own rules for deliberation.	There is individual concern about the ethical aspects of testing someone (myself or others).
<i>Adequacy</i>			
Construct under-representation	There is a concern about aspects of the agreed construct not being included among the communities affected by the uses of the test, deciding democratically.	There is a concern for a community about aspects of the agreed construct not being included, decided through their own rules for deliberation.	There is an individual concern about aspects of the agreed construct not being included.
Construct irrelevant interaction	There is a concern about interactions with the test that are spurious to the construct being targeted, among the communities affected by the uses of the test, deciding democratically.	There is a concern for a community about interactions with the test that are spurious to the construct being targeted, decided through their own rules for deliberation.	There is an individual concern about interactions with the test that are spurious to the construct being targeted.
Purpose misfit	There is concern about the test not being well prepared for the purpose it was supposed to fulfill, among the communities affected by the uses of the test, deciding democratically.	There is concern for a community about the test not being well prepared for the purpose if was supposed to fulfill, decided through their own rules for deliberation.	There is an individual concern about the test not being well prepared for the purpose if was supposed to fulfill.
<i>Preferability</i>			
Failed test expectation	There is concern about a side expectation not related with its purpose that are not being achieved, among the communities affected by the uses of the test, deciding democratically.	There is concern for a community about a side expectation not related with its purpose that are not being achieved, decided through their own rules for deliberation.	There is an individual concern about a side expectation not related with its purpose that are not being achieved.

A Procedural Description of the Experiential Approach to Validation

Now that we have clarified the concept of validity in the experiential approach, we can now explain a procedure for the experiential approach to validation. I will use the general sketch in Kane's argument based approach (1992, 2013), but will also include the participation of the communities affected by the use of the test in the appraisal phase, sketching an argument in two different ways, one starting with a focus on the invalidates found in the study of user experiences and writing an argument in a *reductio ad absurdum* mode of argumentation, and a second in the more classical IUA structure of an argument, so the network of propositions that emerge from the study of user experiences is "assembled" (Addey et al., 2020). Table 6 describes the procedure. It includes a Declaration phase, a Validation/development phase, an Appraisal phase, and an Evaluation phase.

Table 6

Procedural Description of the Experiential Approach to Validation

Declaration
1. Define the test, its purposes, expected interpretations, expected procedures of use, experiential requirements, and goals. Define the targeted population of examinees. 2. Define content domain and format options to explore during test development. 3. Identify risks and opportunities (potential consequences) in the considered procedures of use. 4. Identify who could be potentially affected by the uses of the test and define the communities to be included in the appraisal phase. 5. Define a strategy to represent those communities in the validation effort. 6. Explain how the researcher is going to respond to the P.A.A.P questions of test validation: partiality, appropriateness, adequacy, and preferability. Define what is going to be considered invalid for each of those questions.
Validation/development
7. Respond to the partiality question. 8. Do a preliminary study of the experiences of prospect examinees. Explore modes of assessment that make sense to the purposes of the test and examinee's experiences. 9. Develop one draft version of the test. Make different stakeholders take the draft version and study their assessment experiences.

10. Identify invalidity hypotheses that can be made from the study of the assessment experiences in the previous step.
11. For each of the invalidity hypotheses, understand its scope. Who could be potentially affected by each of those invalidities?
12. Decide which hypotheses to study in more detail.
13. For each selected invalidity hypothesis, continue with further inquiry to decide if the invalidity hypothesis is true or not. Include countermeasures in the test to solve those invalidities.
14. Iterate on a new version of the test and come back to step 7 or move to step 15.

Appraisal

15. After having a version of the test that is accepted by the test developer, start an appraisal phase by sharing the details of the validation process and its results to the communities affected by the uses of the test.
16. Ask the communities what their acceptability criteria would be to say the test is valid for its uses and interpretations. Problematize with them the definitions for sources of invalidity.
17. Start and document the conversation with those communities about the argument presented. Ask the communities about their opinions and any problematization necessary in the argument. Share with them relevant information so they can include themselves in the validity research.
18. Present the invalidity hypotheses studied to the communities. Make them choose and prioritize those, and any other they can raise. Iterate on the new invalidity hypotheses to be studied and adjust the tests if necessary.

19. After having an argument with enough information for the communities to decide, deliberate with them if it is valid or not to interpret and use the test. Document their deliberation. 20. Write an explanation for the Universal Audience (an Interpretation and use argument), using all the research done in the previous steps. Write explanations for each of the communities affected by the uses of the test, and include tailored explanations for each cluster of individuals within that may require a different explanation.
Implementation
21. Present evidence, conversation, and deliberation from the communities to the mandate of a test (in the case the mandate is not the communities), for them to proceed with further action. 22. In case the test is accepted, execute the test and capture information about the assessment experience happening in its official instance.
Evaluation
23. Evaluate the test from the perspective of the P.A.A.P. questions, as a validity researcher and with the communities. 24. Take ownership about instantiations of the test that were not valid. Provide compensatory measures for those. 25. Define what elements of the evaluation can be relevant for future instantiations of the test.

The *reductio ad absurdum* mode of argumentation in the explanations about test validity is the initial proposal to generate an experience motivating and caring with prospect examinees and their communities, as their voices can be heard. By acknowledging the communities' concerns, I expect this to elicit an experience of self-determination and effective Dialogue on the communities with the test designer. Theories of motivation would suggest that this would elicit a positive experience in the communities (Deci & Ryan, 2000). A candid disclosure with the communities about unsolved issues, the possibility to iterate and change the test, and the room for deliberation among the communities is also part of an experience that looks empowering rather than feeling authoritative, vertical, or patronizing.

Table 7 presents a mock example of how an invalidity hypothesis can be studied and resolved. The table includes fields to clarify how each invalidity hypothesis was accepted by a community and solved with further evidence.

A proactive study of the concerns of the communities can serve as well for the test developer to anticipate negative feedback from the public when there is still a chance to do something. A simplification of the argument to a backlog list of sources of invalidity is a straightforward way to explain the validity of a test to an audience by showing that the test is imperfect in tolerable levels for the communities affected by the uses of the test after some democratic deliberation.

It is important to remember that this is just an initial suggestion on how to proceed, and the whole validation procedure must be negotiated with the communities that we want to target in the validity argument. It is also important to listen to the types of concerns the communities raise, and be open-minded to change the validation procedure

itself when there are signals of bad test experiences, including the experience of understanding why this test is valid.

Table 7

Mock Example of an Experiential Approach to validation for a Specific Source of Invalidity

Potential invalidity	Test-taking experience evidence	Who is affected?	Further studies to confirm or refute	Conclusion	Next steps
Test anxiety (Construct-irrelevant variance)	videos of examinees' faces show anxiety for 80% of the population. 70% of the population mentions anxiety when asked about "how do you feel" in the middle of the exam. 80% recognize anxiety after completing the test. 30% of interactions in social media mention anxiety. A focus group with students agree anxiety might be negatively affecting performance.	Up to 80% of population.	Studies show anxiety can be a prerequisite to high performance, but it is not the case for all examinees. An experiment was done to moderate anxiety and see differences in performance. Results show anxiety affects performance of the first half of the exam and about 5% of the raw score in the test.	Anxiety is affecting around 60% of the population, with a potential impact of 5% of the total raw score due to this effect.	Design of interventions to provide emotional support for people who might feel anxious. Allow retesting.

Final Remarks

Incorporating feminist epistemologies and rhetorical argumentation to define the validity of interpretation and uses of a test is an exciting path to consider. Some authors have already considered topics like a participatory evaluation in test validation (Gergen & Dixon-Roman, 2014); however, there has not been a prescriptive proposal for test validation centered around user experiences and connected with participatory research methods. The experiential approach here proposed is an attempt to connect the ideas coming from UX research and human-centered, “user obsessed” design paradigms with the validation practices and theories in Educational Measurement.

Overcoming the Toulmin model (Toulmin, 1958) seems to require a better account of the examinees' realities, which have usually been represented "on average," ignoring the intricacies of the test and its users. Situated Knowledges (Haraway, 1988) and Tindale's rhetorical turn (2004) in argumentation are an interesting path to overcome this problem, by owning the political dimension inherent in Educational Testing by bringing the experiences of many actors and communities. A "thicker" version of the generalization part of Kane's argument structure for a test.

This rhetorical turn in informal logic and argumentation matches well with the concept of Situated Knowledges (Haraway 1988, Lang 2010). Under this view, validation is a process done with many audiences (including the universal audience), explaining why the interpretation and uses of the test are valid, in general, and in specific for instances of test use. This view assumes that a correct study of validity requires the active participation of the communities as agents in the validation process, studying the

characteristics of a test in a situated way and oriented by the experiences of the persons inhabiting those places (direct examinees and people affected by uses)

One limitation in Kane's argument-based approach (1992, 2013) is that it considers the validation process of a test to the generation of a product (the argument) without any input whatsoever coming from the users or the communities affected by consequences of the test use. It is not participatory. In addition to this, the argument-based approach considers a unique IUA for a targeted population that may require a different structure in the argument for different sub-groups within. Kane's argument-based approach only speaks to the universal audience (using Tindale's vocabulary). The concept of Situated Knowledges and its implementation via the rhetorical argumentation turn seems to solve both problems. Further experiences of using the experiential approach will tell its capacity.

Sheneman and Robinson (2020) reported the case of a test used in the signed language interpreting profession that serves as an example of the need for Situated Knowledges and participatory validation. The authors present the case for how deaf communities have little to no incidence in the construct definition of licensure for language interpreters. The absurdity of the situation (deaf people are the primary user of an interpreter and non-deaf people are defining the constructs to say when an interpreter is correctly communicating with a deaf person) shows how the concept of Situated Knowledges can become a better starting point when developing a test and validating it. In fact, Sheneman and Robinson based their critical analysis on the concept of Situated Knowledges.

In the experiential approach, I take a different position about Messick's ideas on declaring that a test is appropriate or not (Messick, 1989). In his view, a test is appropriate if the positive consequences of using a test outweigh the negative consequences. In the experiential approach, a test is appropriate if the affected communities' values align with the test's implicit values. I include consequences differently, in that the validation researcher considers potential consequences (named in this approach as risks and opportunities) to define who is potentially affected by the uses of the test, instead of trying to impose one single view about the final utility of a test. The communities should be able to change the implicit utilitarian stance in Messick's reasoning if they want to solve those moral decisions in different ways, with the moral philosophies more representative of their cultures.

This new stance suits better the re-centering around user experiences, as we can anticipate a value rejection by studying the reactions before the official use of a test. This stance can be considered up to a certain point a post-modern ethical stance, in which no specific moral philosophy is recommended but the ones preferred by the communities. This stance makes appropriateness a negotiated, situated characteristic of a test in the context of use (its interpretations and the procedures in which it will be used).

How do you make sense of your assessment experience explored at the beginning of this paper? How does it relate to the concepts and ideas presented in this paper? Does it make sense for the experiential approach in contrast with your assessment experiences? I am taking the risk of you responding no, but I imagine you will be more convinced than discouraged with this exercise.

The current historical context in educational measurement makes the study of user experiences more relevant and its connection with test validation in an integrated way as well as test development. We have the chance in our hands to create better assessment experiences. This chance started from the criticism of the lack of connection and touch with examinees in test validation, but it is also a path for a different way to enjoy our own experience as practitioners in the testing industry.

A good test experience is not limited to engagement, it needs to extend to a community level by effectively explaining why a test is valid. Standardized tests have created a strong negative memory and rejection in many people around the globe. However, we have a chance now to amend this past if we put explicit efforts into casting better assessment experiences that are pleasurable, empowering, and can express the empathy and respect of its developers. The future of our field depends on this.

CHAPTER 5: EMPIRICAL STUDIES ON THE EXPERIENTIAL APPROACH

Test Design and Validation in the Experiential Approach

As the Experiential Approach is sketched in a more theoretical level, it becomes important to see its capabilities in practice. The Experiential Approach is conceived for both test design and validation, and the previous three chapters sketched mostly the validation part of the approach. I will explain now the connections with test design, and describe in more detail the two empirical studies proposed for this dissertation, as proof-of-concept for the Experiential Approach, in both test design and validation.

Bringing Game Design into Test Design

As described in the introduction chapter, the experiential approach was born from the comparison between game design and test design. Authors in game design have suggested that an effective way to design a game is to concentrate in the experience that is elicited by the constructed device. In contrast, test design does not care much about the experience that is elicited by a test, as long as this experience relates with the validity of the interpretations and uses of the test.

However, tests can and need to be delightful (Wodzak, 2021). From a phenomenological perspective, a test designer may have other goals in terms of the experience that he/she/they want to elicit, but still that designer will require to center the design around the elicited experiences. Sireci and Araneda (2021) used the term “Positive Assessment” to refer to the will in a test designer to elicit positive assessment experiences.

The practices in game design can shed some light on how exactly we can make this happen. This principle of concentrating and designing for the experience that is elicited, is usually put in practice through user experience research (UXR), looking at the

reactions of users to different parts of a videogame for example, or just playtesting a prototype. This motivated the conceptual framework for Assessment Experience explained in Chapter Two, so what we understand as user experience in the context of a test is made explicit and understood better.

In Test design and development, we also have the challenge of creating a test that can be interpreted and used validly. This motivated chapters three and four so we can propose a way to validate a test that is synergic with a design and development strategy that is intensive in UXR. We will need to do a lot of User Experience research if we center our design in the users, and that opens an opportunity for us to leverage on that research that is already planned to be done. The experiential approach to validation presents a few ideas for how to take advantage of this situation.

In the case of videogames, the Gaming Industry has embraced many of the design principles of Agile methodologies, creating cycles of development in an unstructured and open-ended way. This is the design part of the experiential approach. Similar to game design, the Testing Industry has embraced similar practices. In the case of tests, sometimes we have clear the type of experience we want to elicit, but we do not know how to do it for each person in our targeted audience. The iterative process in dialogue with users is the design counterpart of the validation part of the experiential approach.

Interestingly, the model proposed in Chapter two can be used to explicit a type of experience that we may intentionally want to elicit. Similarly to what Evidence-Centered Design did by explicating different models to explain valid inferences, the Experiential Approach focuses on how we can explain inferences by concentrating in the Experience that the test should elicit. By describing the types of interaction, sense-making and stages

in an Assessment Experience, the Experiential Approach can tackle the Design problem in Testing.

A test in the experiential approach is understood as a device that elicit an experience, and thus we can define a test from what it is meant to elicit. I will name the description of expected elicited experience as the *Experience Specification* of the test.

This dissertation then will continue with two empirical studies, as proof-of-concepts of the Experiential Approach for both test design and validation. First, I will do a proof-of-concept for the design part, and then a proof for the validation part. I will explain the connection of those studies with the research questions of this Dissertation and present each idea with a Methodology section.

Empirical Studies and Operational Research Questions

As explained in the Introduction, the three previous chapters correspond to the theoretical basis and literature review of the experiential approach. The dissertation proposal includes as well two “conceptual hypotheses,” to be studied in empirical studies. Those hypotheses are:

- a) The experiential approach will provide valuable information to improve test design and validation practices above and beyond information provided by traditional methods.
- b) By implementing aspects of the experiential approach, we can evaluate its feasibility and practicality.

The Experiential Approach and its ideas may require years of research to fully understand and correctly evaluate the approach. Nevertheless, in this study I will conduct

two proof-of-concept studies that can approximate the feasibility of the Experiential approach, and what are the practical limitations that arise in context of use of the approach.

The overall approach to do this sort of “Evaluation” of the Experiential approach is not necessarily empirical. I will not operationalize variables to capture a value of the approach. I will provide instead an evaluative judgement of the approach after doing the proof-of-concepts, that each one will have their own research questions and goals. The conceptual hypotheses here serve as a meta-objective across the empirical studies presented in this Dissertation.

With respect to the two proof-of-concept studies, one will test the design part of the approach, the other will test the validation part of the approach. For the first one, I will analyze the responses of students who took a diagnostic test designed using a simplified version of the experiential approach, in the context of a class given by Professor Lisa Keller on Regression Analysis. For the second empirical study, I will do an analysis of the experiences shared in Twitter by examinees in Chile about the test used for college admission. I will analyze the tweets generated by the public in the specific instance of the test that was done during the COVID-19 pandemic, in order to see the potential sources of invalidity on that instantiation of the test, and refute or fail to refute the concerns in the public at that time.

The judgement about the value above and beyond other methods will come from an examination of the situations and outcomes in the proof-of-concepts, in contrast with what I know from experience about alternative approaches. I will not make this question empirical. This could be empirical if, for example, we have two teams build a test with

different approaches and compare user experiences for each version of the test.

Unfortunately, those types of studies go far beyond the capacities I have as a doctoral student.

The feasibility and practicality of the Experiential Approach will be judged by the difficulties found in the process of executing the two proof-of-concept studies. That type of information will not be directly shared in the papers explaining the results of the studies, but shared in the Discussion section. One topic of relevance is related with the failed attempts of other studies that were originally meant to be part of this Dissertation. I will mention those in the Discussion section, so the salient problems of the Experiential Approach can be fully discussed.

The first empirical study, involving a test for a diagnostic purpose in a class, is based on an attempt to use the experiential approach. This study helps to respond the conceptual frameworks as it provides an example of a test, done end-to-end with this Experiential “spirit,” in a simplified version of the approach, apt for classroom assessment. The adjustments in the approach to fit the needs of the specific context in which the test is meant to be used also provide interesting insights on the practical limitations of the Experiential Approach.

The second empirical study about Tweets in Chile provides valuable information about the Validation part of the approach. Once again, in a simplified version of the approach, the challenges that will arise from this study, the topics that are mentioned, the research that is suggested, and all the political and technical challenges on its execution will provide valuable insights for both conceptual hypotheses in this Dissertation. Chile does not have a good tradition on Test Validation; however, the few studies available

about the test are known by the author, so a judgement of the value above and beyond other methods can be provided.

In terms of research questions, the first study will include the following “operational” research questions:

1. What are the student’s preferences regarding their assessment experiences?
2. How congruent is the initial level of skill and knowledge of Professor Keller’s class relative to the content of the Regression Class she will teach?
3. Did the Assessment Experience elicit good experiences for the students?

For the second study, I will respond the following “operational” research questions:

1. What topics mentioned in the tweets can be considered potential sources of invalidity?
2. What validity argument could respond to the concerns raised by users on Twitter?
3. What parts of the validity argument are more important to the users? How do different stakeholders prioritize the study of the invalidity hypotheses discovered through the analysis of the Tweets?

Now I will proceed to explain in more detail the method sections of each of the empirical studies in this Dissertation.

First empirical study. Centering Test Design around the Experiences Elicited by a Test in Classroom Assessment.

As I was sketching the studies to be part of this Dissertation, it was clear that I needed an opportunity to design some test using the experiential approach and the different parts previously explained: the conceptual framework for assessment experience, the ideas of feminist epistemologies, the P.A.A.P. minimum etc. To do all the previous can be very time consuming. To consider the many potential iterations with users, plus any need to have IRB approvals to even start to do some research, made me try on something simple as a proof-of-concept. A test for a class fit with my needs.

In order to make a proof-of-concept of the experiential approach to test design, I contacted professors in my program offering classes during the Spring 2022 semester, so I could create for them a test that could be useful for their class. One professor responded to the call, showing interest in creating a diagnostic test for her Regression Class. We had a series of calls and iterations, which are described next.

This study aims to present a novel way to design a test that makes an explicit effort to cast a specified experience on the examinees. In this novel approach, we will introduce the concept of *experience specification*. An Experience specification makes explicit the type of experience we want to elicit with a test, using the conceptual framework for assessment experience explained in chapter Two.

In terms of research questions, we aim to study the initial state of Professor Keller's class with respect to the student's preferences regarding their assessment experiences, and to see if the test created using an experiential approach elicited good experiences for them.

The following sections present the idea of an experience specification, an experiential approach to validation, the process taken with Professor Keller to create the test, and the results obtained with the test.

Experience Specification

As a reminder, the conceptual framework for assessment experience [Chapter 2] considers three aspects or principles, based on Dewey's view about Experience: a principle of interaction, a principle of continuity, and a principle of sense-making. There are different categories proposed in the framework for each of those principles. Particularly for the types of interactions, the framework considers five: cognitive, emotional, expressive, fluent, and physical.

The idea of an experience specification is to make explicit how we expect the experience to happen when mapped into this framework. Each of those principles can have a specification, like the content specifications that a test can have, targeting specific types of experiences. See Table 8 for a summary of the categories considered for each of the dimensions/principles in Araneda's framework.

Table 8

Categories in Araneda's Framework for Assessment Experiences.

Principle of Interaction				
Cognitive	Emotional	Expressive	Fluent	Physical
The examinee interacts with the test in a cognitive level. He/She/they react using their cognitive capacities.	The examinee interacts with the test in an emotional level. He/She/they react experiencing certain types of emotions.	The examinee interacts with the test in an expressive level. He/She/they react experiencing the need to express themselves in an opinion or other forms of engagement with the test as a device.	The examinee interacts with the test in an fluent way. He/She/they react experiencing actions that do not happen in a conscious level.	The examinee interacts with the test in an physical level. He/She/they react experiencing physical sensations the need to do certain body movements.
Principle of sense-making				
Free-stream	Immediate	Integrative	Co-integrative	Dialogical
The examinee does not make sense of the full stream of information in Experience.	The examinee barely makes sense of the stream of information in Experience.	The examinee makes sense of the stream of information in Experience, being able to explain it as a story or narration.	The examinee makes sense of the stream of information in Experience, being able to explain it as a story or narration, including the experiences of peers or other people around the examinee.	The examinee makes sense of the stream of information in Experience, being able to explain it as a story or narration, including the information and Dialogue with the test developer about the assessment Experience.
Principle of continuity				
Ante	Pre	Zero	Post	Long-term
Experiences lived even before knowing that the examinee will take a test.	Experiences lived in anticipation to test a test.	The experience itself of taking a test.	The experiences after the test and their results were given.	The long-term experiences related with consequences associated with a test.

Experiential Approach to Validation

Also, as part of the development of this test, we tapped into some of the ideas of the experiential approach to validation [Chapter 4]. This approach to validation bases its work on the parsimonious study of the experiences of many users. The experiential approach considers four questions that guide the validation effort: partiality, appropriateness, adequacy, and preferability. We will refer to those four questions as the P.A.A.P. minimum. See Table 9 for an explanation of each of the questions considered in the validation effort.

Table 9
Questions in the experiential approach to validation.

Guiding questions	Description of the question
Partiality	In what ways is this validity argument partial and situated?
Appropriateness	Is it appropriate (ethical) to use this test in the intended way according to its purposes? To interpret it and use it in certain procedures.
Adequacy	Is it adequate (scientifically correct) to use this test in the intended way according to its purposes? To interpret it and use it in certain procedures.
Preferability	Why do I prefer this test versus other tests that are also Appropriate and Adequate?

Iterative Procedure for Test Construction

In this section I will describe the collaboration process with Professor Keller, that considered three calls to iterate over versions of different parts of the Assessment Experience.

First call: In our first call, we declared our intentions about the test to be constructed. We responded to a few questions about the test we wanted to build. First, we asked ourselves the purpose of the test. Professor Keller mentioned at least four potential purposes: first, identify the state of the students relative to pre-requisite knowledge for the Regression class, such as concepts in probability theory and statistics. Second, the test could have a more concrete purpose for identifying the content to be taught in class, so she could include content that is less known by the students and is still a pre-requisite for the class. Third, if we could have parallel information about the learning styles or preferences of the students, that would be a nice thing to have. Fourth and last, the test ideally should be helpful to provide feedback to the class.

Regarding the interpretations to be done with a score coming from the test, Professor Keller was particularly interested in having interpretations targeting statistical reasoning or inferential ways of thinking and specific content and techniques. When asked about procedures in which we could use test results, Professor Keller mentioned two: identifying content to be added/covered in class and Remedial actions for the class.

In terms of experiential requirements, professor Keller mentioned that she preferred the test to be relaxed, with no high stakes, and become a learning experience for the students. An opportunity to reflect on the material. Also, the test should be able to be done at home or in a safe space, as comfortable as possible, and with full access to materials.

In terms of higher societal goals, Professor Keller mentioned an overall idea for setting a tone about engaging with the content rather than just completing assignments. A learning experience that conceives thinking and learning in higher value than just

knowing how to do a task. Regarding risks related to the test, we mentioned that if we hinder future learning experiences because of the assessment experience with this test, we would be under a scenario of a negative consequence of using the test.

We also started to ask ourselves about the P.A.A.P. validity questions in the experiential approach. In terms of partiality, it was for the researcher to declare its positionality. It was clear that, as this being a proof-of-concept of ideas that the researcher also proposed, there is an incentive to make the exercise succeed. The fact that I was a student of the Regression class with a different professor may affect the way he conceives this diagnostic test.

Regarding the appropriateness of the test, we considered that the test would be appropriate if it at least does not interfere with the students' learning experience. If, for example, we see or suspect that the test would be a harmful experience for them, or limit their future learning experiences related to Statistics, then the test would not be appropriate.

Regarding the adequacy of the test, content should target the requirements of the Regression Class. Items that can directly ask for previous content can be a straightforward way to accomplish adequacy. In terms of reliability, the number of students that would take the test makes it difficult to produce criteria for the level of reliability necessary for the test. Regarding the preferability of the test, the more comfortable they feel sharing their mastery levels during the assessment experience, the better for the students.

Given all the previous considerations, we concluded the test format should be online and allow anonymity of the students when responding. We considered this anonymity a prerequisite to accomplishing the experiential requirement of comfort that we imposed for the test.

After this call, the I decided to define the following assessment experience, specified in Tables 10, 11 and 12; using the conceptual framework for Assessment Experience. You will find in Appendix A one copy of the first version of the test and intro survey created. In the planned Assessment Experience there will be two sections. One for an introductory survey asking background information and presenting a mock example of an item so we can study the experience elicited by that item format; and a second part with the actual diagnostic test to be used for the class.

Table 10

Experience Specification in the First Version of the Assessment Experience

Overall Assessment Experience

Principle of continuity					
	Ante	Pre	Zero	Post	Long-term
Specification	No expected casted experience here.	Students are expected to experience the assessment in a way that can elicit a comfortable experience, and a candid sharing about their knowledge in the targeted topics.	Students are expected to experience questions that can tap on the knowledges that are a pre-requisite for the Regression class. Again, students should feel comfortable during this process, being invited to engage in the content rather than value procedural knowledges without further understanding.	Students are expected to feel that the class is reactive to their needs. Put their learning experiences first.	Students are expected to receive an effective and positive learning experience relative to the contents of the class (Regression analysis).
Execution (draft v1)	No particular action done.	A survey was designed and delivered, asking to the students general questions about them, and questions about the item format to be used.	A test was designed and delivered, including many sections: an introductory explanation, a video from professor Keller, selection of key concepts and two questions for each: one open-ended and one presenting a definition of the concepts and asking which concepts in the definition were not recognized. Final feedback and star rating available at the end of the test.	Results delivered to professor Keller only.	No particular action done.

Table 11

**Experience Specification in the First Version of the Assessment Experience for the
Introductory Survey.**

Introductory Survey

Principle of Interaction					
	Cognitive	Emotional	Expressive	Fluent	Physical
Specification	Students are expected to identify if the item format to be used in the test (presented in a mock item in the survey) is well suited to the diagnostic purpose in their opinions.	Students are expected to share their knowledge feeling comfortable, not pressured, not judged.	Students are expected to end with a positive opinion about the survey, and future assessment experience in a diagnostic test.	Students are expected to share candidly their preferences when tested.	Students are expected to use a laptop or a cellphone to respond to the questions, in a comfortable way on a physical level.
Execution (draft v1)	A mock item was included in the survey. Questions asking about preferred item types were also included.	The survey was designed to be responded anonymously.	The act itself of having a survey before the test is expected to elicit the specified expressive interactions.	The responses of the test were anonymous. This was put in place to elicit candid responses.	A high-quality delivery system for surveys was used, helpful to present a test in a laptop or a cellphone..

Table 12

Experience Specification in the First Version of the Assessment Experience for the Diagnostic Test.

Principle of Interaction					
Specification	Cognitive	Emotional	Expressive	Fluent	Physical
	Students are expected to recall their knowledges targeted by the test.	Students are expected to feel comfortable while responding to the test.	Students are expected to have a positive opinion about the assessment experience in general and Professor Keller in particular.	Students are expected to share their knowledge candidly.	Students are expected to use a laptop or a cellphone to respond to the questions, in a comfortable way on a physical level as well (usability).
Execution (draft v1)	Each content to be covered in the test included two questions: one open ended asking directly about the concept, and one presenting a definition of the concept and asking the Students to select all the words that they did not recognize.	The test was responded anonymously. A video at the beginning of the test from professor Keller included some messages cheering the students up before the test started.	A video at the beginning of the test from professor Keller was put to explain the purpose of the test and enhance the bond between professor and students.	The test was responded anonymously.	A high-quality delivery system for surveys was used.
Principle of sense-making					
Specification	Free-stream	Immediate	Integrative	Co-integrative	Dialogical
	[Given by the interaction principle]	Students are expected to share any specific situation while doing the test that may be affecting the validity of the interpretations and uses.	Students are expected to make sense of the diagnostic experience and think though the validity of the test done.	No expectation in a co-integrative level.	Students are expected to experience the test as an action from Professor Keller to empathize with them and enhance their connection with their professor.
Execution (draft v1)		A question was included in-between sections asking about any situation that may be interfering their assessment experience.	An open-ended question at the end was included, asking for feedback to the students. A 5-star rating question was also included, asking the students to evaluate the assessment experience.		A video in which professor Keller explains the purpose of the test was included at the beginning of the test.

Second call: in the second call with Professor Keller, we reviewed the first draft version of the diagnostic test. We started to respond to the P.A.A.P. validity questions in the experiential approach after having Professor Keller herself experience the test.

We continued previous conversations, asking ourselves again about the P.A.A.P. questions in the experiential approach to validation, and seeing if the current draft achieved the expectations in terms of validity. Regarding the appropriateness of the test, we felt the draft test and initial survey were not triggering traumatic experiences for the students. Regarding the Adequacy of the test, the item format and questions seemed reasonable. Regarding the Preferability of the test, Professor Keller did feel uncomfortable with the item format. On one side, asking an open-ended question might not be comfortable, forcing the students to do a full scan of their knowledge and work on a written response.

Also, presenting a definition made sense to Professor Keller for the second item format. However, she suspected that asking about what they did not recognize would be less comfortable than asking about what they did recognize. By asking in this "positive" way, the ignorance of a concept can be deduced from the options not taken, protecting the students and their self-perceptions.

Given this, we decided to iterate and change the draft to a second version, in which the mock item for the second item format switched from asking what concepts in the definition they do know, instead of asking for the ones they do not know. After this, we sent the first survey to the students, allowing one extra iteration in the design of the primary diagnostic test.

Third iteration after receiving results from the students in the entry survey: 12 students

responded to the survey. When asked what item format they preferred, they chose multiple-choice items over open-ended questions. More information will be given in the results section.

Data Collection

In total, we separated data collection in two stages: an introductory survey and diagnostic test. For the introductory survey, twelve students responded to the survey. For the diagnostic test, ten students responded. Annexes A and B have copies of both survey and test. Both tests were part of the class with Professor Keller, so no informed consent was necessary. No information about demographics was asked to the students, so their anonymity could be maintained as the class was small.

Data Analysis

I analyzed first the results from the initial survey asked to the students. I calculated the mean and standard deviation of the responses of the students to the question asking for their preferences regarding item formats. This analysis is meant to justify the decisions taken in the process of constructing the Assessment experience in a way that connects with its Users, making it more experiential. This analysis also corresponds to the first research question of the study.

For the diagnostic test, I calculated the rate of non-selection of different options across questions. This will be our metric to detect if a topic is not well understood by the class, and to understand the state of the class relative to the contents of the class.

Finally, I analyzed how students rated their assessment experiences, and what type of written feedback they provided. I categorized their written feedback by the

number of stars they provided, and discuss if in their comments there is justification to say that the test elicited good experiences for them, and how valid are the interpretations and uses coming from this test.

The previous analyses allow us to respond to the specific research questions of the study:

1. What are the student's preferences regarding their assessment experiences?
2. How is the initial level of skill and knowledges of Professor Keller's class relative to the content of the Regression Class she will teach?
3. Did the Assessment Experience (both entry survey and diagnostic test, both created using an experiential approach) elicit good experiences on the students?

Results

The Result section is divided in three parts. One for the results on the question asking about Student's preferences for Item Type, one presenting the results of the feedback obtained by students about the mock item we included in the entry survey, and one last section with the results of the diagnostic test.

Item Preferences for Item Type.

To the question "What type of questions in a test do you find less stressful to respond to?" in the entry survey, the students ranked item formats as summarized in Table 13, which presents the average and standard deviation of the rank assigned by the students to each option. The lower the rank, the higher priority they gave to the option.

Table 13

Summary of preferences (ranks) for item formats that can lead to more comfortable experiences (N = 12).

Item Type	Mean	SD
Multiple choice with one correct answer	1.67	1.03
Written response - short	2.50	0.76
Multiple choice with many answers	2.58	1.11
Written response - short or long	3.25	0.92

Note: Lower average means higher rank.

Feedback on Item Format of Mock Item.

About the mock item included, a problem related to the platform and multiple choice multiple select (MCMS) items was detected thanks to the students' feedback. One option in the platform's settings was off, blocking students from selecting more than one option in the MCMS question. One Student reported the following comment:

"The first question felt too ambiguous and open-ended, the second question moved on before I could select all of my responses."

Many other students identified a situation with the entry survey, in that the multiple choice item was not allowing students to select more than one option. There was

a flag I did not activate on time. Thanks to the report of the students, that technical problem could be solved.

Many other students also reported a preference towards the multiple-choice question. The following comments help to clarify this preference in some students:

“I feel that I understand the concepts and I could manage them in class, but it is difficult to define them.”

“I prefer the second one [Multiple choice question]. If some terms in the definition can be overstriking which can make me understand better.”

“I prefer the second way to answer the question [Multiple choice question]. Although I can find the description of any concept on Google or from other resources, sometimes I do not necessarily fully understand what this concept means. If it is a multiple-choice question, I must understand the meanings of all the words in the question before I can choose the correct answer. I think this kind of learning is more effective.”

The second comment cited can be interpreted as a challenge regarding the construct. Defining a concept and understanding a concept are two different things from that student's perspective. The fourth comment cited points out that the multiple-choice question forces the student to question itself for each sub-concepts required to refine a target concept, which is not elicited when we ask somebody to define a concept in their own words. This observation is, in some way, a challenge in terms of construct under-representation in the end.

The second item format asking for the concept they could recognize in a definition was a great success. Students either liked the item format and evaluated it positively or were indifferent to the format of the question. The cited comments previously, plus the preference for multiple-choice multiple select items over an open question with a potential long response, made us decide to keep the multiple-choice multiple select questions and nothing else in the final version of the test. We also identified timely the technical problem with the mock item and had multiple selections available in the diagnostic test.

It is interesting to see how the participation of the students provided profound insights into the validity of the test. The language used by the students is not technical. However, with a bit of effort, we can locate their criticism in the language of validity theory.

Results of the Diagnostic Test

We asked six questions, presenting the definition of the following concepts: sampling distribution, p-value, null and alternative hypothesis, confidence interval, central-limit theorem, and statistic. The questions can be found in Appendix B. For each question, multiple options were given so students could identify the concepts in the definition they felt they understood or recognize. As for the purpose of the test, we calculated the percentage of “ignorance” on each of the options, so we could see what were the concepts less mastered by the class. The results are presented in Table 14. P1 to P6 represent the question, and the concept after is the option that was made available in the question.

Table 14**Percentage of students not selecting an option for the question****“What words or ideas do you recognize from the following definition?”****(N = 10)**

Question and concept in definition	Percentage not selecting	Question and concept in definition	Percentage not selecting
P4_one_sided_confidence_intervals	60%	P6_stated_alternative_hypothesis	20%
P3_viewpoint	50%	P2_observed_results	10%
P6_prescribed	50%	P2_null_hypothesis	10%
P6_characterize	50%	P3_statement	10%
P1_frequencies	40%	P3_population	10%
P2_rejection_point	40%	P4_lower_bound	10%
P3_reasonable_doubt	40%	P4_upper_bound	10%
P4_computed	40%	P5_normal distribution	10%
P1_probability_distribution	30%	P6_observed_data	10%
P2_alternative_hypothesis	30%	P6_distinguish_null_from_alternative	10%
P3_claim	30%	P1_statistic	0%
P4_parameter	30%	P1_sample	0%
P5_sample_means	30%	P1_population	0%
P5_population_distribution	30%	P2_level_of_significance	0%
P5_sufficient	30%	P3_argument	0%
P6_quantity_derived	30%	P4_estimates	0%
P6_data_set	30%	P4_confidence_level	0%
P2_hypothesis_test	20%	P5_distribution	0%
P6_statistical_hypothesis_testing	20%	P5_sample_size	0%
P6_quantify	20%		

Note: P1 asks for sampling distribution, P2 for p-value, P3 for null and alternative hypothesis, P4 for confidence interval, P5 for central-limit theorem, and P6 for the concept of statistic.

As for the feedback received about the diagnostic test, Table 15 summarizes the number of students that marked different star rating levels and their comments about the assessment experience they had.

Table 15
Student Feedback about the Diagnostic Test

Number of Stars given to the test (1 to 5)	Number of students	Comments written by students on each category
5	7	“Good way for me to review the info I should know,” “I enjoyed it and it was nice to remember and revise the concepts.”, “Great experience! User-friendly,” “Very easy to go through!,” “Awesome!”, “This was a great assessment experience, thank you very much!”
4	2	“"recognize" vs "understand and can explain well" are very different levels of familiarity (depending on what conclusions are meant to be drawn from the responses).”, “I do not know what is the purpose of this survey. Is that help us to review the definitions? I am not feeling bad during the survey though.”
3	1	“For me, with almost no foundation, it was quite scary to think of doing a project. Because I really had no clue, I did not know where to start, and even though the data was right in front of me, I was very confused.”

The more negative comments about the assessment experience can help us have some ideas about the validity of the interpretations and uses of the test. One Student made

a very interesting challenge to the stem of the question, that is making equal two things (recognize and understand) that are indeed different. The Student is even aware of the validity problem in the question due to the ambiguity in the stem.

The last comment is a more serious concern to be considered. Even with the efforts we made to think about the appropriateness of the test, we still crossed a line with at least one student, creating an experience that may interfere with his/her future learning. As the student comments, the test will still create a bad experience for students with low or no foundation in the questions, even if the definitions are explicitly provided. The fact that the student now can become more aware that is very behind the implicit expectation in the test, expecting from the students to, at least, be able to read the definition; poses a big pressure on the students. How does this affect the mood of the student? His/her/their confidence? As for this student, we may suspect that the given responses are underestimating the real understanding of that student.

Conclusions for this study will be given together with the second empirical study.

Second empirical study: A Study about Invalidity Hypotheses mentioned by Users on Twitter.

User Experience Research (UXR) has proven to be an interesting new area of interest for test developers, in which we can find valuable information about the quality of an experience elicited by a test (Araneda, 2022a) [Chapter 2] and of the validity of interpretation and uses of a test (Araneda, 2022c) [Chapter 4]. Critical scholars have mentioned the need for making validation participatory (Gergen & Dixon-Román, 2014), and one step in this direction is – to the very least – to include their experiences in the validation process.

User Experiences can be considered a source of evidence at some point. In the American Educational Research Association, American Psychological Association, and National Council on Measurement in Education *Standards for Educational and Psychological Testing* (2014, referred to hereafter as the *United-States Standards*, or *US Standards*), evidence coming from response processes can be considered part of a more comprehensive source of evidence coming from user experience research. Current views about response process data support this (Zumbo & Hubley, 2017). Many methods used in UX research, like the ones we use in educational testing, do not limit their scope to cognitive interactions, but extend to other types of interactions between an examinee and a test.

Twitter offers an excellent opportunity to study user experiences. Users are avid to share their thoughts, emotions, and reactions to different products, and educational tests are not the exception. On Twitter, we will find students, parents, teachers, proctors, and many other stakeholders looking for a co-construction of their experiences and a

dialogical engagement with test developers, making comments or asking questions to the accounts available in this social network of test developers.

The COVID-19 pandemic also brought a particular need to look at this layer of evidence when it comes to studying the validity of interpretations of a score. With new, unexplored situations happening all over the world in all evaluation practices, insights coming from the experiences lived from the within can bring valuable information about the efficacy of measures taken to navigate the pandemic and gauge in a better way the results obtained during this time.

An experiential approach to validation (Araneda, 2022c) [Chapter 4] is an approach in which a validity researcher constructs an interpretation/use argument (IUA; Kane, 2013a) from the emergent information coming from user experiences. In this approach, the validity researcher writes an argument first using a *reductio ad absurdum* mode of argumentation. In this mode of argumentation, we assume that all the assumptions in an argument are incorrect, and we disprove that those are incorrect with further evidence. The argument then responds to a list of potential sources of invalidities, arisen by a validation researcher from the study of user experiences, to be disconfirmed later. The experiential approach considers an appraisal phase with the communities' inputs, writing an IUA in the more classical format à-la-Toulmin (Toulmin, 1958), summarizing all the results found in the process.

This study aims to make an initial attempt to use this approach by studying user experiences on Twitter, finding invalidity hypotheses, and trying to sketch a validity argument that could serve as a research agenda that respond to the concerns raised by the

students and other stakeholders on Twitter. This study is checking the initial stages of the experiential approach, related to hypothesis generation.

Background for the Study: College Admissions Testing in Chile During COVID

Chile offered a unique opportunity to test this approach to validation. During the COVID-19 pandemic, the test used for their college admission system – named Prueba de Transición, acronym PDT in Spanish – was applied to students even when there was no vaccination available. The admission system in Chile takes its decisions exclusively based on the scores generated by the PDT battery of tests, and two scores that depend on high-school GPA. This means that, without the scores of the test, the admission decisions can only be based on high-school GPA. This pushed authorities to take the test in person despite the risks that the COVID-19 pandemic posed, putting students in a significant risk decision: to do the test and risk their health, or to wait a year and apply to college later.

The correct interpretation of the scores of that year is In question. A sudden shift in the composition of students doing the test, with students from private schools jumping from 11% to 17%, show that the decision process of taking or not the test was mediated by socioeconomic status (DEMRE, 2021). Table 16 shows the series of percentage of private schools in a window of five years in the past from that instance of the test in this study (a nearly 70% increase). We are unsure why and how this jump happened or if we should examine other aspects of the score interpretation.

The situation in Chile is representative of many other places in the world and shows why an experiential approach is an approach that deserves consideration. The type

of evidence that it studies is novel, and the approach balances in a better way the confirmationist bias of test developers with the refutationist bias of the public.

Table 16

Evolution in the participation of private schools in the total population of examinees taking the Chilean university admissions test.

Year	Percentage
2021	17.2%
2020	11.7%
2019	11.4%
2018	10.5%
2017	10.4%

We will study the tweets generated by users during the period in which the test was prepared, taken, and when the results were received. We will use the stages in an assessment experience mentioned by Araneda (2022a) [Chapter 2] to structure the analysis of the tweets. Next, we will identify invalidity hypotheses from the topics present in the tweets, and sketch a validity argument from the general topics that emerge from the study. We will include as well a survey, asking students for priorities along the invalidity hypotheses detected, to finally present a visualization of the sketch of the validity argument that includes the priorities of the communities.

Method

Data

We connected to Twitter API to retrieve tweets emitted between May 2020 and March 2021. The PDT tests were taken by students in two occasions, one starting on January the 4th 2021 and a second starting on January the 8th 2021. We separated the tweets in three phases of the assessment experience: pre-experience (from May the 1st 2020 to December the 31st 2020), zero-experience (from January the first, 2021 to January 12th 2021), and post-experience (from January the 23th, 2021 to March the 15th, 2021). We selected two types of tweets: ones having the phrase “Prueba de Transición” on them, and the ones directed to the account @demre_uchile, the official account of the developer of the test.

We found 7,115 tweets for the pre-experience, 3,218 for the zero experience, and 1,541 for the post-experience. Table 17 summarizes the number of tweets found for each type of search and period in the assessment experience.

Table 17

Number of tweets found.

Starting date	End date	Stage in the assessment experience	Number of tweets found	
			‘@demre_uchile’	‘Prueba de Transición’
05/01/2020	12/31/2020	Pre	1298	5817
01/01/2021	01/12/2021	Zero	1635	1583
01/13/2021	03/15/2021	Post	346	1195

I analyzed a random sample of 10% of the tweets in their original language (Spanish) and categorized them on topics, using Grounded theory (Strauss & Corbin, 1994). Next, I wrote invalidity hypotheses for each of the topics found when we can imagine that the matter mentioned by the user can come from a source of invalidity.

Asking for priorities and sketching a validity argument

With a list of topics in hand, I created a survey to be shared with different groups that can be considered affected by the uses of the test. The people who responded to this survey is a different group than the people who wrote the analyzed tweets. Table 18 presents a summary of the demographics of the respondents. I contacted the respondents using my personal social media accounts, and by sharing the survey in forums of undergraduate students at my former university in Chile (Universidad de Chile).

Table 18

Demographics of Survey Respondents (Total N = 30)

Type of respondent	N	Percentage.
Student who took the test the year of the study (January 2021).	5	17%
Former student who took the test in a different occasion from the one analyzed in the study.	13	43%
General Public indirectly involved with the test on a regular basis (Teachers for example).	4	13%
General Public not involved with the test on a regular basis.	8	27%

In the survey, I asked respondents to rank-order their preference of topics / invalidity hypotheses detected on the first step to be further researched. I included a question as well for them to identify to which group they belong. Responses were anonymous. Annex C presents this survey sent to participants (written in Spanish). We received thirty responses for the survey. We transformed each question asking for priorities in a Likert scale from one to five, and calculated an average rating for each invalidity hypothesis.

Next, I visualized the sketch the validity argument that emerges from the concerns shared by students and the general public on Twitter, by putting together invalidity hypotheses that correspond to the same theme, device and P.A.A.P. question; creating overarching hypotheses that summarize a group of similar invalidity hypotheses. I clustered invalidity hypotheses by considering to what claim, written as an affirmative phrase, could use the research related to invalidity hypotheses as Warrants.

I included a color scale for each overarching hypothesis, proportional to the average of average invalidity hypotheses rankings that belong to the same overarching hypothesis. This enhanced sketch can serve future validity researchers to guide their work, when evaluating the validity of the interpretations of test scores generated the year of the pandemic.

Results

After analyzing the sample of tweets, we found 123 invalidity hypotheses, ranging across different topics and parts of the admission system. Figure 2 presents the means and standard deviation of the 123 invalidity hypotheses ordered from highest to lowest.

Figure 2

Mean and standard deviation of priorities given by respondents to 123 invalidity hypotheses, ordering from highest to lowest

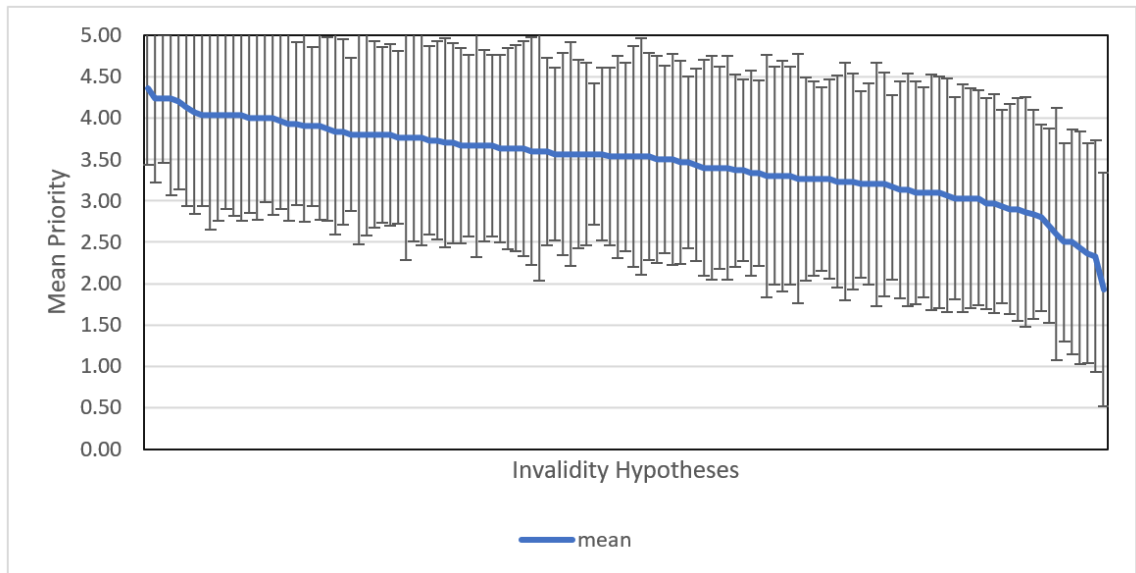


Table 19 presents the top seven invalidity hypotheses ranked as priority. Tables 20 to 25 present the full list of invalidity hypotheses, divided by the device to which each invalidity hypothesis corresponds. For each of the invalidity hypotheses, we present to which P.A.A.P. question each corresponds, and we include the average of priority given by the participants.

Table 19
Top seven rated Invalidity Hypotheses

Invalidity hypothesis	Mean	Std.
Test prep for the math test was not aligned with the actual content asked in the math test	4.37	0.93
Students who presented depression symptoms due to the context of the pandemic saw their capacity to prepare for the test hindered, affecting their performance and the equity of the results	4.23	1.01
Online classes affected the normal representation of high-school GPA	4.23	0.77
Problems with the matching algorithm blocked students to be accepted in their programs	4.23	1.17
Students had to leave their studies because of economical struggles related to the pandemic, limiting their available time to prepare	4.20	1.06
Students could not register to take the test due to problems in the payment system	4.13	1.20
There was one item in the history test that did not have a correct answer	4.07	1.23

Note: These are translations from Spanish.

Table 20**Invalidity hypotheses related to Test preparation Materials**

P.A.A.P.	Invalidity Hypothesis	Mean	Std.
		Rating	Rating
Adequacy	Test prep for the math test were not aligned with the actual content asked in the math test	4.37	0.93
Adequacy	Official test prep materials did not comply to Test specifications	3.97	1.07
Adequacy	One question in the prep materials for the history test had a wrong key	3.90	1.16
Adequacy	Some test prep programs do not cover the content really covered in the tests	3.70	1.26
Adequacy	A practice online version of the test had some problems in certain browsers	3.33	1.24
Appropriateness	Students that presented depression symptoms due to the context of the pandemic saw their capacity to prepare to the test hindered, affecting their performance and the equity of the results	4.23	1.01
Appropriateness	Students from private schools have been dedicating to preparation of the test far longer than students from public schools with less time and resources to pair this.	4.03	1.10
Appropriateness	Classes in person provided an advantage to students vs online classes	3.80	0.92

Note: These are translations from Spanish.

Table 21
Invalidity hypotheses related to Test Battery

P.A.A.P.	Invalidity Hypothesis	Mean Rating	Std. Rating
<i>History Test</i>			
Adequacy	There was one item in the history test that did not have a correct answer	4.07	1.23
Adequacy	Many questions in the History test did not have a correct response according to some examinees	4.00	1.14
Adequacy	One item in the history test related to energy use did not have a legend, confusing students	4.00	1.23
Adequacy	PDT History test did not respected the table of specifications	3.80	1.32
Appropriateness	Due to misinformation related to the history test, students were put aside of the admission process, impacting the overall admission process on programs that require the history test	4.03	1.27
Appropriateness	Some students saw affected their performance in an item that called Pinochet's regime as "military government" instead of dictatorship.	3.10	1.42
<i>Language Test</i>			
Adequacy	Language test does not measure writing skills, having students with good results in the test but poor performance on writing later in their college programs	3.83	1.12
Adequacy	Some texts in the language test have had too much exposure in test prep materials or the test itself	3.10	1.27
Preferability	Students would prefer more "fun" texts when those are long	3.13	1.41

<i>Math Test</i>			
Adequacy	The math test is not capable of recognizing response process accordingly, making the results not robust	3.67	1.09
Adequacy	Students do not see a correspondence between the amount of effort put into study to the math test versus their results in the test	3.17	1.12
<i>Science Test</i>			
Appropriateness	Questions asking about refraction index in the science test had a problem of opportunity to learn	3.57	1.10
<i>Test Battery</i>			
Adequacy	Students are becoming less prepared to college along the years.	3.63	1.22
Adequacy	Critical thinking should be part of the test battery but it is not part of it	3.57	1.36
Appropriateness	Access to technology, correlated with SES, explain good performance on students	3.73	1.20
Appropriateness	There was not enough transparency about the correct responses and its explanations	3.70	1.21
Appropriateness	Students that meet their negative expectations have an impact in future decisions and instances of the test	3.53	1.22
Appropriateness	The test represent to students the end of a life stage and is meaningful for them	3.50	1.28
Appropriateness	Students that have low scores can experience "existential crisis" or other types of psychological distress	3.40	1.35
Appropriateness	Content domain should have been adapted to the special situations of the covid pandemic	3.27	1.11
Appropriateness	Test difficulty went down and this is the real explanation behind reduction of gaps between types of school.	3.03	1.38

Appropriateness	The test is considered as a "dehumanizer" machine	3.03	1.33
Appropriateness	Magnet schools were specially affected by covid conditions in their performances in the test	2.50	1.36
Partiality	Challenge to the administrative standpoint of test creators, and their lack of connection with classrooms	3.77	1.48
Partiality	The test is responding to a neoliberal project in education	3.03	1.30
Partiality	Changes in the test respond to a societal / political decision to leave poor students ignorant by making the test easier for them	2.37	1.33
Partiality	Changes in the test are driven by left policies	1.93	1.41
Preferability	Students do prefer the new item format, that focuses on competences vs content acquisition	3.57	1.14
Preferability	The new test is considered a "patch solution" to the problems in the admission system and with the old test PSU	3.40	1.22
Preferability	Students would have preferred an online test versus one in person	2.87	1.38

Note: These are translations from Spanish.

Table 22**Invalidity hypotheses related to Test procedures and Protocols**

P.A.A.P.	Invalidity Hypothesis	Mean Rating	Std. Rating
Adequacy	Unclear rules related to close contact with covid infected people led students to confusion, affecting their performance or withdrawing from taking the test.	3.53	1.07
Adequacy	Waiting times to enter the rooms for taking the test affected the performance of students	3.37	1.16
Adequacy	Protocols were not enough to contain the number of infected students and covid cases were created because of the in-person nature of the test	3.23	1.28
Adequacy	The number of instances was not enough for students to have enough dates to show their performances	3.10	1.35
Adequacy	Cellphones created circumstances that affected test performance in the room	2.83	1.26
Adequacy	Limitations on what the students can have during the test affect their performance in the test	2.50	1.20
Adequacy	Some students performance were affected by the use of masks	2.33	1.40
Appropriateness	Rural students were more adamant to take the take in person, as their travel to the city meant a higher risk of exposure	3.67	1.18
Appropriateness	Group 1 was in disadvantage compared to Group 2 due to test leaks	3.60	1.38
Appropriateness	Some students saw affected their performance for problems with the assignation or changes in their test locations	3.50	1.14
Appropriateness	Locations in which there was an expectation to protests had students performing less than normal	3.27	1.17

Table 23
Invalidity hypotheses related to Information Delivery Systems

P.A.A.P.	Invalidity Hypothesis	Mean	Std.
		Rating	Rating
Adequacy	Delays in the call-center service led to students not being able to take the test	4.03	1.38
Adequacy	Mismatches between published weights by universities and the official website misled students in their admission process	3.93	1.17
Adequacy	Students living in districts that had covid restrictions were affected in their performances or misled by lack of information	3.90	0.96
Adequacy	Confusion around SES information before pandemic or during pandemic led student to have limited access to funding	3.83	1.23
Adequacy	Problems in the communication channels and help desk made Information not timely given and affected student performance	3.73	1.14
Adequacy	Limitation to access to one score simulator affected the choice of programs in the admission process of students	3.57	1.04
Adequacy	Misunderstandings in the logic of hsgpa calculation misled students in their decision making during high school years	3.57	1.22
Adequacy	Lack of information about an increase in special entry programs led students to under utilize those benefits	3.50	1.25
Adequacy	Some students were not aware of the existence or meaning of puntaje ranking, affecting their admission decisions, and planning	3.30	1.47

Adequacy	An apparent change in the policies for fellowships misled students that now had to pay but had an exception in the past	3.27	1.51
Adequacy	A confusion about how different PDT tests were compared to PSU tests led students to use test prep materials that were inadequate	3.13	1.31
Adequacy	Expectations on the test having a similar situation to the previous year made students be less prepared than usual.	3.03	1.22
Adequacy	The misunderstanding about the transitional nature of the test impacted the performance of the students.	2.93	1.17
Appropriateness	People accepted in the PACE program did not understand their status from the information provided in the website	3.77	1.04
Appropriateness	Students not always understood the rules regarding the use of old scores, distorting their application process	3.40	1.30
Appropriateness	Some examinees are not technology savvy and had problems to find information online	3.27	1.23

Note: These are translations from Spanish.

Table 24**Invalidity hypotheses related to Test Score Interpretations**

P.A.A.P.	Invalidity Hypothesis	Mean Rating	Std. Rating
Adequacy	Stress and lack of preparation due to the covid pandemic affected student performance	3.93	0.98
Adequacy	Anxiety affected student performance in the tests	3.63	1.13
Adequacy	Test scores are not predictive for student performance in college	3.60	1.57
Adequacy	Students that feel they are not prepared have extra pressure when doing the test and then under-performing	3.53	1.14
Adequacy	The threat over the health of students affected their performances in the test	3.47	1.22
Adequacy	The perception of being part of an "experiment" with the transition test led students to do not take the test or affected performance	3.43	1.17
Adequacy	Lack of responses to the problems of examinees led to affected performances due to dialogical effects	3.37	1.10
Adequacy	Students with problems to navigate the uncertainties about the future can see their test performance affected, moderated by anxiety	3.33	1.12
Adequacy	Students with low expectations on their success for college admission saw their performance affected	3.30	1.32
Adequacy	Students that fear about their emotional state after doing the test may see their performance affected	3.20	1.21
Adequacy	Students that used glasses saw impacted their performance due to the use of masks	3.20	1.47

Adequacy	Students with higher expectations on themselves can see their performance affected	3.20	1.35
Adequacy	Students that did not sleep well see their performance affected	3.10	1.40
Adequacy	Some students took the test in a state of apathy	2.90	1.27
Adequacy	Students using strategies with smart random chance still had good chances to enter to college	2.43	1.41
Appropriateness	Spaces of bullying or cyber bullying can impact the self-perception and motivation of some students, affecting their test performance	3.80	1.21
Appropriateness	Negative expectations on a student can create a self-fulfilling prophecy. Job expectations can moderate this.	3.57	0.86
Appropriateness	Breakfast taken the morning of the test affect student performance	3.20	1.13
Appropriateness	Personal connections with the exact date doing the test can affect student performance	2.80	1.13
Preferability	A score calculated based on a normative logic blocks students to know how many responses they need to have to achieve certain score, some students would prefer to know this in advance.	3.47	1.04
Preferability	Dichotomous scoring in the test are less preferable to partial scoring	2.97	1.27

Note: These are translations from Spanish.

Table 25**Invalidity hypotheses related to Admission System Rules and Procedures**

P.A.A.P.	Invalidity Hypothesis	Mean	Std.
		Rating	Rating
Adequacy	Online classes affected the normal representation of high-school GPA	4.23	0.77
Adequacy	Problems with the matching algorithm blocked students to be accepted in their programs	4.23	1.17
Adequacy	Students could not register to take the test due to problems in the payment system	4.13	1.20
Adequacy	Password recovery is not supported for some browsers and blocked students in their application process	4.03	1.13
Adequacy	Students underestimate the value of true vocation in their admission decisions, leading to poor career choices and future drop outs	4.00	1.02
Adequacy	The amount of alternative entry gates for college admission was not enough for the needs specific of the covid pandemic that year	3.80	1.13
Adequacy	Schools use test pre materials to put grades, which can make high school gpa and test results spuriously correlated	3.63	1.25
Adequacy	Measures taken to limit the impact of problems with the test in the admission system were insufficient	3.57	1.04
Adequacy	The change of name affected student's preparation or performance in the test	2.60	1.52
Appropriateness	Students had to leave their studies because of economical struggles related to the pandemic, limiting their available time to prepare	4.20	1.06

Appropriateness	Problems with student ids blocked students to take the test	4.03	1.22
Appropriateness	Some students did not receive their NEM scores timely, affecting their admission process	4.03	1.27
Appropriateness	The overall impact of the pandemic highly affected the equity of the admission that year	4.00	1.17
Appropriateness	Information about financial situation was not always correctly input by students and affected admission decisions	3.90	1.12
Appropriateness	Problems with information about funding affected admission decisions	3.87	1.11
Appropriateness	Life events during high school years have an impact in high school GPA, which has a permanent effect on students that hinder their chances in the future.	3.80	1.06
Appropriateness	The admission system did not adapt to the circumstances of the pandemic in their admission process that year	3.80	1.10
Appropriateness	Foreigner students had problems to enter in the process due to their lack of Chilean ids	3.77	1.25
Appropriateness	Students had too many things in their hands during their last year. That, combined with the covid context, hindered their chances to prepare to the test	3.77	1.30
Appropriateness	It was not appropriate to change the conversion tables of hsgpa and ranking that year	3.67	1.35
Appropriateness	Students struggling with pandemic related situations had problems to realize the limit dates for registration to the test	3.67	1.15
Appropriateness	Teacher preparation to guide students is a moderator for student performance and successful application	3.67	1.09
Appropriateness	Students from lgbt communities saw their performances affected by the backlash in social media against the use of social names	3.53	1.33

Appropriateness	The fee to take the test creates a disadvantage for lower SES Students from older cohorts that do not have any benefit to waive the payment	3.53	1.43
Appropriateness	The reduction of the weight in the test scores affected students that had low high school gpa and had an expectation about compensating with the test scores	3.53	1.25
Appropriateness	Students opted-out and refused to take the test because the protocols and conditions were not safe enough in the eyes of students and their families	3.40	1.35
Appropriateness	Students opted out after they registered, not receiving a refund	3.30	1.39
Appropriateness	The inclusion of hsgpa in its last year created inequity in the admission process	3.30	1.32
Appropriateness	Students that manifested themselves through violence should have been made responsible	3.23	1.43
Appropriateness	It was not appropriate to do the test during the pandemic	3.07	1.41
Appropriateness	Students that required to do in person paperwork saw themselves in a risky situation in terms of their health	2.97	1.33
Appropriateness	dates defined for registration were not the best ones for students, that usually have money at the beginning of the month	2.90	1.35
Appropriateness	The date chosen to take the test conflicted students	2.70	1.18
Partiality	Reflection about the role of older higher education institutions in the construction of the country through their work in the admission system and test administration	3.63	1.30
Partiality	Unclear roles in the admission process between demre and government confused students and affected their sense- making	3.23	1.30

Preferability	Some examinees complained about the expectation generated by the announcement of 30% PDT as minimum, understood as maximum.	3.60	1.13
Preferability	Some students are not ok with the speed of the changes in the admission system	3.27	1.20

Note: These are translations from Spanish.

For each of the invalidity hypothesis, I associated a positive overarching claim that would be the claim in a validity argument that can take any research about the invalidity hypothesis as warrant or rebuttal. This argument based on the overarching hypothesis, would be one that, in theory, can represent and give response to the multiple concerns from the public. Table 26 present the list of overarching hypotheses, with the aggregated priority considering the priorities of the invalidity hypotheses within. Annex D presents the detailed list connecting overarching claims with specific invalidity hypotheses.

Table 26
Results by Overarching Hypotheses

Device	P.A.A.P.	Overarching Hypothesis	Agg. Prior ity	Std. Dev. of means
Admission system	Adequacy	The rules and procedures and technology in the Admission system are Adequate for the selection of Students for Higher Education	3.80	0.51
	Appropriateness	The Admission System abides to the values of the communities affected by the uses of the test	3.23	--
	Appropriateness	The Admission System abides to the law	3.30	--
	Appropriateness	The problems arising from the pandemic were resolved successfully	3.43	0.42
	Appropriateness	The Admission System propends an Equitable Access to Higher Education	3.69	0.41
	Appropriateness	The Admission System propends to a Fair Access to Higher Education	3.79	0.25
	Partiality	About the roles of institutions in the test development and admission process	3.43	0.28
	Preferability	The structure of the Admission System is preferable to other alternatives	3.43	0.24
History Test	Adequacy	The History test was adequate for its purposes	3.97	0.12
	Appropriateness	The History test was appropriate for its purposes	3.57	0.66
Information	Adequacy	Information regarding the tests and the admission system is accessible for all students	3.48	0.26
	Adequacy	Information regarding the tests and the admission system was correctly and timely delivered	3.52	0.36

Language Test	Adequacy	The Language test was adequate for its purposes	3.47	0.52
	Preferability	The Language test was preferable in comparison to other ways to test	3.13	--
Math Test	Adequacy	The Math test was adequate for its purposes	3.42	0.35
Science Test	Adequacy	The Science test was adequate for its purposes	3.57	--
Test Battery	Adequacy	The test battery covers the domains relevant for college admission decisions	3.60	0.05
	Appropriateness	The test battery promotes a Fair Access to higher education	3.10	0.85
	Appropriateness	The test battery adapted to the special needs generated by the pandemic	3.27	--
	Appropriateness	The test battery promotes an Equitable Access to higher education	3.32	0.37
	Appropriateness	The test battery promotes a positive assessment experience	3.37	0.23
	Partiality	About the political influences on the decisions related to the test	2.44	0.55
	Partiality	About the role of test developers and their connection to the classroom	3.77	--
	Preferability	The in person format of the test is preferable over online formats	2.87	--
	Preferability	The test battery is preferable over the old test PSU	3.48	0.12
Test prep	Adequacy	Official test prep materials were adequate so student could prepare to the test	3.85	0.38
	Appropriateness	All students had a fair chance to prepare to the test	3.80	--

Test procedures	Appropriateness	Official test prep materials promoted a Fair Access to Higher Education	4.13	0.14
	Adequacy	Protocols and procedures promoted the best possible performance of students to take the test	2.90	0.44
	Adequacy	Protocols and procedures were adequate to navigate the challenges the COVID-19 pandemic created	3.05	0.51
Test Scores Interpretation	Appropriateness	Protocols and procedures promoted a fair access to higher education	3.51	0.18
	Adequacy	Test score interpretations are meaningful and substantive, over and beyond what random response strategies can provide to students in the test	3.02	0.82
	Adequacy	Test score interpretations can untangle engagement, self-efficacy and other sources of Construct-irrelevant variance related to Student's psychological states	3.32	0.21
	Adequacy	Test score interpretations were still adequate under the specific circumstances of the COVID-19 pandemic	3.53	0.37
	Appropriateness	Test score interpretations promote a Fair access to Higher Education	3.23	0.51
	Appropriateness	Test score interpretations promote an Equitable access to Higher Education	3.57	--
	Preferability	Test score calculation and interpretation are preferable versus other alternative options	3.22	0.35

Figure 3 has a scheme with all the overarching hypotheses, and their relationships between each other. As you can see, the data coming from Twitter shows clearly that the validity of interpretations and uses depends on an intricate ecosystem of devices, for which the public has many concerns. Figure 4 provides a heat map for the most rated hypothesis, showing that test preparation materials should have a big part in test validation work from the point of view of the public.

Figure 3

Scheme of Validity Argument with Overarching Hypotheses that can Respond to the Concerns of Users on Twitter that year.

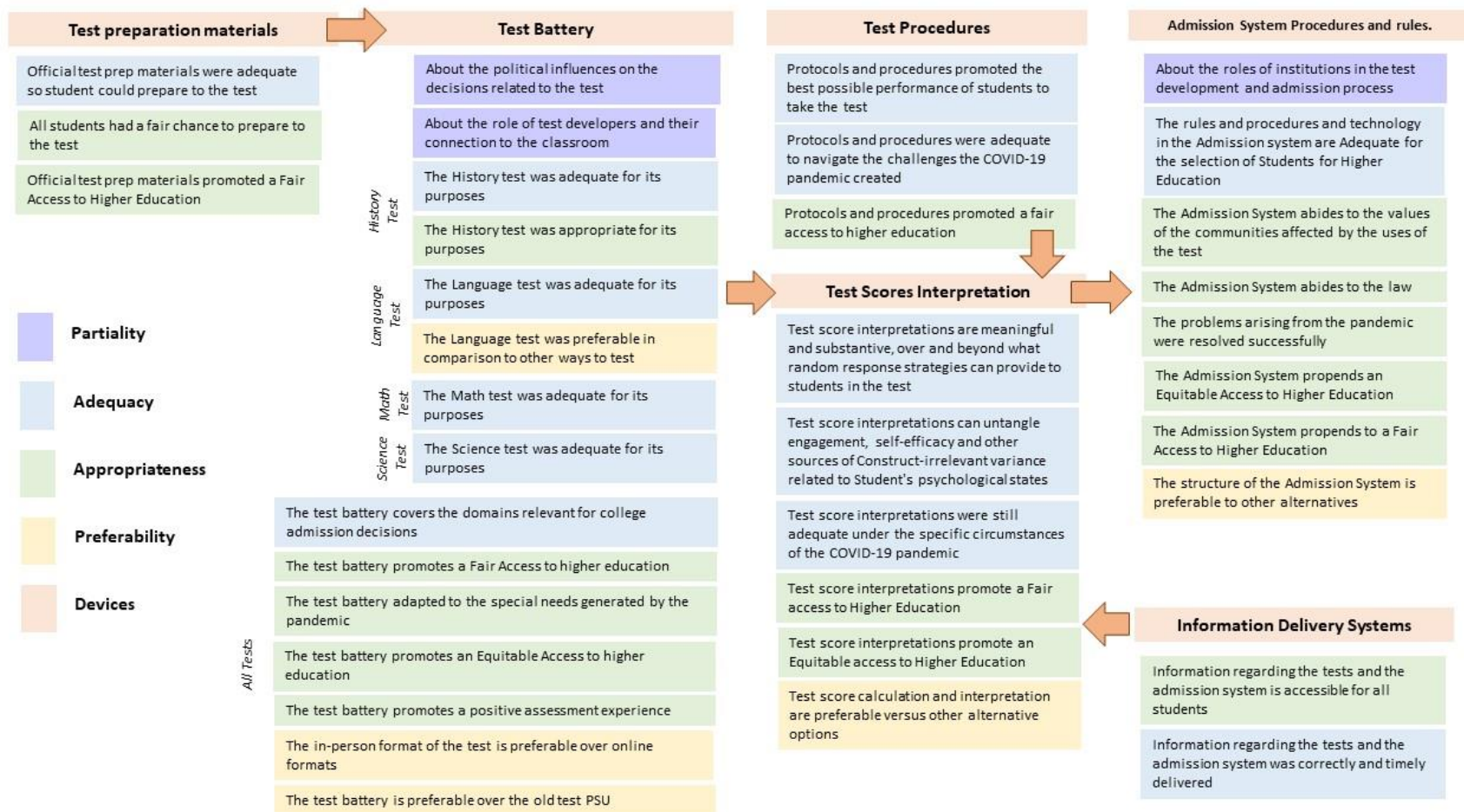
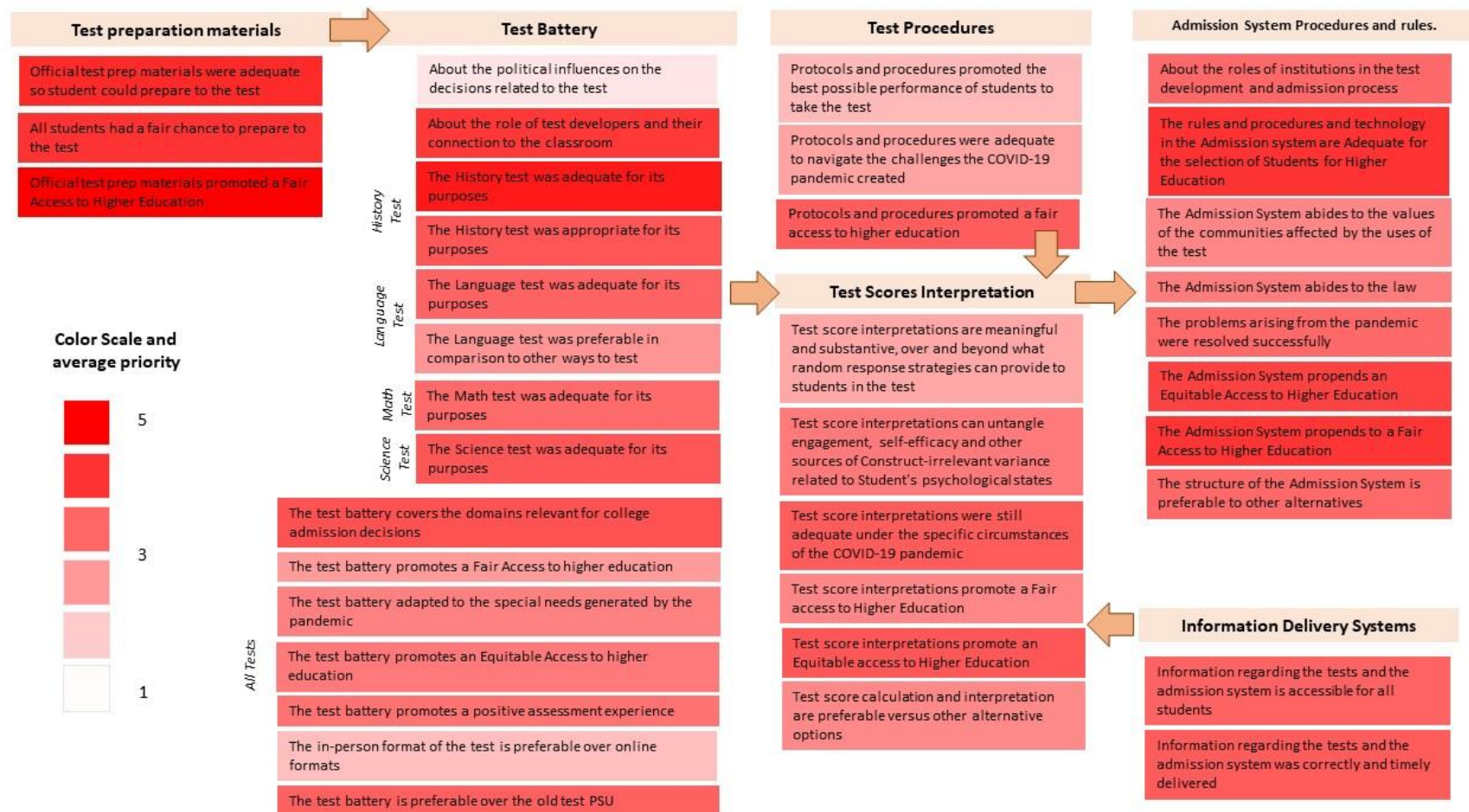


Figure 4

Heatmap on the Scheme of Validity Argument with the Priorities of the Public for that year's score interpretations and uses.



Conclusion

Both empirical studies have shown very interesting insights about the feasibility of the Experiential Approach. On one side, for a test design approach based on experiences, we could show that by enriching an assessment experience with entry surveys we can adapt the test on the fly to match the group of student's experiences and foster a more positive experience. The insights coming from the feedback and rating in different parts of the assessment experience showed in practice how the knowledge about the assessment experiences elicited by a test can help to interpret in a better way the scores generated by a test. Also, we could find a simplified version of the Experiential approach under low budget.

On the other side, for the validation part of it, we could see how different it can look an argument based on the experiences and opinions of the general public versus the type of argument that would arise from an expert. Considerations about Appropriateness, Partiality and Preferability, beyond Adequacy and the more conservative notion of Validity were necessary to catalogue the variety of comments related to the test, even when the object of criticisms was not directly the test.

In fact, this study corroborates a vision of assessment technologies as ecosystems. It is difficult to say anything about a test if we cannot show evidence as well about procedures, test materials, information systems, and the final use of them in the admission system. Either as a pre-requisite, or because the test will need to change to adapt to the pre-requisites of other devices, a comprehensive validation argument emerges from the experiences of users.

It is interesting to see what was more important for the public. Research about test preparation materials was ranked higher than other topics. The two top overarching hypotheses were (1) Official test prep materials promoted a Fair Access to Higher Education, and (2) The History test was adequate for its purposes. The situation with the History Test indeed requires much more detail on that year, given that a lot of students complained on Twitter about not knowing they needed to take the test (the previous year, because of some protests, it was waived), plus other complains that really make yourself as an expert to question the quality of that test. The public as well saw this.

Both Empirical studies show then that a simplified version of the experiential approach is feasible, and that the general expectation for having valuable insights coming from the re-centering on user experience was achieved.

CHAPTER 6: DISCUSSION

Discussion

The Experiential approach to test design and validation has many promises, however it was not clear before doing the empirical work proposed in this Dissertation if those promises were going to be satisfied.

With respect to the first conceptual hypothesis of this Dissertation, (a) The experiential approach will provide valuable information to improve test design and validation practices above and beyond information provided by traditional methods; we can say that both empirical studies showed that user's input was incredibly insightful. In fact, users may be underestimated in general about their capacity to understand issues related to the validity of the interpretations and uses of a test.

The first empirical study showed that it is possible to ask for student's feedback and adapt a test to the specific context and people taking the test, and their feedback was that "Valuable information" we imagined users could bring. This is beyond traditional methods, that usually do not consider having user's input in any stage of test development. The use of a mock item is also an innovation facilitated by the experiential approach, that could be conceived because of the shift in how we conceive a test (as a device that is helpful to elicit experiences).

The second empirical study showed interesting and valuable insights about score interpretation of the test PDT done during the COVID-19 pandemic. The "Valuable information" came from the detailed and situated account of what happened in that instance of the test. Some problems may have been guessed by an expert, however many of the hypotheses raised were somehow unexpected and very specific. In that regard, the

experiential approach did show the capacity to break the bias in a validity researcher that usually sketches a validity argument from the research that has already been made.

About the second hypothesis (b) By implementing aspects of the experiential approach, we can evaluate its feasibility and practicality; the first empirical study showed that it is feasible to have a simplified version of the experiential approach; however a co-construction with examinees directly may require some smartness to speak about the test without giving it in advance.

One aspect that was indeed more complicated to implement was to design a validity argument around the experience elicited. In the context of the diagnostic test of the first empirical study, this would have implied including an extra section explaining why the diagnostic test was valid. In the context of the second empirical study, we would have needed an argument with a lot of research already done, to then see how the public react to the argument with evidence on it. Even though in both studies we tried to target and check on the experience elicited by the test, we could not check on the experience elicited by the argument. This is, in fact, the last step of the experiential approach and that part was not tested because of the practical complications and extra complexity that would have entailed. This is indeed a limitation of the Experiential approach, in that writing a validity argument with warrants and rebuttals included is already a difficult task, to check on the experience elicited makes it even more cumbersome.

In the second empirical study, it is important as well to mention that the volume of information was superior to the capacity of realistically review each tweet. This forced me to sample from the total of tweets found. The use of natural language processing (NLP) may be a way to facilitate the work of analyzing such an amount of data, however

the training of those methods still will require humans in the loop, and the use of pre-existing libraries may fail to capture the type of information that is truly relevant for a test. The fact that we could get so many insights from public information makes the experiential approach very practical. In the cases of test with less engagement in social media, quality service surveys can compensate and provide the type of information needed to have at least one experiential study.

Also, about the survey asking for priorities, the amount of invalidity hypothesis and the specific knowledge needed to even understand those hypotheses posed many problems. The relative low number of respondents ($N = 30$) and the composition of the people that responded (more representative of the public than the specific cohort that took the test on January 2021) shows at least two limitations when consulting the communities for feedback.

Overall, the Experiential Approach showed to be a novel approach to test design, development and validation that can help practitioners to overcome some of the limitations in classical approaches to test validation and the concepts of validity implied by them. The proof-of-concepts served as “baby-steps” for the Experiential approach, helpful to identify its most immediate weaknesses to be improved.

To more fully critique the experiential approaches, I include two sections for that are important for understanding the overall results of this Dissertation. One section will mention some failed studies, that give information of the practical limitations of the Experiential approach. Another section will explore some questions posed during the writing process of this document, that are worth mentioning here as the questions and answers are interesting.

Failed studies for this Dissertation

One study was related to a potential collaboration with a Student organization called “Student Bridges.” This organization helps students from underrepresented groups in the Amherst Area to navigate the complexities of college admission decisions. I started some conversations with them to do a proof-of-concept of the design part of the experiential approach with them. Unfortunately, the extension of that project was far longer than what I realistically could do for this Dissertation and the deadlines I originally have planned for this. In some initial conversations with people in that organization, they considered useful to create a test that could help students to understand better their vocations. To create such a test in the format of a game or gamified test was considered interesting as well for Student Bridges. However, my initial estimation was that to build such a test would take at least a year (maybe two), and unfortunately I did not have the time at the moment I had to take a decision.

Another failed study was planned together with Professor Craig Wells with his class on Statistics (Stats 2 class at UMass). We started a similar process to the one done with Professor Lisa Keller in the first empirical study, however the scope of the project was bigger. Professor Wells wanted to have continuous evaluations through the semester for his class. As an attempt to create engaging assessment experiences, I created a series of surveys and videos asking students to detect the problems in some mock of an incorrect response process solving problems similar to the ones Professor Wells was asking to his students.

We had two instances of those diagnostic tests, however the students did not engage in them. Only one student responded the first test; no one responded to the

second. This is an important insight about the experiential approach. Sometimes we overestimate the value of our innovations. This is a hard limitation on the experiential approach, in that users may show an initial positive opinion about an idea, but low engagement later due to a limited time budget later. At least, the one student that engaged in the test mentioned the tests were helpful.

A third study was related to the experiences of students taking the Massachusetts Adult Proficiency Test (MAPT). We made a call to students to participate in focus groups about the test, having a low rate of response to the call. Only five persons showed interest in participating, far from the original twenty we were considering. Once again, the experiential approach requires motivated users. The call included money compensation, but not even that was effective to motivate students to participate.

Questions from the committee during the writing process

Many questions were posed by the committee during the writing process of this document. One provocative question raised by one of the members was the following: Should tests be fun? All tests? Whose definition of fun? Given the focus on User Experience and the origins of the Experiential Approach in Game design, the Experiential approach can be interpreted to be either helpful only in the context of game-like tests, or that the tests created using this approach will be fun.

This led to a clarification, in that the Experiential Approach is not necessarily trying to elicit a fun experience all the time. What it may be more general and characteristic of the approach is that the tests created through the experiential approach should elicit **positive experiences**. That can or cannot be “Fun.” The methods to achieve

a positive experience are like the ones used in Game Design, but the methods are not meant to elicit Fun only. Now, what is a positive experience in the context of testing? What entails in the cultural, spatial, temporal contingency we live? This is an exciting question that may require further research in the future, to understand what configures a positive assessment experience from the perspectives of positive psychology.

Another interesting question raised was about what part of the classic validity argument do I see in the Experiential approach framework changing or impacted in particular? Content analysis/evaluation? Test design choices? Administration conditions?

First, the mode of argumentation provides a huge impact in how validity arguments are written. Chapelle for example, in the context of TOEFL and the writing of a validity argument there, she complained that an Interpretation and Use Argument (IUA) usually is defined from the research that has already been done (Chapelle et al., 2010). The Experiential Approach provides a rationale for question generation, making that less dependent to the pre-conceptions of a Validation Researcher. That is a change in how Validation has been done so far.

The second change is that the evidence gathered later in the process is always contingent to a problem reported in user experience reports. That makes the type of evidence gathered much more unpredictable, and “Situated,” in the sense of adapted to the specific needs that a user will face when using a test in specific contexts of use. So, for each source of evidence, the approach gives a little twist. Content evidence, for example, will start to include challenges about the content that defines a test coming from the communities. Response process will require to include better detail on how past experiences influence cognitive process. Internal Structure is only demanded when there

are traces in the experiences, always providing a narrative to interpret the statistics in advance. Relationship with other variables, will be driven by what the communities consider as valid criteria or other sort of variables. Consequences, will change to general ethical questionings, including consequences when the communities have that a consequentialist moral philosophy. In summary, the use of an experiential framework as a theoretical lens to re-interpret what Test Validity is, and How Test Validation is done, gives a little twist on everything we do in Test Development and Validation.

The following list summarizes the lessons learned throughout the writing process of this Dissertation about the Experiential Approach:

- The closer you get to the test users (the people that take the test), the more likely you can expose them; creating a need for greater controls on the research to be done and creating further limitations in the work to be done with them.
- The Experiential approach requires having an initial stage of understanding of users, ideally in a participatory way, before even start defining what the test is.
- The Experiential approach requires the attention of users, hence the topics to be asked or the innovations to be tested must be something of their interest in order to have them in the loop.
- Tests are embedded in an intricate network of other devices, that all together confirm an ecosystem that can be called an “assessment technology.” Test Validation from an Experiential point of view requires to analyze many devices, as the experience lived by examinees does not distinguish between those as something different from the test.

- The volume of information that Users can generate is enormous. It requires more innovative work on how to handle that amount of data.
- The Experiential approach gives an extremely detailed view of what may be wrong with a test. The public may also speak about what is right, but at least in social media people may tend to share what is negative and more consequential to them. This can become a problem if the political context is not receptive to those criticisms.
- To be proactive and look for the inputs from the communities may require explaining technical concepts or introducing participants to vocabulary related to the test they may ignore. This poses a limit to how much we can rely on such input, as users may not be able to even understand the problems that other users report.

One last limitation of the Experiential Approach is that, by asking users, the approach can gain a tone that can be considered confrontational. This is particularly salient when we listen to what users on Twitter have to say. In the same way experts may have a “confirmationist” bias when validating a test, the public may have a “refutationist” bias and denounce problems that may not be critical or even real. This confrontational tone requires a society in which freedom of speech is extensive, and dissensus is not punished. This is a characteristic of American Society where this approach could see its first glimpse; however, it is important to acknowledge how situated is this characteristic, not present in many countries around the world.

In summary, the Experiential approach can be defined in the end as the re-centering of test design, development and validation around the experiences elicited by

devices on users. Our review of Feminist epistemologies and the work done in our empirical studies shows *that User Experience Research is central to Construct Validity*. The Experiential Approach extends the work of Kane and the argument-based approach, bringing the ideas of Rhetorical Argumentation (Tindale, 2004) into the table. A new view about the appropriateness of the test, as a correspondence with the values of the communities affected by the uses of the test, offers an interesting solution to the separation between Fairness and Validity. The inclusion of the Preferability question allowed us to include new criteria for test design and validation that can help us to create delightful and positive assessment experiences successfully.

In terms of future research related to the Experiential approach, there are many open paths from the work presented here. One natural extension of the work can be to create a Game-based Assessment using the Experiential approach. Another future path of work is related to the conceptual framework for Assessment Experience. There is a need to validate it in a more empirical level, check for example if there is agreement in some human observers about the types of interactions found etc.

One very interesting piece of research that arises from the experiential approach is to confirm the hypothesized dialogical effect of a validity argument in construct validity. How does it change student performance if a test-taker is informed of the validity of the interpretations and uses? Does it make a difference if we explain it in a better way?

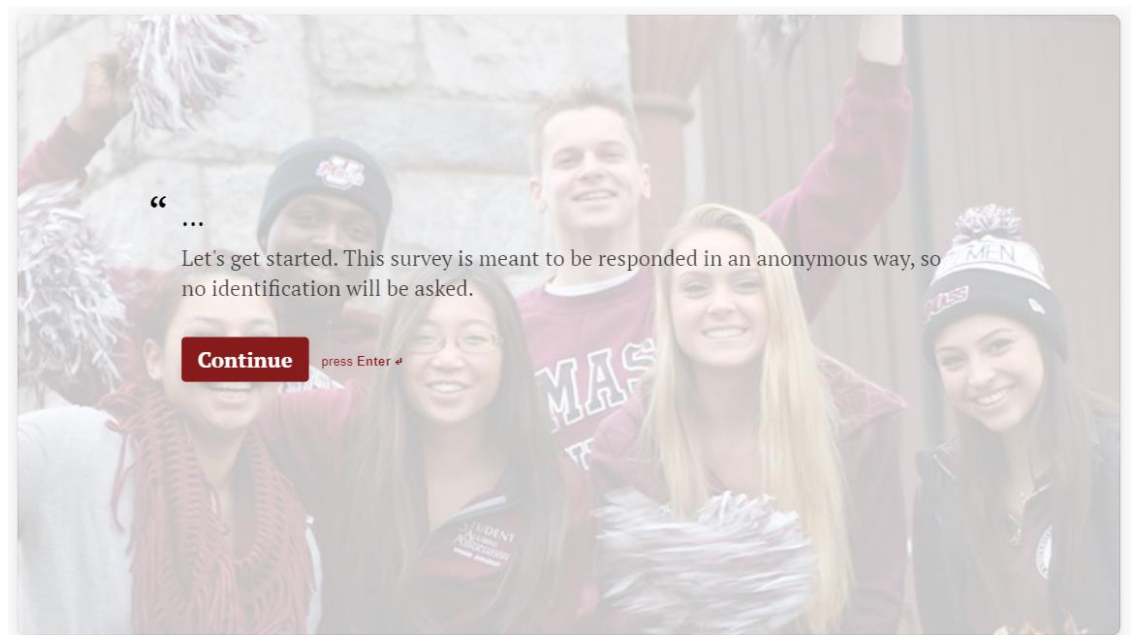
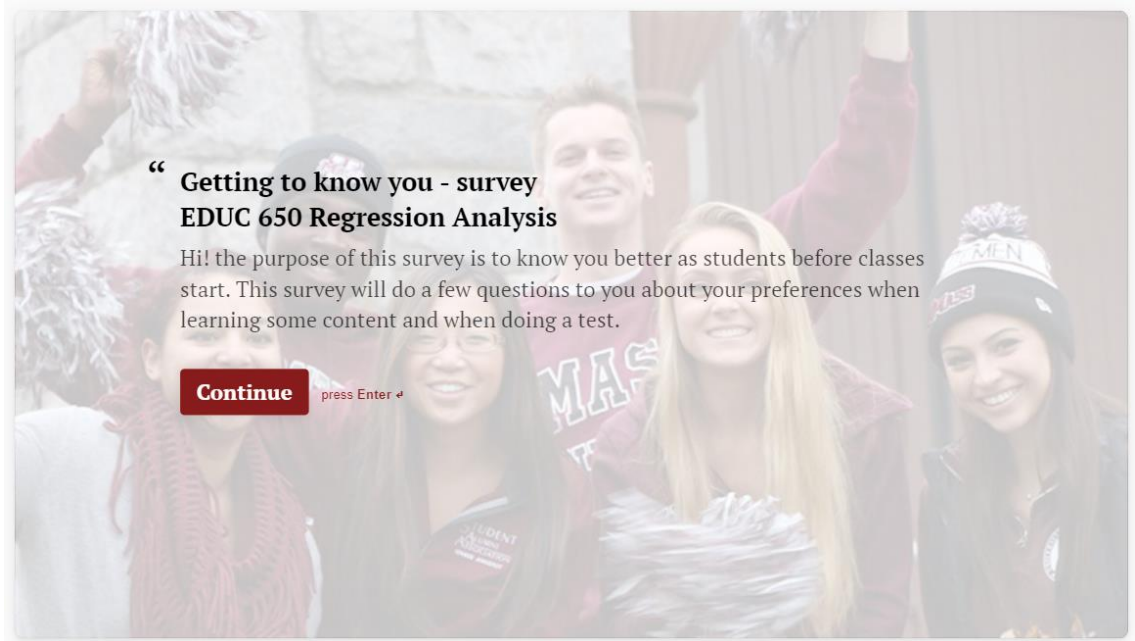
Another interesting path is in the connections of the Experiential approach and Critical Race Theory. The experiential approach is more passive on detecting structures of oppression, in that it waits for the experiences to report those to then start doing some research about them. What other alternatives to the experiential approach exist that can

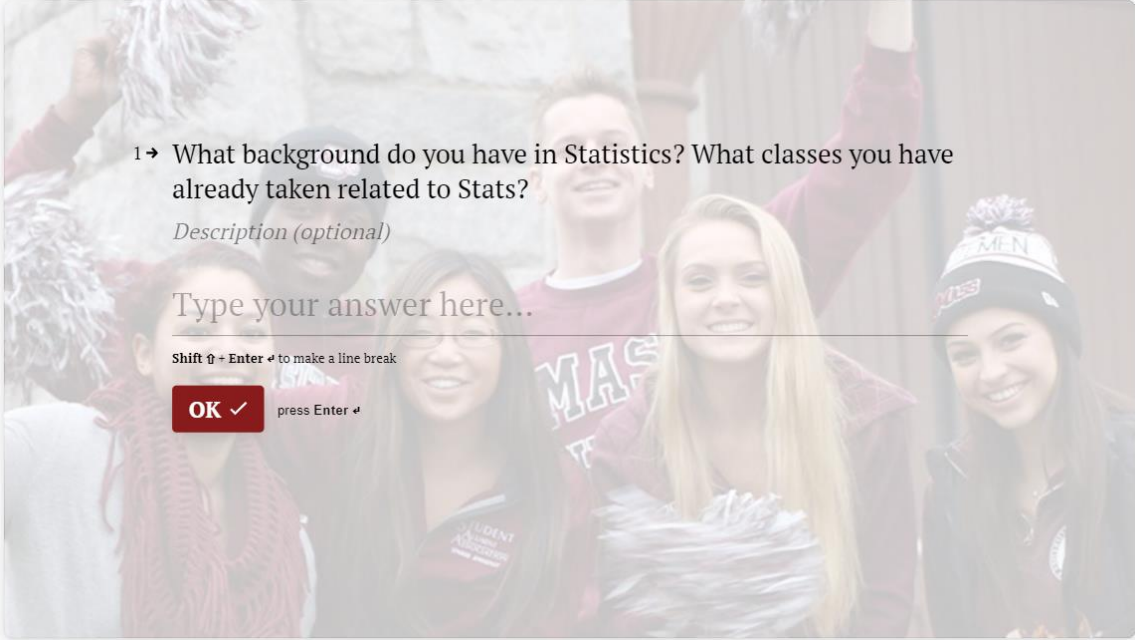
match better with Critical Race Theory? I imagine that, in the future, we may end up speaking about experiential approaches in plural, as there may be one for each context.

I hope this Dissertation can open a conversation and encourage practitioners to include more voices and Standpoints in their work. I hope this can lead us to a better and more enjoyable practice in Educational Testing, that is empathetic with test users and their communities. The future of our field will depend on this.

A positionality statement is included in Annex E.

ANNEX A: INTRODUCTORY SURVEY





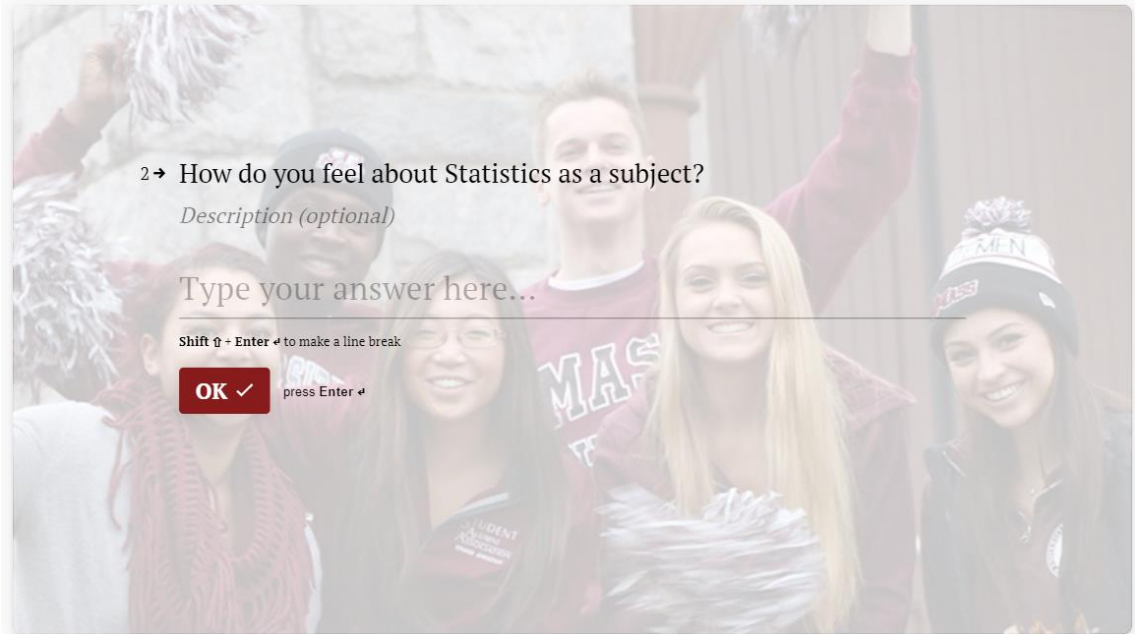
1 → What background do you have in Statistics? What classes you have already taken related to Stats?

Description (optional)

Type your answer here...

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OK ✓ press Enter ↵



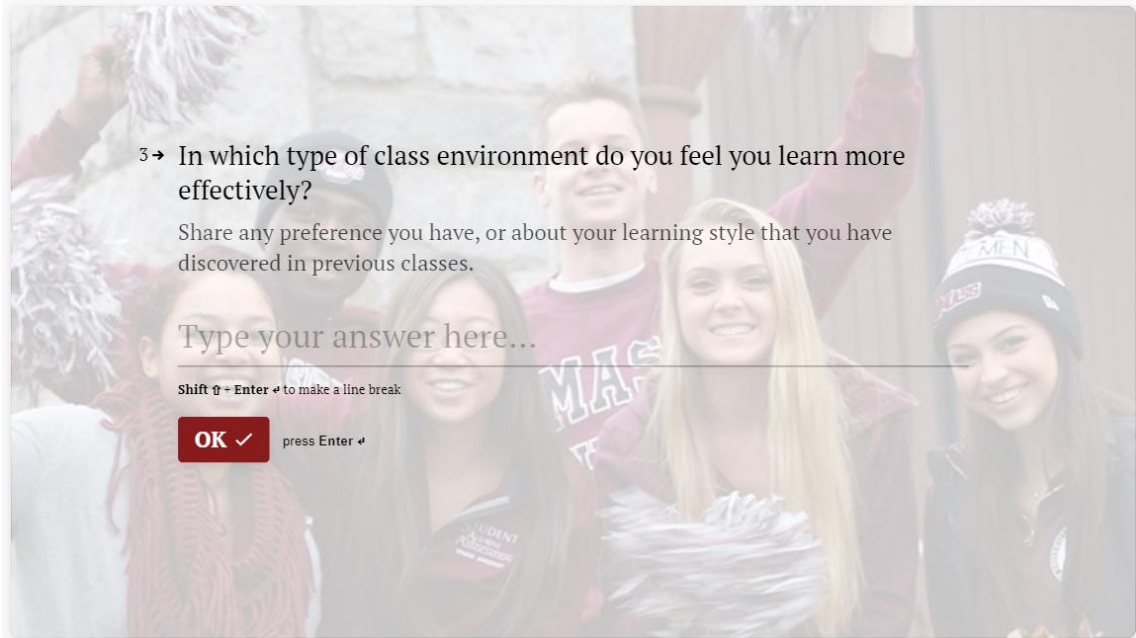
2 → How do you feel about Statistics as a subject?

Description (optional)

Type your answer here...

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OK ✓ press Enter ↵



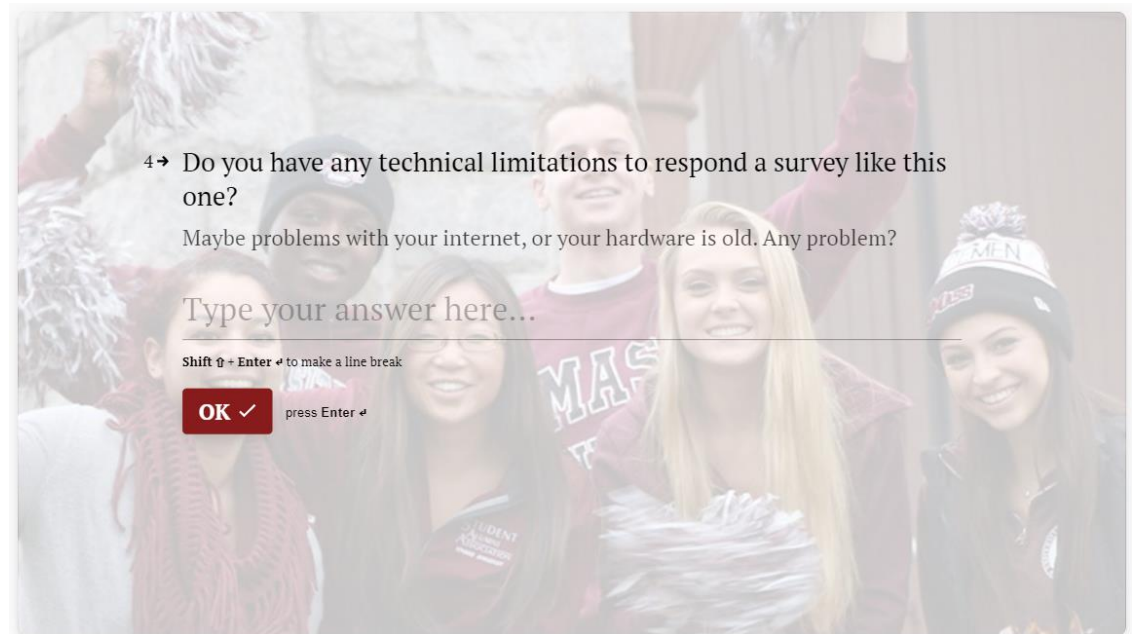
3→ In which type of class environment do you feel you learn more effectively?

Share any preference you have, or about your learning style that you have discovered in previous classes.

Type your answer here...

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OK ✓ press Enter ↵



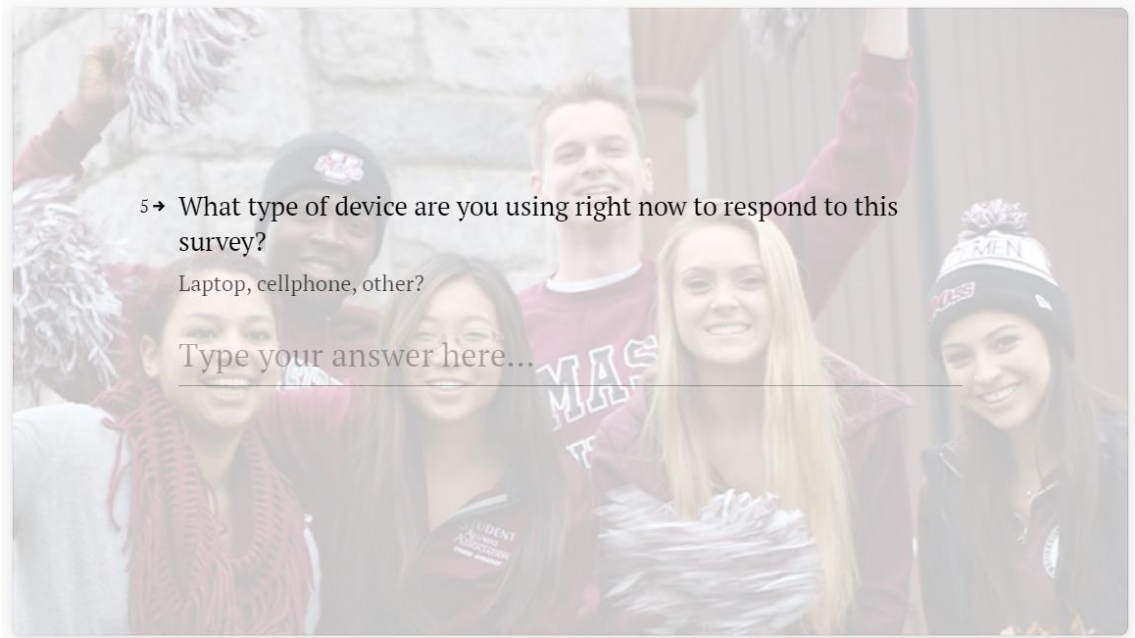
4→ Do you have any technical limitations to respond a survey like this one?

Maybe problems with your internet, or your hardware is old. Any problem?

Type your answer here...

Shift ⌘ + Enter ↵ to make a line break

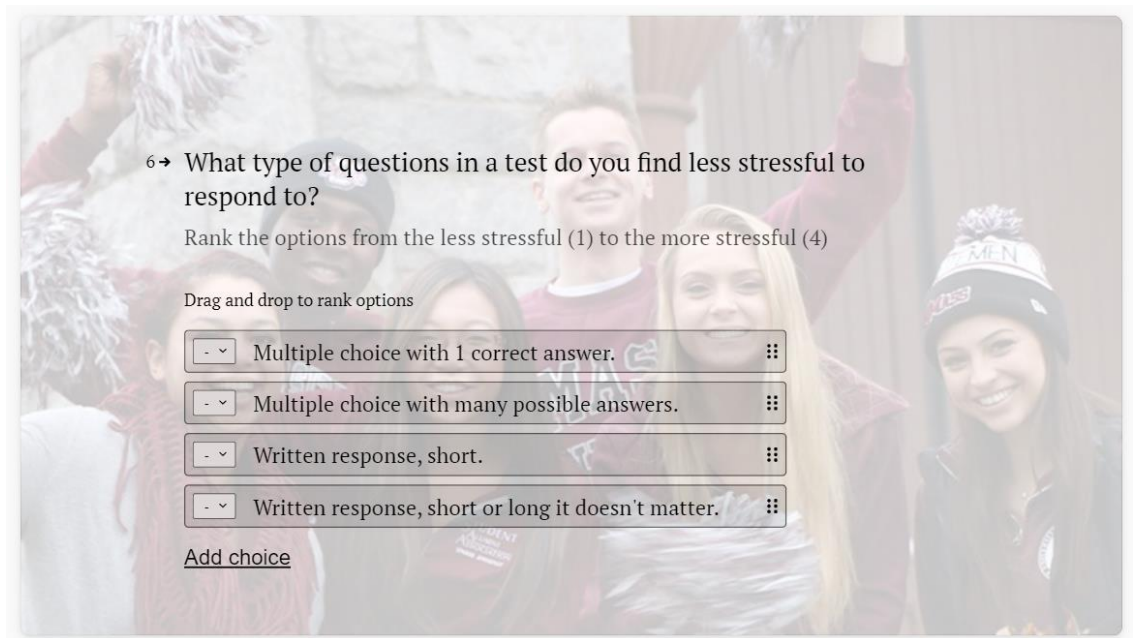
OK ✓ press Enter ↵



5→ What type of device are you using right now to respond to this survey?

Laptop, cellphone, other?

Type your answer here...



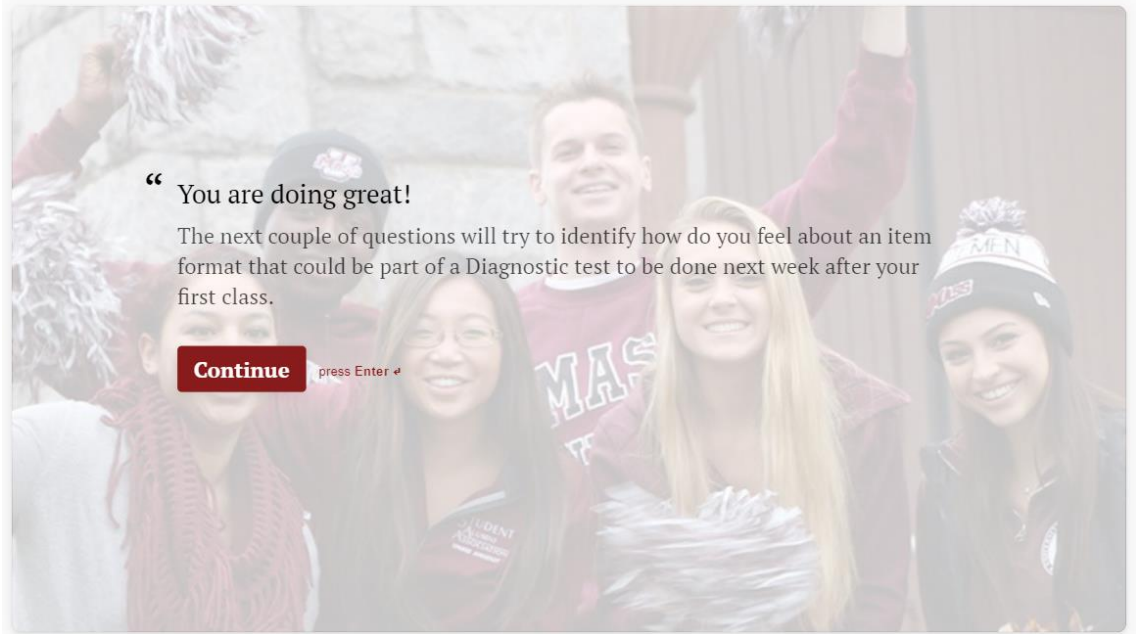
6→ What type of questions in a test do you find less stressful to respond to?

Rank the options from the less stressful (1) to the more stressful (4)

Drag and drop to rank options

-	Multiple choice with 1 correct answer.	1
-	Multiple choice with many possible answers.	2
-	Written response, short.	3
-	Written response, short or long it doesn't matter.	4

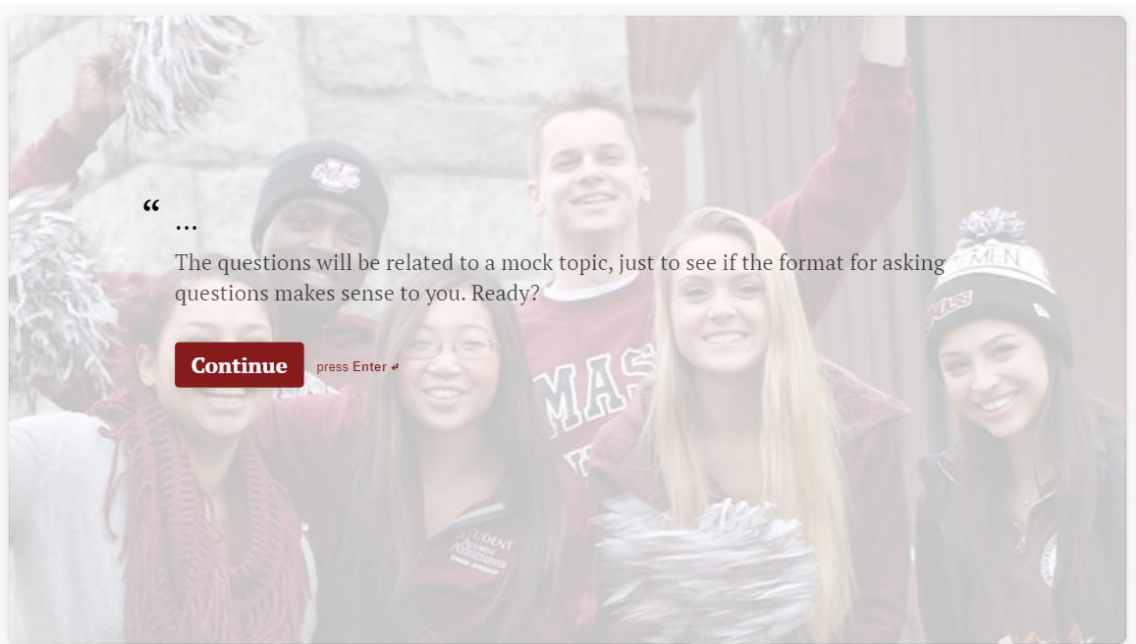
[Add choice](#)



“ You are doing great!

The next couple of questions will try to identify how do you feel about an item format that could be part of a Diagnostic test to be done next week after your first class.

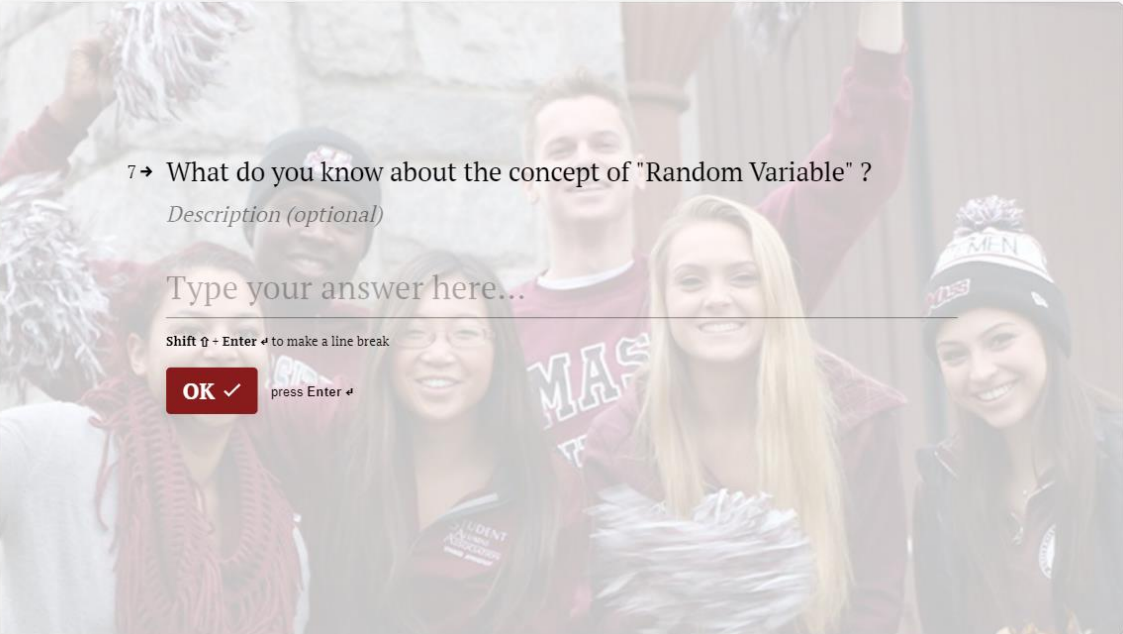
Continue press Enter ↵



“ ...

The questions will be related to a mock topic, just to see if the format for asking questions makes sense to you. Ready?

Continue press Enter ↵



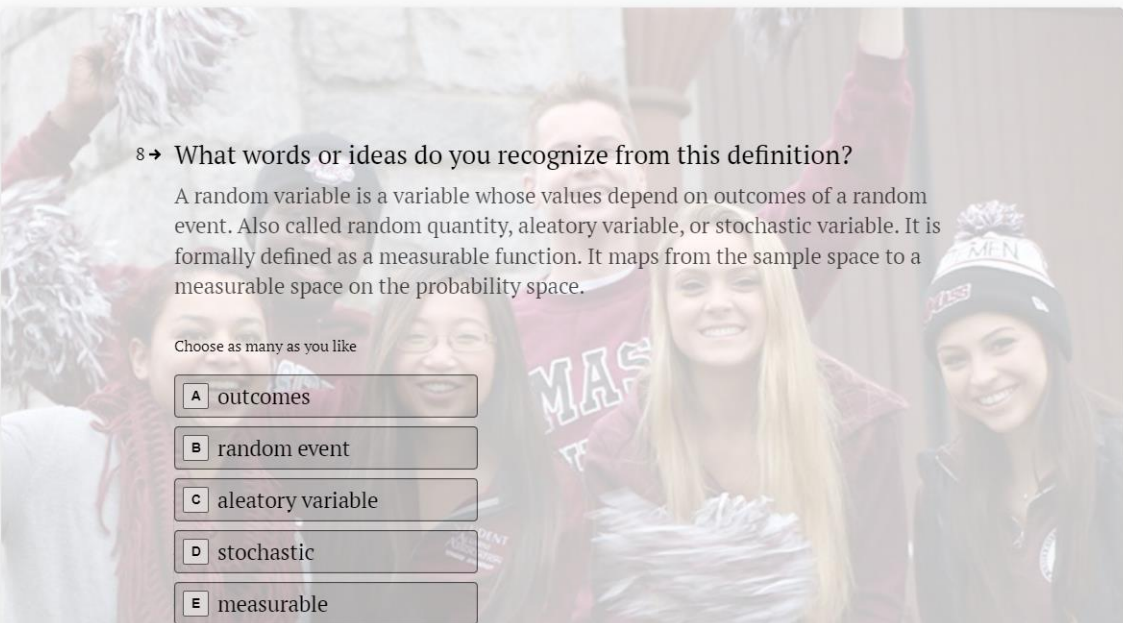
7 → What do you know about the concept of "Random Variable" ?

Description (optional)

Type your answer here...

Shift ⌘ + Enter ↵ to make a line break

OK ✓ press Enter ↵

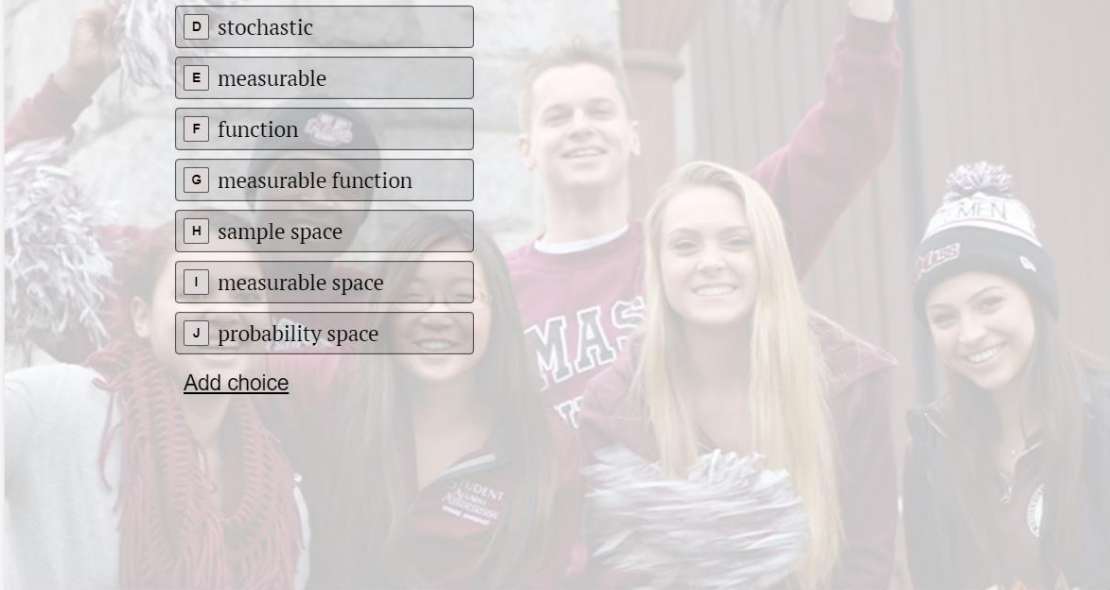


8 → What words or ideas do you recognize from this definition?

A random variable is a variable whose values depend on outcomes of a random event. Also called random quantity, aleatory variable, or stochastic variable. It is formally defined as a measurable function. It maps from the sample space to a measurable space on the probability space.

Choose as many as you like

- ☐ A outcomes
- ☐ B random event
- ☐ C aleatory variable
- ☐ D stochastic
- ☐ E measurable



☐ D stochastic

☐ E measurable

☐ F function

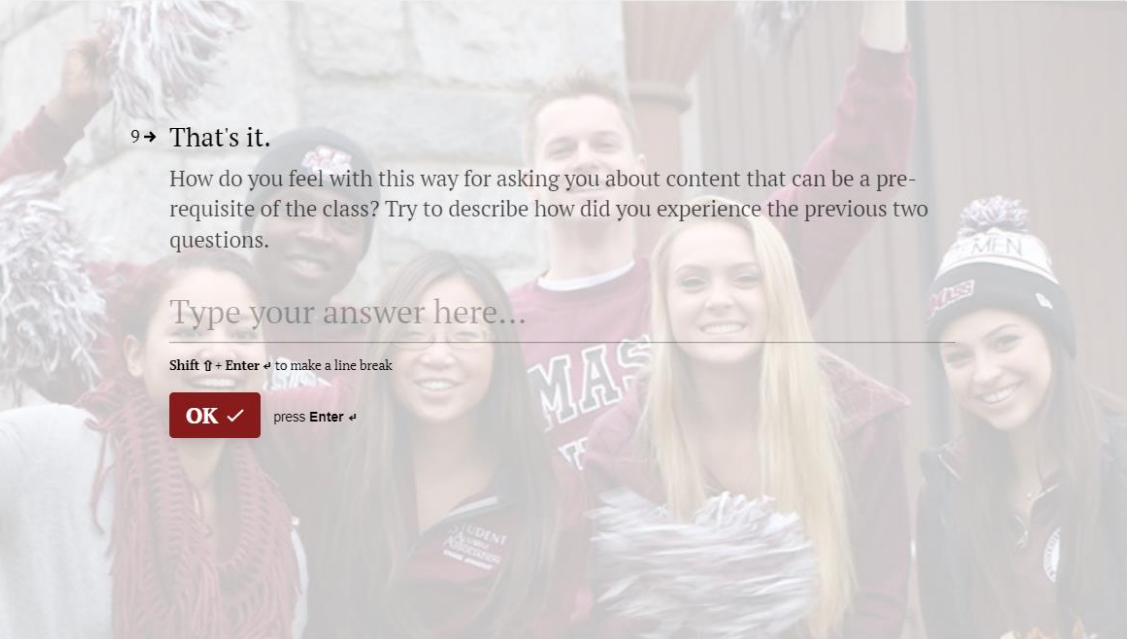
☐ G measurable function

☐ H sample space

☐ I measurable space

☐ J probability space

[Add choice](#)



9→ That's it.

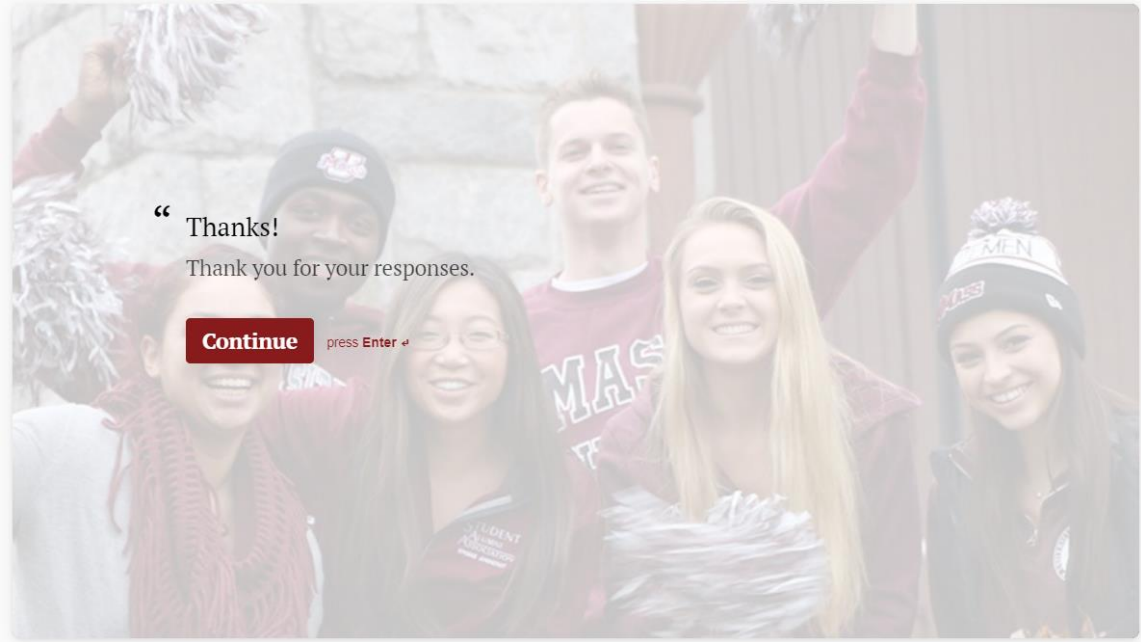
How do you feel with this way for asking you about content that can be a pre-requisite of the class? Try to describe how did you experience the previous two questions.

Type your answer here...

Shift ⌘ + Enter ↵ to make a line break

OK ✓

press Enter ↵



ANNEX B: DIAGNOSTIC TEST

1 → Diagnostic Test

EDUC 650 Regression Analysis

Hi! the purpose of this survey is to gather information about your baseline knowledge relative to the contents to be seen in class.

How are you feeling today?

Type your answer here...

“

...

This survey is meant to be anonymously responded, so no identification will be asked.

The results of this survey/test will be used by professor Keller to adapt the content of the class to your needs.

Continue

press **Enter** ↵

“ Video from professor Keller

Here is a video from professor Keller to cheer you up!



Link to Video: <https://youtu.be/dgVC8bgsbxg>

“ Time and start.

Let's get started!

There is no limit to the time you can spend responding to the questions.

Press Continue to begin the test!



2 → What words or ideas you do recognize from the following definition?

A sampling distribution is a probability distribution of a statistic obtained from a larger number of samples drawn from a specific population. The sampling distribution of a given population is the distribution of frequencies of a range of different outcomes that could possibly occur for a statistic of a population.

Source: <https://www.investopedia.com/terms/s/sampling-distribution.asp>

Choose as many as you like

☐ A probability distribution

☐ B statistic

☐ C sample

☐

Choose as many as you like

☐ A probability distribution

☐ B statistic

☒ C sample

☐ D population

☐ E frequencies

[Add choice](#)

3→ What words or ideas you do recognize from the following definition?

What Is P-Value?

In statistics, the p-value is the probability of obtaining results at least as extreme as the observed results of a statistical hypothesis test, assuming that the null hypothesis is correct. The p-value is used as an alternative to rejection points to provide the smallest level of significance at which the null hypothesis would be rejected. A smaller p-value means that there is stronger evidence in favor of the alternative hypothesis.

<https://www.investopedia.com/terms/p/p-value.asp>

Choose as many as you like

☐ A observed results

☐ B hypothesis test

Choose as many as you like

☐ A observed results

☐ B hypothesis test

☐ C null hypothesis

☐ D rejection point

☐ E level of significance

☐ F null hypothesis

☐ G alternative hypothesis

[Add choice](#)

press **Enter** ↵

Choose as many as you like

HYPOTHESES

Choose as many as you like

- ☐ A viewpoint
- ☐ B statement
- ☐ C population
- ☐ D argument
- ☐ E reasonable doubt
- ☐ F claim

Add choice

5 → What words or ideas you do recognize from the following definition?

In statistics, a confidence interval (CI) is a range of estimates for an unknown parameter, defined as an interval with a lower bound and an upper bound (notwithstanding one-sided confidence intervals, which are bounded only on one side). The interval is computed at a designated confidence level. The 95% confidence level is most common, but other levels (such as 90% or 99%) are sometimes used.

https://en.wikipedia.org/wiki/Confidence_interval

Choose as many as you like

- ☐ A estimates
- ☐ B parameter

Choose as many as you like

- ☐ A estimates
- ☐ B parameter
- ☐ C lower bound
- ☐ D upper bound
- ☐ E one-sided confidence intervals
- ☐ F computed
- ☐ G confidence level

Add choice

“Awesome.
Let's move to a new concept. This is the last round of concepts.

Continue press Enter ↵

6→ Is everything going well so far? Are you having any technical trouble while responding to this test? Or maybe are you feeling bad and this is interfering with your responses to the test?

Description (optional)

Type your answer here...

Shift ⌵ + Enter ↵ to make a line break

OK ✓

press Enter ↵

“ Ok. Let's continue.

Description (optional)

Continue

press Enter ↵

7 → What words or ideas you do recognize from the following definition?

The central limit theorem (CLT) states that the distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population's distribution. Sample sizes equal to or greater than 30 are often considered sufficient for the CLT to hold.

https://www.investopedia.com/terms/c/central_limit_theorem.asp

Choose as many as you like

☐ A distribution

☐ B sample means

☐ C normal distribution

☐

https://www.investopedia.com/terms/c/central_limit_theorem.asp

Choose as many as you like

☐ A distribution

☐ B sample means

☐ C normal distribution

☐ D sample size

☐ E population's distribution

☐ F sufficient

[Add choice](#)

8 → What words or ideas you do recognize from the following definition?

A test statistic is a statistic (a quantity derived from the sample) used in statistical hypothesis testing. A hypothesis test is typically specified in terms of a test statistic, considered as a numerical summary of a data-set that reduces the data to one value that can be used to perform the hypothesis test.

In general, a test statistic is selected or defined in such a way as to quantify, within observed data, behaviors that would distinguish the null from the alternative hypothesis, where such an alternative is prescribed, or that would characterize the null hypothesis if there is no explicitly stated alternative hypothesis.

https://en.wikipedia.org/wiki/Test_statistic

Choose as many as you like

Choose as many as you like

☐ A quantity derived

☐ B statistical hypothesis testing

☐ C data-set

☐ D quantify

☐ E observed data

☐ F distinguish the null from the alternative hypothesis

☐ G prescribed

☐ H characterize

☐ I stated alternative hypothesis

[Add choice](#)

9 → That's it.

Thank you very much for your responses! As an optional last question, give us your feedback about your overall assessment experience.

Type your answer here...

Shift ⏏ + Enter ↵ to make a line break

OK ✓

press Enter ↵

10 → That's it.

Thank you very much for your responses! As an optional last question, give us your feedback about your overall assessment experience.



ANNEX C: SURVEY TO PRIORITIZE INVALIDITY HYPOTHESES (IN SPANISH)

Cuéntame sobre tu experiencia: un estudio sobre hipótesis de invalidez mencionadas por usuarios en Twitter.

Hola, mi nombre es Sergio Araneda. Soy un candidato a Doctor de la Universidad de Massachusetts Amherst y quiero hacer un estudio sobre la Prueba de Transición (mejor conocida como PDT, que reemplazó a la prueba PSU ese año), y como la pandemia del COVID-19 afectó la validez de la interpretación y usos del test tomado en Enero de 2021.

Analizamos los temas mencionados por los usuarios en Twitter, y creamos una lista de potenciales problemas de la prueba a partir de este análisis. Queremos que nos ayudes a priorizar los estudios que encontramos, con el fin de hacer nuestra investigación más participativa.

¡Gracias de antemano por responder esta encuesta!



sondaxius@gmail.com (no compartidos) [Cambiar de cuenta](#)



¿A qué grupo perteneces?

- ☐ Soy un estudiante que rindió la PDT en Enero 2021
- ☐ Soy un estudiante que rindió PDT, PSU o PAA en el pasado, pero no en Enero 2021
- ☐ Soy un estudiante de enseñanza media preparándome para rendir la PAES este año (2022)
- ☐ Soy un estudiante de enseñanza media, aún no preparándome para rendir la prueba PAES (1ero a 3ero medio)
- ☐ Soy profesor/a de enseñanza media
- ☐ Soy profesor/a de un preuniversitario
- ☐ Soy parte de una oficina de admisión en una institución de educación superior
- ☐ Soy padre o madre de un estudiante que rindió la PDT o PSU en los últimos cinco años
- ☐ Soy padre o madre de un estudiante que cursa la enseñanza media
- ☐ Público General
- ☐ Otro: _____



¿Cuál es tu edad?

Tu respuesta _____



[Siguiente](#)



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Cuéntame sobre tu experiencia: un estudio sobre hipótesis de invalidez mencionadas por usuarios en Twitter.

 sondaxius@gmail.com (no compartidos) [Cambiar de cuenta](#)



*Obligatorio

Materiales de preparación

Las siguientes hipótesis corresponden a potenciales problemas con los materiales de preparación para la PDT ese año.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021.

*

Urgente Importante Relevante No muy relevante Irrelevante

Una versión online de un ensayo de la prueba tuvo problemas en algunos navegadores.

☒☐☐☐☐

Los materiales de preparación oficiales no respetaron las tablas de especificación de las pruebas.

☐☐☐☐☐

Una pregunta en los materiales de preparación de la prueba de historia tenía un error en su alternativa

☐☐☐☐☐

Algunos preuniversitarios no cubren realmente el contenido de la prueba.

☐☐☐☐☐

Los materiales de preparación de la prueba no estuvieron alineados con los contenidos que finalmente se preguntaron en la prueba de matemática.

☐☐☐☐☐

Las clases en persona proveyeron una ventaja por sobre los alumnos que tuvieron clases online.

☐☐☐☐☐

Los estudiantes de escuelas privadas pudieron dedicarse a la preparación de la prueba por mucho más tiempo que sus pares en escuelas públicas.

☐☐☐☐☐

Los estudiantes que presentaron síntomas de depresión debido al contexto de la pandemia vieron afectada su capacidad para preparar la prueba, afectando su performance y la equidad de los resultados.

☐☐☐☐☐[Atrás](#)[Siguiende](#)

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Información sobre la prueba

Las siguientes hipótesis corresponden a potenciales problemas con la forma en que información sobre la prueba PDT fue comunicada durante el periodo Abril 2020 - Marzo 2021.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021. *

Urgente Importante Relevante No muy relevante Irrelevante

Limitaciones en el acceso al simulador de puntajes ranking afectó la elección de programas en proceso de admisión de los estudiantes.

☐☐☐☐☐

Los estudiantes confundieron el nombre de la prueba PDT con la prueba PSU, llevándolos a usar materiales de preparación de la prueba que eran inadecuados.

☐☐☐☐☐

Demoras en los servicios de call center llevaron a algunos estudiantes a no poder tomar su prueba.

☐☐☐☐☐

Las expectativas de que la prueba presenta situaciones similares a la del año anterior hizo que los estudiantes estuvieran menos preparados que usualmente.

☐☐☐☐☐

La falta de información sobre el aumento en los ingresos especiales hizo que los estudiantes sub-utilizaran estos beneficios.

☐☐☐☐☐

Discordancias en los pesos de los puntajes de las pruebas PDT entre el sitio web oficial de la postulación y los sitios web específicos de cada carrera llevaron a los estudiantes a tomar decisiones incorrectas en su proceso de admisión.



Malos entendidos en la lógica para el cálculo del puntaje NEM llevó a estudiantes a tomar decisiones equivocadas durante sus años de enseñanza media.



Algunas personas aceptadas en el programa PACE no entendieron su estado desde la información provista en el sitio web.

☐☐☐☐☐

Problemas en los canales de comunicación y mesa de ayuda hicieron que ciertas informaciones no llegaran a tiempo y afectan la performance de los estudiantes en la prueba.

☐☐☐☐☐

Algunos estudiantes no estaban al tanto de la existencia del puntaje ranking, afectando sus decisiones de admisión.

☐☐☐☐☐

Estudiantes que
viven en
distritos que
tenían
restricciones de
movilidad
relacionadas
con la pandemia
vieron sus
performances
afectadas o
influidas por
falta de
información.



Los estudiantes
no siempre
entendieron las
reglas
relacionadas al
uso de puntajes
de años
anteriores,
distorsionando
su proceso de
postulación.



Un malentendido sobre la naturaleza transitoria de la prueba pudo haber impactado la performance de algunos alumnos.

☐☐☐☐☐

Un aparente cambio en las políticas de becas llevó a algunos estudiantes a ser sorprendidos de que debían pagar por la prueba.

☐☐☐☐☐

Algunos examinados no tienen suficiente manejo de tecnologías, teniendo problemas para encontrar información online.

☐☐☐☐☐

Confusiones
sobre
información
relacionada con
el nivel socio-
económico
afectado
durante la
pandemia llevó
a algunos
estudiantes a
tener acceso
limitado a
fuentes de
financiamiento.



[Atrás](#)

[Siguienie](#)



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Procedimientos en la toma de prueba

Las siguientes hipótesis corresponden a potenciales problemas con la forma en que las pruebas fueron tomadas en Enero de 2021.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021. *

	Urgente	Importante	Relevante	No muy relevante	Irrelevante
Los celulares crearon circunstancias en las salas donde se tomaba el test, afectando la performance de los alumnos presentes.	☐	☐	☐	☐	☐
Las limitaciones sobre los útiles que los alumnos pueden traer para dar la prueba afecta su performance.	☐	☐	☐	☐	☐

Los protocolos no fueron lo suficientemente buenos para contener el número de alumnos infectados con covid, y potencialmente esto generó contagios debido a que la prueba fue dada en persona.

☐☐☐☐☐

El número de instancias para dar la prueba no fue suficiente para los estudiantes.

☐☐☐☐☐

Falta de claridad respecto a las reglas sobre personas que fueron contacto estrecho de personas infectadas con covid confundieron a los alumnos, afectando su performance o incentivándolos a que no tomarán la prueba.

☐☐☐☐☐

Los altos tiempos de espera para entrar a las salas afectó la performance de los estudiantes.

☐☐☐☐☐

Los lugares de
rendición en
donde había una
expectativa de
protestas
(respecto al año
anterior) vio a
sus estudiantes
afectados en
sus
performances al
dar la prueba.

☐☐☐☐☐

Estudiantes de
escuelas rurales
fueron más
reacios a tomar
la prueba en
persona, debido
a que un viaje a
la ciudad los
expone a un
mayor riesgo.

☐☐☐☐☐

Los estudiantes
vieron afectadas
su performances
en la prueba
debido a la
asignación o
cambios de
lugar de
examinación.

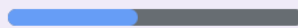
☐☐☐☐☐

Los estudiantes
vieron afectada
su performance
debido al uso de
mascarillas.

☐☐☐☐☐

[Atrás](#)

[Siguiente](#)



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Batería de pruebas

Las siguientes hipótesis corresponden a potenciales problemas con todas las pruebas PDT, tomadas en Enero 2021.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021. *

	Urgente	Importante	Relevante	No muy relevante	Irrelevante
La ansiedad de los alumnos afectó su performance en la prueba.	☐	☐	☐	☐	☐
Pensamiento crítico debería ser parte de la batería de pruebas para la admisión universitaria.	☐	☐	☐	☐	☐
La falta de respuestas a los problemas de los examinados llevó a que sus performances en la prueba se vieran afectadas por efectos dialógicos.	☐	☐	☐	☐	☐

Los estudiantes
están siendo
menos
preparados para
la universidad con
el paso de los
años.

☐☐☐☐☐

La dificultad de la
prueba es baja, y
esta es la real
explicación detrás
de la reducción
de gaps entre
tipos de escuela.

☐☐☐☐☐

Acceso a la
tecnología,
correlacionado
con situación
socio-económica,
explica la buena
performance de
algunos alumnos
en la prueba.

☐☐☐☐☐

El tipo de desayuno que toman los estudiantes afecta su performance en la prueba.

☐☐☐☐☐

El dominio de contenidos debió ser adaptado a las situaciones especiales relacionadas con la pandemia.

☐☐☐☐☐

El grupo 1 estuvo en desventaja respecto al grupo 2 de la prueba, debido a las filtraciones de la prueba en el grupo 1.

☐☐☐☐☐

Los liceos emblemáticos se vieron particularmente afectados por las condiciones especiales de la pandemia.

☐☐☐☐☐

Los estudiantes que reciben bajos puntajes pueden experimentar “crisis existenciales” y otros tipos de estrés psicológico.

☐☐☐☐☐

Los estudiantes que les va mal en la prueba y ya tenían expectativas negativas, tienen un impacto en sus futuras decisiones de admisión y de postulación en el futuro.

☐☐☐☐☐

No hubo suficiente transparencia sobre las respuestas correctas de las preguntas en las pruebas PDT, ni de las explicaciones detrás de cada respuesta correcta.

☐☐☐☐☐

Los creadores de la prueba no tienen conexión con las salas de clases.

☐☐☐☐☐

Los cambios en la prueba han sido motivados por políticas de izquierda.

☐☐☐☐☐

Los cambios en la prueba corresponden a una decisión social/política de dejar a los alumnos pobres ignorantes al hacer la prueba más fácil.

☐☐☐☐☐

La prueba de transición PDT fue una solución parche a los problemas en el sistema de admisión de la antigua prueba PSU.

☐☐☐☐☐

El test es considerado una máquina "deshumanizante"

☐☐☐☐☐

El test responde a una visión neoliberal de la educación.

☐☐☐☐☐

El test representa para los estudiantes el fin de una etapa de sus vidas, volviendolo un momento significativo.

☐☐☐☐☐

Los estudiantes prefieren el tipo de preguntas nuevas, focalizadas en competencias en vez de adquisición de contenidos.

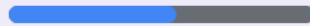
☐☐☐☐☐

Los estudiantes hubieran preferido tener una prueba online en vez de una en persona.

☐☐☐☐☐

[Atrás](#)

[Siguiente](#)



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Problemas específicos para ciertas pruebas

Las siguientes hipótesis corresponden a potenciales problemas que son específicos para alguna de las pruebas PDT, tomadas en Enero de 2021.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021. *

Urgente Importante Relevante No muy relevante Irrelevante

El test de lenguaje no mide habilidades de escritura, llevando a algunos alumnos que tienen buen puntaje en la prueba a tener mal desempeño en la universidad.

☐☐☐☐☐

Algunos textos en la prueba de lenguaje habían tenido demasiada exposición en materiales de preparación de la prueba, o en versiones anteriores de esta.

☐☐☐☐☐

Algunos
estudiantes
hubieran
preferido tener
textos más
entretenidos
cuando estos son
muy largos.

☐☐☐☐☐

Algunos
estudiantes no
ven una
correspondencia
entre el esfuerzo
hecho para
preparar la
prueba de
matemática y los
resultados que
obtuvieron en
esta.

☐☐☐☐☐

La prueba de matemática no es capaz de reconocer procesos de respuesta, volviendo sus resultados menos representativos de las reales capacidades de los alumnos.

☐☐☐☐☐

Preguntas sobre índice de refracción en la prueba de ciencias tienen un problema de oportunidad de aprender (no todos los alumnos tuvieron una chance justa de aprender este concepto).

☐☐☐☐☐

Varias preguntas en el test de Historia no tenían una respuesta correcta de acuerdo a algunos examinados.

☐☐☐☐☐

Una pregunta en el test de historia relacionado con uso de energía no tenía una leyenda en su gráfico, confundiendo a los estudiantes.

☐☐☐☐☐

La prueba PDT de Historia al parecer no respecto a la tabla de especificaciones.

☐☐☐☐☐

Hubo una pregunta en la prueba de historia que al parecer no tenía respuesta correcta.

☐☐☐☐☐

Debido a la falta de información relacionada con la prueba de historia, algunos estudiantes no pudieron postular debido a que los programas volvieron a requerir la prueba de historia y los estudiantes no se registraron para darla.

☐☐☐☐☐

Algunos
estudiantes
vieron afectada
su performance
por una pregunta
que llamaba al
régimen de
Pinochet
"gobierno militar"
en vez de
"dictadura".

[Atrás](#)[Siguiende](#)

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Puntajes obtenidos

Las siguientes hipótesis corresponden a potenciales problemas con los puntajes obtenidos luego de rendir las pruebas PDT en Enero de 2021.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021. *

Urgente Importante Relevante No muy relevante Irrelevante

Expectativas negativas de los estudiantes respecto a sus puntajes en la prueba crean una profecía autocumplida. Las expectativas laborales pueden estar moderando este efecto.

☐ ☐ ☐ ☐ ☐

Conexiones personales con la fecha en que se da la prueba pueden afectar la performance de algunos estudiantes.

☐ ☒ ☐ ☐ ☐

Algunos
estudiantes
tomaron la
prueba en un
estado de
apatía.

☐☐☐☐☐

Bullying en
persona o online
puede tener un
impacto en la
autopercepción
y motivación de
algunos
estudiantes,
afectando su
performance en
la prueba.

☐☐☐☐☐

Stress y falta de
preparación
debido a la
pandemia covid
afectó la
performance de
los estudiantes.

☐☐☐☐☐

Los estudiantes
que no pudieron
dormir bien
antes de la
prueba vieron
sus
performances
afectadas.

☐☐☐☐☐

Los estudiantes
que le temen a
su estado
emocional luego
de dar la prueba
pueden ver su
performance
afectada.

☐☐☐☐☐

Los estudiantes
que sienten que
no están
preparados para
la prueba
pueden tener
una presión
extra al darla,
afectando su
performance.

☐☐☐☐☐

Los estudiantes que usaron lentes al dar la prueba puede que hayan visto su performance afectada de manera especial por el uso de mascarillas.

☐☐☐☐☐

Los estudiantes que usaron una estrategia de azar inteligente tuvieron de todas maneras una buena posibilidad de entrar a la universidad.

☐☐☐☐☐

Los estudiantes que tienen altas expectativas en ellos mismos pueden ver sus performances afectadas.

☐☐☐☐☐

Estudiantes con bajas expectativas de éxito en la admisión universitaria pueden ser sus performances afectadas.

☐☐☐☐☐

Estudiantes con problemas para navegar las incertezas sobre su futuro pueden ver sus performances afectadas, moderadas por ansiedad.

☐☐☐☐☐

Los puntajes no son predictivos de la performance de los estudiantes en la universidad.

☐☐☐☐☐

La percepción de ser parte de un "experimento" con la transición de la prueba llevó a ciertos estudiantes a no tomarla, o ver su performance afectada.



La amenaza sobre la salud de los estudiantes afectó su performance en la prueba.



Un puntaje
calculado con
una lógica
normativa
impide que los
estudiantes
sepan cuántas
respuestas
correctas
necesitan tener
para obtener
cierto puntaje.
Algunos
estudiantes
preferirían saber
esto de
antemano.

☐☐☐☐☐

El puntaje
dicotómico (0 o
1) es menos
preferido por los
estudiantes que
los puntajes
parciales.

☐☐☐☐☐[Atrás](#)[Siguiente](#)

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Sistema de Admisión

Las siguientes hipótesis corresponden a potenciales problemas del sistema de admisión en general, que usa los puntajes obtenidos en las pruebas PDT rendidas en Enero de 2021.

Por favor selecciona la opción que representa de mejor manera tu opinión personal sobre la relevancia de los siguientes problemas potenciales con la prueba PDT tomada en Enero de 2021. *

Urgente Importante Relevante No muy relevante Irrelevante

Las medidas tomadas para limitar el impacto de los problemas de la prueba en el sistema de admisión fueron insuficientes.

☐☐☐☐☐

Las clases online afectaron la representación normal del puntaje NEM.

☐☐☐☐☐

El sitio web para la recuperación de contraseñas no estaba soportado por ciertos navegadores, bloqueando a algunos estudiantes en su proceso de selección de carreras.

☐☐☐☐☐

Problemas con el algoritmo de selección bloquearon a algunos estudiantes para ser aceptados en los programas a los cuales postularon.

☐☐☐☐☐

Las escuelas a veces colocan notas usando materiales de preparación para la prueba, lo cual puede generar una correlación espuria entre el puntaje NEM y los puntajes en las pruebas PDT.

☐☐☐☐☐

Algunos estudiantes no pudieron registrarse debido a problemas en el sistema de pagos.

☐☐☐☐☐

Los estudiantes subestiman la importancia de aspectos vocacionales en sus decisiones de admisión, llevándolos a tomar malas decisiones y desertar tempranamente en sus programas.

☐☐☐☐☐

La cantidad de vías de admisión alternativas no fue suficiente para las necesidades específicas del año de la pandemia de COVID-19.

☐☐☐☐☐

El cambio de nombre de la prueba afectó la preparación de los estudiantes, y posterior performance en la prueba.

☐☐☐☐☐

Las fechas definidas para el registro de los alumnos a la prueba no fueron las mejores, debido a que los alumnos usualmente tienen dinero cercano al inicio de cada mes.

☐☐☐☐☐

Alumnos
extranjeros
tuvieron
problemas para
entrar en el
proceso de
admisión debido
a la falta de
identificaciones
chilenas.

☐☐☐☐☐

La información
sobre la
situación
financiera de los
alumnos no
siempre fue
correctamente
ingresada,
afectando sus
decisiones de
postulación.

☐☐☐☐☐

No fue
apropiado
realizar el test
durante la
pandemia.

☐☐☐☐☐

No fue apropiado cambiar la tabla de conversión de notas y puntaje ranking ese año.

☐☐☐☐☐

Eventos en la vida de los alumnos durante su tiempo en la enseñanza media pueden tener un impacto permanente en el puntaje NEM de un alumno, afectando para siempre su probabilidad de éxito en el ingreso en el futuro.

☐☐☐☐☐

El impacto total de la pandemia afectó altamente la equidad de la admisión ese año.

☐☐☐☐☐

Problemas con la información relacionada al financiamiento afectó las decisiones de los alumnos en su postulación.

☐☐☐☐☐

Problemas con las identificaciones de los alumnos bloquearon a algunos de ellos para tomar la prueba.

☐☐☐☐☐

Algunos alumnos no recibieron su puntaje NEM a tiempo, afectando su proceso de postulación.

☐☐☐☐☐

Estudiantes de la comunidad lgbt+ vieron su performance en la prueba afectados por el rechazo en redes sociales sobre el uso de nombres sociales para personas trans.

☐☐☐☐☐

Algunos estudiantes tuvieron que dejar sus estudios debido a los problemas económicos relacionados con la pandemia, limitando su tiempo disponible para prepararse.

☐☐☐☐☐

Los estudiantes se vieron sobrecargados durante su último año de la enseñanza media. Esto, combinado con el contexto de la pandemia, afectó su capacidad para prepararse para la prueba.

☐☐☐☐☐

Los estudiantes
que no quisieron
dar la prueba
luego de
registrarse no
recibieron
devolución de su
cuota de
inscripción.

☐☐☐☐☐

Algunos
estudiantes
decidieron no
dar la prueba
debido a que los
protocolos y
condiciones para
dar la prueba no
eran
suficientemente
seguras en los
ojos de esos
estudiantes y
sus familias.

☐☐☐☐☐

Los estudiantes
que tuvieron
problemas
relacionados con
la pandemia
tuvieron además
problemas para
estar al tanto de
las fechas claves
en el proceso de
registro del
sistema de
admisión.

☐☐☐☐☐

Los estudiantes
que se
manifestaron de
forma violenta
debieron haber
sido hechos
responsables por
sus actos.

☐☐☐☐☐

Los estudiantes
que debían
realizar trámites
en persona se
vieron en
situaciones de
riesgo de su
salud.

☐☐☐☐☐

La preparación de los profesores para guiar a sus alumnos en el sistema de admisión es un moderador de la performance de los estudiantes y de su éxito en sus postulaciones.

☐☐☐☐☐

El sistema de admisión no se adaptó a las circunstancias de la pandemia.

☐☐☐☐☐

La fecha escogida para realizar la prueba generaba problemas para los alumnos.

☐☐☐☐☐

La cuota de inscripción crea una desventaja para alumnos de estrato socioeconómico bajo de cohortes antiguas, que no tienen beneficio para tener esta cuota cubierta.

☐☐☐☐☐

La inclusión de las notas de cuarto medio en el puntaje NEM creó desigualdad en el proceso de admisión.

☐☐☐☐☐

La reducción del peso de los puntajes de las pruebas en los puntajes ponderados de las carreras afectó a los estudiantes que tenían expectativas de compensar con estos puntajes su bajo puntaje NEM.



Reflexión sobre el rol de las instituciones de educación superior en la construcción del país a través del trabajo hecho sobre el sistema de admisión y la administración de la prueba.



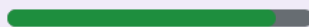
La falta de claridad en los roles de DEMRE y el gobierno en el sistema de admisión confundió a los estudiantes y afectó su proceso para hacer sentido de sus experiencias de evaluación.

☐☐☐☐☐

Algunos examinados alegaron sobre la expectativa generada en el anuncio sobre el peso mínimo de las pruebas PDT de un 30%, que fue entendido como un máximo.

☐☐☐☐☐

Algunos estudiantes no se sienten cómodos con la velocidad de los cambios en el sistema de admisión.

☐☐☐☐☐[Atrás](#)[Siguiente](#)

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Eso es Todo.

Muchas gracias por responder esta encuesta!

¿Cómo fue tu experiencia al responder esta encuesta?

	1	2	3	4	5	
Muy Mala	☐	☐	☐	☐	☐	Muy Buena

¿Tienes algún feedback que quieras compartir?

Tu respuesta



[Atrás](#)

[Enviar](#)

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ANNEX D: LIST OF OVERARCHING HYPOTHESES

Device	P.A.A.P.	Overarching Hypothesis	Invalidity Hypothesis	Mean	Standard deviation
Admission system	Adequacy	The rules, procedures and technology in the Admission system are Adequate for the selection of students for Higher Education	Measures taken to limit the impact of problems with the test in the admission system were insufficient Online classes affected the normal representation of high-school GPA Password recovery is not supported for some browsers and blocked students in their application process Problems with the matching algorithm blocked students to be accepted in their programs Schools use test pre materials to put grades, which can make high school gpa and test results spuriously correlated Students could not register to take the test due to problems in the payment system Students underestimate the value of true vocation in their admission decisions, leading to poor career choices and future drop outs The amount of alternative entry gates for college admission was not enough for the needs specific of the covid pandemic that year The change of name affected student's preparation or performance in the test	3.80	0.512739
Admission system	Appropriateness	The Admission System abides to the law	Students opted out after they registered, not receiving a refund	3.30	--
Admission system	Appropriateness	The Admission System abides to the values of the communities affected by the uses of the test	Students that manifested themselves through violence should have been made responsible	3.23	--
Admission system	Appropriateness	The Admission System propends an Equitable Access to Higher Education	dates defined for registration were not the best ones for students, that usually have money at the beginning of the month Foreigner students had problems to enter in the process due to their lack of Chilean ids	3.69	0.406299

Admission system	Appropriateness	The Admission System propends to a Fair Access to Higher Education	Information about financial situation was not always correctly input by students and affected admission decisions Problems with information about funding affected admission decisions Students had to leave their studies because of economical struggles related to the pandemic, limiting their available time to prepare Teacher preparation to guide students is a moderator for student performance and successful application The fee to take the test creates a disadvantage for lower SES Students from older cohorts that do not have any benefit to waive the payment Life events during high school years have an impact in high school GPA, which has a permanent effect on students that hinder their chances in the future. Problems with student ids blocked students to take the test Some students did not receive their NEM scores timely, affecting their admission process Students from lgbt communities saw their performances affected by the backlash in social media against the use of social names The reduction of the weight in the test scores affected students that had low high school gpa and had an expectation about compensating with the test scores	3.79	0.250111
Admission system	Appropriateness	The problems arising from the pandemic were resolved successfully	It was not appropriate to do the test during the pandemic It was not appropriate to change the conversion tables of hsgpa and ranking that year Students had too many things in their hands during their last year. That, combined with the covid context, hindered their chances to prepare to the test Students opted-out and refused to take the test because the protocols and conditions were not safe enough in the eyes of students and their families Students struggling with pandemic related situations had problems to realize the limit dates for registration to the test Students that required to do in person paperwork saw themselves in a risky situation in terms of their health	3.43	0.419582

			The admission system did not adapt to the circumstances of the pandemic in their admission process that year The date chosen to take the test conflicted students The inclusion of hsgpa in its last year created inequity in the admission process The overall impact of the pandemic highly affected the equity of the admission that year		
Admission system	Partiality	About the roles of institutions in the test development and admission process	Reflection about the role of older higher education institutions in the construction of the country through their work in the admission system and test administration Unclear roles in the admission process between demre and government confused students and affected their sense-making	3.43	0.282842
Admission system	Preferability	The structure of the Admission System is preferable to other alternatives	Some examinees complained about the expectation generated by the announcement of 30% PDT as minimum, understood as maximum. Some students are not ok with the speed of the changes in the admission system	3.43	0.235702
History Test	Adequacy	The History test was adequate for its purposes	Many questions in the History test did not have a correct response according to some examinees One item in the history test related to energy use did not have a legend, confusing students PDT History test did not respected the table of specifications There was one item in the history test that did not have a correct answer	3.97	0.115470
History Test	Appropriateness	The History test was appropriate for its purposes	Due to misinformation related to the history test, students were put aside of the admission process, impacting the overall admission process on programs that require the history test Some students saw affected their performance in an item that called Pinochet's regime as "military government" instead of dictatorship.	3.57	0.659966
Information	Adequacy	Information regarding the tests and the admission system was correctly and timely delivered	A confusion about how different PDT tests were compared to PSU tests led students to use test prep materials that were inadequate An apparent change in the policies for fellowships misled students that now had to pay but had an exception in the past Confusion around SES information before pandemic or during pandemic led student to have limited access to funding	3.52	0.361738

			Delays in the call-center service led to students not being able to take the test Expectations on the test having a similar situation to the previous year made students be less prepared than usual. Lack of information about an increase in special entry programs led students to under utilize those benefits Limitation to access to one score simulator affected the choice of programs in the admission process of students Mismatches between published weights by universities and the official website misled students in their admission process Misunderstandings in the logic of hsgpa calculation misled students in their decision making during high school years Problems in the communication channels and help desk made Information not timely given and affected student performance Some students were not aware of the existence or meaning of puntaje ranking, affecting their admission decisions, and planning Students living in districts that had covid restrictions were affected in their performances or misled by lack of information The misunderstanding about the transitional nature of the test impacted the performance of the students.		
Information	Appropriateness	Information regarding the tests and the admission system is accessible for all students	People accepted in the PACE program did not understand their status from the information provided in the website Some examinees are not technology savvy and had problems to find information online	3.48	0.258915
Language Test	Adequacy	The Language test was adequate for its purposes	Students not always understood the rules regarding the use of old scores, distorting their application process Language test does not measure writing skills, having students with good results in the test but poor performance on writing later in their college programs Some texts in the language test have had too much exposure in test prep materials or the test itself	3.47	0.518545
Language Test	Preferability	The Language test was preferable in comparison to other ways to test	Students would prefer more "fun" texts when those are long	3.13	--
Math test	Adequacy	The Math test was adequate for its purposes	Students do not see a correspondence between the amount of effort put into study to the math test versus their results in the test	3.42	0.353553

Science test	Appropriateness	The Science test was adequate for its purposes	The math test is not capable of recognizing response process accordingly, making the results not robust	3.57	--
Test Battery	Adequacy	The test battery covers the domains relevant for college admission decisions	Questions asking about refraction index in the science test had a problem of opportunity to learn Critical thinking should be part of the test battery but it is not part of it	3.60	0.047140
Test Battery	Appropriateness	The test battery adapted to the special needs generated by the pandemic	Students are becoming less prepared to college along the years. Content domain should have been adapted to the special situations of the covid pandemic	3.27	--
Test Battery	Appropriateness	The test battery promotes a Fair Access to higher education	Magnet schools were specially affected by covid conditions in their performances in the test There was not enough transparency about the correct responses and its explanations	3.10	0.848528
Test Battery	Appropriateness	The test battery promotes a positive assessment experience	Students that have low scores can experience "existential crisis" or other types of psychological distress Students that meet their negative expectations have an impact in future decisions and instances of the test The test is considered as a "dehumanizer" machine The test represent to students the end of a life stage and is meaningful for them	3.37	0.229330
Test Battery	Appropriateness	The test battery promotes an Equitable Access to higher education	Access to technology, correlated with SES, explain good performance on students Test difficulty went down and this is the real explanation behind reduction of gaps between types of school.	3.32	0.365655
Test Battery	Partiality	About the political influences on the decisions related to the test	Changes in the test are driven by left policies Changes in the test respond to a societal / political decision to leave poor students ignorant by making the test easier for them The test is responding to a neoliberal project in education	2.44	0.554109
Test Battery	Partiality	About the role of test developers and their connection to the classroom	Challenge to the administrative standpoint of test creators, and their lack of connection with classrooms	3.77	--
Test Battery	Preferability	The in person format of the test is preferable over online formats	Students would have preferred an online test versus one in person	2.87	--
Test Battery	Preferability	The test battery is preferable over the old test PSU	Students do prefer the new item format, that focuses on competences vs content acquisition	3.48	0.117851

Test prep	Adequacy	Official test prep materials were adequate so student could prepare to the test	The new test is considered a "patch solution" to the problems in the admission system and with the old test PSU A practice online version of the test had some problems in certain browsers Official test prep materials did not comply to Test specifications One question in the prep materials for the history test had a wrong key Some test prep programs do not cover the content really covered in the tests Test prep for the math test were not aligned with the actual content asked in the math test	3.85	0.378300
Test prep	Appropriateness	All students had a fair chance to prepare to the test	Classes in person provided an advantage to students vs online classes	3.80	--
Test Prep	Appropriateness	Official test prep materials promoted a Fair Access to Higher Education	Students from private schools have been dedicating to preparation of the test far longer than students from public schools with less time and resources to pair this. Students that presented depression symptoms due to the context of the pandemic saw their capacity to prepare to the test hindered, affecting their performance and the equity of the results	4.13	0.141421
Test procedures	Adequacy	Protocols and procedures promoted the best possible performance of students to take the test	Cellphones created circumstances that affected test performance in the room Limitations on what the students can have during the test affect their performance in the test Waiting times to enter the rooms for taking the test affected the performance of students	2.90	0.437162
Test procedures	Adequacy	Protocols and procedures were adequate to navigate the challenges the COVID-19 pandemic created	Protocols were not enough to contain the number of infected students and covid cases were created because of the in-person nature of the test Some students performance were affected by the use of masks The number of instances was not enough for students to have enough dates to show their performances Unclear rules related to close contact with covid infected people led students to confusion, affecting their performance or withdrawing from taking the test.	3.05	0.510990
Test procedures	Appropriateness	Protocols and procedures promoted a fair access to higher education	Group 1 was in disadvantage compared to Group 2 due to test leaks	3.51	0.175066

			Locations in which there was an expectation to protests had students performing less than normal Rural students were more adamant to take the take in person, as their travel to the city meant a higher risk of exposure Some students saw affected their performance for problems with the assignation or changes in their test locations Students using strategies with smart random chance still had good chances to enter to college Test scores are not predictive for student performance in college	3.02	0.824957
Test Scores Interpretation	Adequacy	Test score interpretations are meaningful and substantive, over and beyond what random response strategies can provide to students in the test			
Test Scores Interpretation	Adequacy	Test score interpretations can untangle engagement, self-efficacy and other sources of Construct-irrelevant variance related to Student's psychological states	Anxiety affected student performance in the tests Lack of responses to the problems of examinees led to affected performances due to dialogical effects Some students took the test in a state of apathy Students that fear about their emotional state after doing the test may see their performance affected Students that feel they are not prepared have extra pressure when doing the test and then under-performing Students with higher expectations on themselves can see their performance affected Students with low expectations on their success for college admission saw their performance affected Students with problems to navigate the uncertainties about the future can see their test performance affected, moderated by anxiety The perception of being part of an "experiment" with the transition test led students to do not take the test or affected performance Students that did not sleep well see their performance affected	3.32	0.213437
Test Scores Interpretation	Adequacy	Test score interpretations promote a Fair access to Higher Education		3.23	0.513160
Test Scores Interpretation	Adequacy	Test score interpretations were still adequate under the specific circumstances of the COVID-19 pandemic	Stress and lack of preparation due to the covid pandemic affected student performance Students that used glasses saw impacted their performance due to the use of masks The threat over the health of students affected their performances in the test	3.53	0.371184

Test Scores Interpretation	Appropriateness	Test score interpretations promote a Fair access to Higher Education	Personal connections with the exact date doing the test can affect student performance Spaces of bullying or cyber bullying can impact the self-perception and motivation of some students, affecting their test performance	3.23	0.513160
Test Scores Interpretation	Appropriateness	Test score interpretations promote an Equitable access to Higher Education	Negative expectations on a student can create a self-fulfilling prophecy. Job expectations can moderate this.	3.57	--
Test Scores Interpretation	Appropriateness	The test battery promotes an Equitable Access to higher education	Breakfast taken the morning of the test affect student performance	3.32	0.365655
Test Scores Interpretation	Preferability	Test score calculation and interpretation are preferable versus other alternative options	A score calculated based on a normative logic blocks students to know how many responses they need to have to achieve certain score, some students would prefer to know this in advance. Dichotomous scoring in the test are less preferable to partial scoring	3.22 3.22	0.353553 0.353553

ANNEX E: POSITIONALITY STATEMENT

I am Latino. I am from Chile, born and raised in Santiago. I am part of the global south. My personal experiences with educational assessment started with a test used when I was in kindergarten to be selected in a school. Later, my experiences continued with a test called SIMCE in Chile, used for accountability purposes. Also, my most memorable assessment experiences include a test called PSU, used for college admission in Chile. Chile during the 90s and 00s had the rise of a neoliberal project in which Standardized tests had an enormous role.

One implicit meta-value in this Dissertation is democracy as a political solution to the political problem of conflicting values in a society. This preference has a clear root in the experience of Chilean society and the trauma of living in a the Chile post-dictatorship. I appreciate democracy, and I think the universal audience will require such democratic deliberation in the communities affected by the uses of the test to decide about the appropriateness of using a test or for interpreting its scores in a certain way.

I am Gay. Feminist epistemologies are somehow a substitute for “gay” epistemologies. Feminists have done a much better job than gay scholars in explicating better epistemologies that can dismantle and undress structures of power. As a gay, I have suffered in many ways the oppression for having a non-mainstream sexual orientation, making me susceptible to the suffering of others in the hands of their oppressors.

Because of my sexual orientation and identity, I can understand that diversity is a value on its own. There has been an increasing acceptance of the LGBT+ communities in western cultures; however, it can be accepted in a way that creates a norm and a border

for what type of LGBT+ people is accepted and what type is not. True support for the LGBT+ community is to accept diversity as a value, meaning that each sexual orientation is accepted. Diversity can be considered a second meta-value present in this paper.

I am white. I have been privileged in so many ways. Chile can be a very racist place at some point; however, the discourses about oppression usually focus on socio-economical differences. Latinoamerica has a long tradition of Eurocentric preferences coming from its colonial roots. I have profited from those preferences in many ways, most of the time without noticing them. Being in the United States has helped me to become more aware of racism, hidden in many of the stories that I had in my mind to understand oppression in Chile and beyond.

I am a man. I do have conflicts with my own gender identity, in the sense that I am a man but I refuse to enter in the logics of patriarchy. At some point, I have considered my gender identity as multiple, or patched. Some sort of calico identity, in which I have elements of a man but also of woman. I have explored the idea of identifying as non-binary, however my gender expression is man and I feel comfortable with it so far. I can see how I have profited in the privileges of being a man, and how my own gaze sometimes falls once again in the historical infamies of the patriarchy. I suffer this.

Because I am a man, I do not think I can declare myself as a feminist. Feminism was born by, for and with women, and even if their work can serve other scholars, feminism in its more pure vocation is about and for women. I dare to say I am a fan and supporter of feminisms worldwide.

I am a democratic socialist. Since my undergraduate time, I have engaged in political movements, more recently in Chile, with a big social uprising happening in 2019. I feel myself part of a generation that wants to change the structures of oppression inherited from Pinochet's dictatorship time, deeply ingrained in the laws and culture of my country as an experiment of global neoliberalism. And beyond Chile, I am interested in fighting back other systems of oppression worldwide.

I am writing this paper at an extraordinary moment in history. We had the COVID-19 pandemic during the writing process of this Dissertation. We are also having a massive shift in the professional field of educational measurement, with some existential crises like the crisis that global warming brings upon us all. There is also a shift in power from traditional companies like ETS to new EdTech startups like Duolingo. The decision made by the University of California about not using standardized tests for college admission is a decision that will have ripple effects in the upcoming future, and this paper is fueled by certain need (if not desperation) to find better ways to create tests, and respond to the public in more effective ways.

In a more cynical tone, I am in the pursuit of methods for a sustainable testing industry, acting under the assumption that it is not sustainable to ignore the voices of the oppressed. In many ways the testing industry is a wrecking ship. But, how authentic is then this concern for participation and the inclusion of more voices? I do have two conflicting vocations in my gaze, one political and one practical, concerned also about money and how to find a job. I do not like this, but it is there.

I am writing this paper while I live in the United States. The specific struggles during the period 2019-2022 have influenced the writing of this paper. The election of

current president Biden competing with ex-president Trump, the Black-Lives-Matter movement during 2020, and the specific struggles of this country related to the COVID-19 pandemic have influenced the need to make validity about future instantiations and take in a more conscious way the temporal contingency of a test and its uses and interpretations.

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